NEURO MCQ AL HARRIF

{PART 1}



وَمَا تَوْفِيقِى إِلَّا بِاللَّهِ

استعن بالله ولا تعجز

اللَّهمَّ لا سَهلَ إلَّا ما جَعَلتَه سَهلًا، وأنتَ تَجعَلُ الحَزْنَ إذا شِئتَ سَهلًا

ملحوظة : هذه المذكرة شغل خاص بتيم الحريف ولا نسامح فى سرقتها او اخذ اى اسئلة منها

القناة الخاصة بالحريف https://t.me/ALHARRIF11

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Anatomy

Nerves of head and neck

1- Which of the following nerves arise from pons: {Summar2025}

a- CNX b- CNXII C-CNVII d-CNIX

2- The accessory nerve is {Mid2025}

a. Purely motor b. Purely sensory c. Gives motor innervation to one muscles

d. Gives motor innervation to muscles and sensory innervation of the neck

3-Root value of the lesser occipital nerve is: {Formative2025}

a) C4 b) C2 and C3 c) C2 d) C2, C3 and C4

4- Root of Cervical plexus {Mid2025}

a. From C4-C8 b. From C1 - C4 c. From C1- C5

5-Buccal branch of facial nerve is ....., While buccal branch of mandibular nerve is ..... {Mid2025}

a. sensory – motor b. motor – Sensory

6- Infraorbital Nerve is a branch of {Mid2025}

a. Opthalamic nerve b. Maxillary nerve c. Mandibular nerve d. Facial nerve

7- Inferior limb of ansa cervicalis Supply infrahyoid muscles is …….. {Final2025}

a. Descending cervicalis c2,3 b. Descending hypoglossi

8- Roots of cervical plexus pass anterior to.....muscle. {Mid2024}

a) Scalenus medius. b) Scalenus anterior. c) Scalenus posterior, d) Semispinalis capitis.

9-Maxillary nerve leaves the skull through: {Formative2024}

a) Foramen ovale b) Stylomastoid foramen

c) Foramen spinosum d) Foramen rotundum

10- Loss of sensation of skin over the tip of the nose is due to injury of {Mid2024}

a) Facial N. b) OlfactoryN c) Mandibular N d) Ophthalmic N

11- The only cranial nerve that enters cerebrum directly is: {Mid2024}

a) Optic nerve b) Olfactory nerve c) Trigeminal nerve d) Accessory nerve

12- Which of the following statements is INCORRECT ? The facial nerve: {قسم}{Final2023}

A) Supplies muscles of facial expression B) Supplies buccinator

C) Gives off the greater petrosal nerve D) Contains taste fibres

E) Contains fibres destined for the ciliary ganglion

13- All cranial nerves have peripheral attachment into ventral Aspect of Brain Stem EXCEPT {Final2021}

A- 5th cranial nerve B- 3rd cranial nerve C- 10th cranial nerve D- 4 th cranial nerve

14- Sensory supply of scalp behind the ear are the following EXCEPT {Mid2021}

A- Great auricular nerve (C2 and C3) B- Posterior auricular nerve

C- Greater occipital nerve (C2) D- Third occipital nerve (C3)

15- Concerning the cervical plexus the followings are TRUE EXCEPT {Mid2021}

A- Supplies infrahyoid muscles B- Supplies motor branches to trapezius

C- Provides sensory branches for the diaphragm

D- Provides sensory branches for the front of the scalp

16-What is the motor branch of the cervical plexus that supplies the diaphragm? {Formative2024}

a) Supraclavicular. b) Great auricular. c) Phrenic. d) Lesser occipital.

17- Special taste sensation from the tip of the tongue is carried by: {Mid2024}

a) Vagus. b) Lingual. c) Hypoglossal. d) Chorda tympani.

18-Cutaneous innervation of the forehead is provided by the: {Formative2024}{Mid2023}

a) Supratrochlear. b) External nasal. c) Infraorbital. d) Facial.

19-Nerve supply of the third pharyngeal nerve is: {Formative2024}

a) Facial nerve b) Mandibular nerve c) Glossopharyngeal nerve d) Vagus nerve

20- Regarding the facial nerve the followings are TRUE EXCEPT {Mid2021}

A- Supplies mylohyoid muscle

B- Has some fibers which are distributed in the lingual nerve

C- Transverses the stylomastoid foramen

D- Has branches running superficial to blood vessels in parotid gland

21- The vagus nerve leaves the skull through the : {Mid2021}{قسم}

A- Jugular foramen B- Occipital foramen

C- Foramen rotundum D- Foramen spinosum

22- The foramen magnum transmits the followings EXCEPT {Mid2021}

A- Cerebrospinal fluid B- The posterior spinal arteries

C- The vertebral arteries D- The cranial accessory nerve

23- Mandibular division of trigeminal nerve leaves skull through : {Mid2021}{قسم}

A- Foramen rotundum B- Foramen ovale C- Jugular foramen D- Foramen magnum

24- Pure Motor Cranial Nerves are {Final2021}

A- 5th, 7th, 9th, 10th& 11th B- 1st, 2nd& 8th

C- 3rd, 4th, 6th& 12th D- 3rd, 4th, 7th& 9th

25- The cervical plexus: {قسم}

a) Supplies infrahyoid muscles b) Supplies motor branches to trapezius

c) Provides sensory branches for the diaphragm

d) Provides sensory branches for the front of the scalp

26- The abducent nerve leaves the skull through the {قسم}

a) foramen rotundum. b) jugular foramen. c) inferior orbital fissure

d) superior orbital fissure. e) foramen ovale.

27- The ophthalmic division of the trigeminal nerve leaves the skull through the {قسم}

a) inferior orbital fissure. b) foramen ovale. c) foramen rotundum

d) superior orbital fissure. e) pterygopalatine foramen.

28- The maxillary division of the trigeminal nerve leaves the skull through the {قسم}

a) foramen spinosum. b) foramen rotundum. c) superior orbital fissure.

d) foramen ovale. e) jugular foramen.

29- The oculomotor nerve leaves the skull through the {قسم}

a) inferior orbital fissure. b) foramen rotundum. c) superior orbital fissure.

d) foramen magnum. e) foramen ovale.

30- When testing the sensory innervation of the face, it is important to remember that the skin of the tip of the nose is supplied by which one of the following nerves? {قسم}

a) Zygomatic branch of the facial nerve b) Maxillary division of the trigeminal nerve

c) Ophthalmic division of trigeminal nerve d) External nasal branch of facial nerve

e) Buccal branch of the mandibular division of the trigeminal nerve

31- Which of the following statements about ophthalmic nerve is INCORRECT ? {قسم}

A) supplies the conjunctiva B) supplies the mucous membrane of the nasal cavity

C) Supplies levator palpebrae superioris D) Has a frontal branch E) Is wholly sensory

32- Branches of the oculomotor nerve contain fibres the are : {قسم}

a) sensory to the cornea b) sympathetic to smooth muscle

c) parasympathetic to smooth muscle d) motor to skeletal muscle

33- The glossopharyngeal nerve: {قسم}

a) Is the IXth cranial nerve b) Emerges from the brainstem between pyramid and olive

c) Is attached to brainstem rostral to vagus. d) Contains no special visceral afferent fibres

34- The glossopharyngeal nerve: {قسم}

a) Grooves the back of the jugular tubercle of the occipital bone

b) Gives a branch to the carotid sinus

c) Supplies stylopharyngeus

d) Passes between the inferior and middle constrictors of the pharynx

35-The muscles of facial expression are supplied by the …… nerve

a. Glossopharyngeal b. Hypoglossal c. Mandibular d. Facial

36-The ophthalmic division of the trigeminal nerve

a. Supplies sensory fibres to the eyeball b. Supplies the orbicularis oculi muscle

c. Gives parasympathetic fibres to the lacrimal gland d. Carries sensory fibres for smell

37-Which of the following cranial nerves does not transmit taste information to the brain?

a. Trochlear (4th cranial nerve) b. Facial (7th cranial nerve)

c. Vagus (10th cranial nerve) d. Glossopharyngeal (9th cranial nerve)

38-Regarding the parasympathetic nerve supply to the structures in the head and neck:

a. Parasympathetic fibres in the facial nerve end in the otic ganglion.

b. The pterygopalatine ganglion sends fibres to the parotid gland

c. Parasympathetic fibres in the oculomotor nerve (III) control the lacrimal gland

d. The ciliary ganglion sends fibres to the circular muscle in the iris

39-The cervical plexus is formed by the ventral rami of

a. C1, C2, C3 b. C3, C4, C5 c. C1, C2, C3, C4

d. C3, C4, C5, C6 e. C5, C6, C7, C8, T1

40-Which of the following is NOT supplied by the glossopharyngeal nerve

a. The stylohyoid muscle b. The parotid gland

c. The palatine tonsil d. The stylopharyngeus muscle

41-The hypoglossal nerve supplies the following muscles, except

a. Intrinsic muscles of the tongue b. Hyoglossus c. Genioglossus d. Palatoglossus

42-The superior orbital fissure do not transmit

a. abducent nerve b. the maxillary division of the trigeminal nerve

c. ophthalmic nerve d. trochlear nerve e. oculomotor nerve

43-Which of the following nerves supplying scalp and face is not derived from trigeminal nerve?

a- Great auricular b- Auriculotemporal. c- Zygomaticotemporal. d- infraorbital

44-The buccal branch of the facial nerve is primarily:

A) Sensory B) Motor C) Mixed D) Parasympathetic

45-The buccal branch of the trigeminal nerve (V3) is:

A) Motor B) Sensory C) Autonomic D) Mixed

46-The stylopharyngeus muscle is innervated by:

A) Vagus nerve (CN X) B) Glossopharyngeal nerve (CN IX)

C) Hypoglossal nerve (CN XII) D) Accessory nerve (CN XI)

47-Which of the following correctly matches nerve and function?

A) Buccal branch of CN VII — sensory B) Buccal branch of CN V — motor

C) Buccal branch of CN VII — motor D) Buccal branch of CN V — parasympathetic

48-The recurrent laryngeal nerve is a branch of which cranial nerve?

A) Glossopharyngeal nerve B) Vagus nerve C) Hypoglossal nerve D) Accessory nerve

49-The buccal branch of the facial nerve supplies which structures?

A) Skin of cheek B) Muscles of facial expression around the mouth

C) Tongue D) Upper lip sensation

50-Which cranial nerves originate from the midbrain?

A) CN III and IV B) CN V and VI C) CN VII and VIII D) CN IX and X

51-Cranial nerves V, VI, VII, and VIII arise from the:

A) Midbrain B) Pons C) Medulla D) Cerebrum

52-Which muscle of the pharynx is NOT innervated by the vagus nerve?

A) Superior constrictor B) Middle constrictor

C) Stylopharyngeus D) Inferior constrictor

53-Which muscle of the tongue is NOT innervated by the hypoglossal nerve?

A) Genioglossus B) Hyoglossus C) Styloglossus D) Palatoglossus

54-The palatoglossus muscle is supplied by the:

A) Hypoglossal nerve B) Glossopharyngeal nerve

C) Pharyngeal plexus D) Trigeminal nerve

55-The trochlear nerve enters the orbit through the:

A) Optic canal B) Foramen rotundum

C) Superior orbital fissure D) Inferior orbital fissure

56-The superior oblique muscle is innervated by which cranial nerve?

A) Oculomotor nerve B) Trochlear nerve C) Abducens nerve D) Trigeminal nerve

57-All of the following nerves pass through the superior orbital fissure EXCEPT:

A) Oculomotor nerve B) Trochlear nerve C) Abducens nerve D) Optic nerve

58-Which is the largest cranial nerve?

A) Facial nerve B) Vagus nerve C) Trigeminal nerve D) Glossopharyngeal nerve

59-The accessory nerve is primarily:

A) Sensory B) Motor C) Mixed D) Autonomic

60-Which muscles are supplied by the accessory nerve?

A) Sternocleidomastoid and trapezius B) Masseter and temporalis

C) Lateral rectus and superior oblique D) Platysma and orbicularis oculi

61-The ophthalmic nerve enters the orbital cavity through:

A) Optic canal B) Superior orbital fissure

C) Inferior orbital fissure D) Foramen rotundum

62-The maxillary nerve (V2) leaves the cranial cavity through:

A) Foramen ovale B) Foramen rotundum

C) Superior orbital fissure D) Jugular foramen

63-The mandibular nerve (V3) leaves the skull through which foramen?

A) Foramen rotundum B) Foramen ovale C) Foramen spinosum D) Jugular foramen

64-The ophthalmic nerve (V1) is primarily:

A) Motor B) Sensory C) Mixed D) Autonomic

65-The maxillary nerve (V2) is primarily:

A) Motor B) Sensory C) Mixed D) Autonomic

66-The buccal nerve is a branch of:

A) Posterior division of mandibular nerve B) Anterior division of mandibular nerve

C) Maxillary nerve D) Ophthalmic nerve

67-The abducent nerve (CN VI) innervates which muscle?

A) Medial rectus B) Superior rectus C) Lateral rectus D) Inferior oblique

68-Which of the following is NOT a terminal branch of the facial nerve?

A) Temporal B) Zygomatic C) Buccal D) Lingual

69-The superior laryngeal nerve is a branch of which cranial nerve?

A) Glossopharyngeal nerve B) Vagus nerve C) Hypoglossal nerve D) Accessory nerve

70-The accessory nerve exits the skull through:

A) Foramen ovale B) Jugular foramen C) Hypoglossal canal D) Foramen rotundum

1. C	2. A	3. C	4. B	5. B
6. B	7. A	8. A	9. D	10.D
11.B	12.E	13.D	14.B	15.C,D
16.C	17.D	18.A	19.C	20.A
21.A	22.D	23.B	24.C	25.A
26.D	27.D	28.B	29.C	30.C
31.C	32.D	33.A,C	34.B,C	35.D
36.A	37.A	38.D	39.C	40.A
41.D	42.B	43.A	44.B	45.B

46.B	47.C	48.B	49.B	50.A
51.B	52.C	53.D	54.C	55.C
56.B	57.D	58.C	59.B	60.A
61.B	62.B	63.B	64.B	65.B
66.B	67.C	68.D	69.B	70.B

Blood supply and lymphatic of the head and neck

1-one of the following is Tributaries of the internal jugular vein: {Formative2026}

a) Vertebral vein b) Inferior thyroid vein

c) transverse cervical vein d) Middle thyroid vein

2- Anterior Jugular Vein drain into {Mid2025}

a. External Jugular Vein b. Subclavian Vein

c. Internal Jugular Vein d. Right brachipcephalic

3-Buccal artery is a branch of: {Formative2025}

a) Mandibular artery b) Mandibular part of maximally artery

c) Pterygopalatine part of maxillary artery d) Pterygoid part of maxillary artery

4--the middle meningeal artery is a branch from: {Mid2025}{Summar2025}

a) Mandibular artery b) Internal carotid artery c) Subclavian artery d) Maxillary artery

5-Vertebral artery is a branch of: {Formative2024}{Summar2025}

a) External carotid artery b) Internal thoracic artery

c) Subclavian artery d) Brachiocephalic artery

6-Vertebral artery is a branch of {Mid2025}

a. First part of subclavian b. second part of subclavian c. 3rd part of subclavian

7- Masseteric artery is a branch of {Mid2024}

a) Superficial temporal artery b) Maxillary artery.

c) Lingual artery d) Superior thyroidal artery

8- Posterior communicating artery is a branch of: {Mid2024}

a) Middle cerebral artery b) Posterior cerebral artery

c) Internal carotid artery d) Anterior cerebral artery

9- Which of the following is NOT a branch of the internal carotid artery ? {Final2022} {Formative2022}

A- Anterior cerebral artery B- Posterior communicating artery

C- Superior thyroid artery D- Ophthalmic artery

10- The vertebral arteries arise from which artery {Final2022}{Formative2022}

A- External carotid B- Thyrocervical trunk C- Subclavian D- Common carotid

11-Which is the terminal branch of the internal carotid artery ? {Formative2022}

a) ophthalmic artery b) anterior cerebral artery c) middle cerebral artery

d) b and c e) posterior communicating artery

12- Which of the following statements is INCORRECT, The carotid artery gives off the: {Final2022}

A- Superior thyroid artery B- Transverse cervical artery C- Facial artery

D- Posterior auricular artery E- Ascending pharyngeal artery

13- Costo-cervical trunk is a branch from {Mid2021}

A- Front of external carotid artery B- 1st part of subclavian artery

C- 2nd part of subclavian artery D- 3rd part of subclavian artery

14- All are branches from external carotid artery EXCEPT {Mid2021}

A- Superficial temporal artery B- Ascending cervical artery

C- Lingual artery D- Maxillary artery

15- All are branches of external carotid artery EXCEPT : {Mid2023}

A- Superior thyroid B- Transverse cervical C- Ascending pharyngeal D- Facial

16- Which of the following statements is INCORRECT? The external carotid artery gives off the : {قسم}

A) Superior thyroid artery B) Transverse cervical artery C) Facial artery

D) Posterior auricular artery E) Ascending pharyngeal artery

17- The internal carotid artery: {قسم}

a) Has a number of named branches in neck b) Begins at the level of 6 cervical vertebra

c) Passes along lateral wall of cavernous sinus d) Has posterior communicating branch

18-Is not a branch of internal carotid artery

a. Artery choroidal artery b. Hypophyseal artery c. Anterior cerebral artery

d. Opthalmic artery e. Posterior cerebral artery

19- Sphenopalatine artery is a branch of:

a- Maxillary artery b- Mandibular artery c- Ophthalmic artery d- Facial artery

20-one of the following veins drains into subclavian vein

a. common facial vein b. posterior auricular vein

c. anterior facial vein d. external jugular vein

21- All are branches from external carotid artery EXCEPT

A- Superficial temporal artery B- Ophthalmic artery

C- Lingual artery D- Superior thyroid arteryartery

22- Tributaries of the internal jugular vein include the: {قسم}

a) Middle thyroid vein b) Lingual vein c) Inferior thyroid vein d) Vertebral vein

23- The facial vein: {قسم}

a) Communicates with the external jugular vein

b) Is formed by the confluence of the superior ophthalmic and supraorbital veins

c) Lies in front of the facial artery and pursues a tortuous course

d) Communicates with the pterygoid plexus via the deep facial vein

24- All about facial vein is TRUE EXCEPT {Mid2023}

A- Communicate with cavernous sinus

B- It is anterior to facial artery and has tortuous course

C- Formed by union of supratrochlear and supraorbital vein

D- Ptyrigoid plexus communicate with deep facial vein

25- All of the following are found in the carotid sheath EXCEPT the {Final2022} {Formative2022}

A- Common carotid artery B- Internal carotid artery C- Internal jugular vein

D- Vagus nerve E- Hypoglossal nerve

26-The common carotid artery lies within the:

A) Posterior triangle of the neck B) Submandibular triangle

C) Anterior triangle of the neck, outside carotid sheath D) Carotid sheath

27-The right common carotid artery is a branch of:

A) Arch of aorta B) Brachiocephalic artery

C) Subclavian artery D) External carotid artery

28-The left common carotid artery arises directly from:

A) Brachiocephalic artery B) Arch of aorta

C) Subclavian artery D) Internal carotid artery

29-The common carotid artery bifurcation is typically located at the level of:

A) C1–C2 vertebra B) C3–C4 vertebra C) C5–C6 vertebra D) T1–T2 vertebra

30-Which of the following is the first branch of the external carotid artery?

A) Facial artery B) Superior thyroid artery C) Lingual artery D) Occipital artery

31-All of the following are branches of the external carotid artery EXCEPT:

A) Superior thyroid artery B) Ascending pharyngeal artery

C) Lingual artery D) Inferior thyroid artery

32-Which of the following is NOT a terminal branch of the external carotid artery?

A) Maxillary artery B) Superficial temporal artery

C) Posterior auricular artery D) None of the above

33-The subclavian artery terminates at the:

A) Inner border of the first rib B) Outer border of the first rib

C) Middle of the clavicle D) Sternoclavicular joint

34-All of the followings are branches of the external carotid artery Except:

a. Ascending pharyngeal artery b. Lingual artery

c. Facial artery d. Deep auricular artery

35-Which of the following is NOT a branch from the mandibular part of maxillary Artery

a. Deep auricular artery b. Pharyngeal artery

c. anterior tympanic artery d. middle meningeal artery

36-All of the followings are branches of the internal Carotid Artery Except:

a. hypophyseal artery b. meningeal artery

c. Anterior choroidal artery d. Lingual artery

37-Which of the following is a branch internal Carotid Artery

a. Middle cerebral artery b. Occipital artery

c. Posterior auricular artery d. Maxillary artery

38-Which of the following is NOT a branch of the subclavian artery?

A) Vertebral artery B) Thyrocervical trunk

C) Internal thoracic artery D) Superior thyroid artery

39-All of the following are branches of the first part of the subclavian artery EXCEPT:

A) Vertebral artery B) Thyrocervical trunk

C) Internal thoracic artery D) Costocervical trunk

40-At the outer border of the first rib, the subclavian artery continues as the:

A) Brachial artery B) Axillary artery C) Radial artery D) Thoracic artery

41-Which of the following is a branch of the first part of the subclavian artery?

A) Costocervical trunk B) Dorsal scapular artery

C) Internal thoracic artery D) Superior thyroid artery

42-The right subclavian artery originates from the:

A) Arch of aorta B) Brachiocephalic artery

C) Common carotid artery D) Vertebral artery

43-The left subclavian artery originates from the:

A) Brachiocephalic artery B) Arch of aorta

C) Internal thoracic artery D) Axillary artery

44-The thyrocervical trunk arises from the:

A) First part of subclavian artery B) Second part of subclavian artery

C) Third part of subclavian artery D) Axillary artery

45-The external jugular vein drains into the:

A) Internal jugular vein B) Brachiocephalic vein

C) Subclavian vein D) Superior vena cava

46-The posterior auricular vein contributes to the formation of the:

A) Internal jugular vein B) External jugular vein

C) Anterior jugular vein D) Facial vein

47-The internal thoracic artery is a branch of the:

A) Axillary artery B) Subclavian artery (first part)

C) Carotid artery D) Subclavian artery (third part)

48-The subclavian artery becomes the axillary artery at the level of the:

A) First rib B) Clavicle C) Sternum D) Second rib

49-The vertebral artery is a branch of the:

A) First part of the subclavian artery B) Second part of the subclavian artery

C) External carotid artery D) Axillary artery

50-Which artery from the first part of the subclavian ascends to supply the brain?

A) Internal thoracic B) Vertebral artery

C) Inferior thyroid artery D) Superior intercostal artery

51-The deep cervical artery arises from the:

A) Thyrocervical trunk B) Internal thoracic artery

C) Costocervical trunk D) Dorsal scapular artery

52-Which artery is typically a branch of the third part of the subclavian artery?

A) Suprascapular artery B) Internal thoracic artery

C) Dorsal scapular artery D) Superior intercostal artery

53-The common facial vein drains into the:

A) External jugular vein B) Internal jugular vein C) Subclavian vein D) Vertebral vein

54-The brachiocephalic vein is formed by the union of the:

A) External jugular and vertebral veins B) Internal jugular and subclavian veins

C) Axillary and internal thoracic veins D) Superior thyroid and vertebral veins

55-All of the following are branches of the costocervical trunk EXCEPT:

A) Deep cervical artery B) Superior intercostal artery

C) Inferior thyroid artery D) None of the above

56-All of the following are tributaries of the external jugular vein EXCEPT:

A) Transverse cervical vein B) Suprascapular vein

C) Posterior external jugular vein D) Lingual vein

57-Which of the following drains directly into the internal jugular vein?

A) Posterior auricular vein B) External jugular vein

C) Maxillary vein D) Inferior petrosal sinus

58-The subclavian vein joins with the internal jugular vein to form the:

A) Superior vena cava B) Axillary vein C) Brachiocephalic vein D) External jugular vein

59-The lingual vein drains into the:

A) External jugular vein B) Internal jugular vein C) Subclavian vein D) Facial vein

60-Right and left brachiocephalic veins unite to form the:

A) Inferior vena cava B) Azygos vein C) Superior vena cava D) Internal jugular vein

61-All of the following are lymph nodes of the head and neck EXCEPT:

A) Occipital nodes B) Submandibular nodes C) Popliteal nodes D) Retroauricular nodes

62-The posterior division of the retromandibular vein is formed by the union of:

A) Facial and lingual veins B) Maxillary and superficial temporal veins

C) Occipital and posterior auricular veins D) Facial and maxillary veins

63-The juguloomohyoid node is located near the:

A) Omohyoid muscle B) Posterior belly of digastric

C) Sternocleidomastoid muscle D) Hyoid bone

64-Which of the following is a tributary of the external jugular vein?

A) Facial vein B) Internal jugular vein C) Anterior jugular vein D) Vertebral vein

65-The common facial vein is formed by the union of the facial vein and:

A) Posterior division of the retromandibular vein B) External jugular vein

C) Anterior division of the retromandibular vein D) Lingual vein

66-Which of the following is a tributary of the subclavian vein?

A) Facial vein B) Internal jugular vein

C) External jugular vein D) Superior thyroid vein

67-All of the following are tributaries of the IJV EXCEPT:

A) Lingual vein B) Superior thyroid vein

C) Middle thyroid vein D) External jugular vein

68-Which of the following is a tributary of the brachiocephalic vein?

A) Vertebral vein B) Internal thoracic vein C) Inferior thyroid vein D) All of the above

69-The 1st posterior intercostal vein drains into the:

A) Azygos vein B) Thoracic duct C) Right brachiocephalic vein D) Internal thoracic vein

70-The jugulodigastric node is located:

A) Below the posterior belly of digastric muscle B) Near the omohyoid muscle

C) Under the mandible D) At the angle of the jaw

71-The juguloomohyoid node mainly drains lymph from the:

A) Tonsil B) Tongue C) Submental region D) Occipital scalp

72-The jugulodigastric node primarily drains lymph from:

A) Tongue and tonsils B) Larynx C) Submandibular gland D) Pharynx only

73-The efferent lymphatics from deep cervical nodes drain into the:

A) Subclavian vein B) Jugular lymph trunk

C) External jugular vein D) Thoracic duct only

74-Which of the following nodes is a major node for tonsillar infections?

A) Juguloomohyoid node B) Jugulodigastric node

C) Submental node D) Retropharyngeal node

1. D	2. A	3. D	4. D	5. C
6. A	7. B	8. C	9. C	10.C
11.D	12.B	13.C	14.B	15.B
16.B	17.D	18.E	19.A	20.D
21.B	22.A,B	23.D	24.B	25.E
26.D	27.B	28.B	29.B	30.B
31.D	32.C	33.B	34.D	35.B
36.D	37.A	38.D	39.D	40.B
41.C	42.B	43.B	44.A	45.C
46.B	47.B	48.A	49.A	50.B
51.C	52.C	53.B	54.B	55.C
56.D	57.D	58.C	59.B	60.C
61.C	62.B	63.A	64.C	65.C
66.C	67.D	68.D	69.C	70.A
71.B	72.A	73.B	74.B	75.

The Orbit

1- The levator palpebrae superioris muscle is innervated by the {قسم}{Formative2026}

a) facial nerve. b) trochlear nerve. c) trigeminal nerve.

d) oculomotor nerve. e) abducent nerve.

2-The innervations of the eye and its muscles include: {Summar2025}

a) somatic efferent nerve fibers to the external ocular muscles.

b) visceral (autonomic) efferent nerve fibers from the superior cervical ganglia.

c) somatic efferent nerve fibers to the iris muscles and ciliary muscles.

d) a & b are correct.

3-To rotate the cornea upwards and laterally, we should use: {Summar2025}

a) superior oblique b) inferior oblique c) lateral rectus d) medial rectus

4-The lacrimal gland lies at: {Formative2025}

a) Superomedial part of the orbit b) Lateral part of the orbit

c) Inferior part of the orbit d) Superolateral part of the orbit

5-Which of the following muscles rotates the cornea upward and laterally ? {Mid2025}

a. superior oblique b. inferior oblique c. Medial rectus d. Lateral rectus

6-Which of the following muscles elvates the eyelids {Mid2025}

a. levator palbebrae superioris b. superior rectus

7- The Abducent nerve supplies the muscle: {Mid2024}

a) Lateral rectus. b) Superior tarsal. c) Superior rectus. d) Inferior oblique.

8- The main nerve supply to the extraocular muscles {Final2024}

A- oculomotor nerve B- Trochlear C- Abducent D- Trigeminal

9-Which of the following muscles rotates the cornea downward and laterally?

a. superior oblique b. inferior oblique c. Medial rectus d. Lateral rectus

10-Function of lateral rectus: {Formative2024}

a) Medial rotation b) Lateral rotation c) Adduction d) Abduction

11-What is responsible for adjusting the lens {Formative2024}

a) Choroid b) Ora serrata c) Iris d) Ciliary muscle

12- The inferior oblique muscle of the eye is innervated by : {Mid2021}{قسم}

A- Abducent nerve B- Oculomotor nerve C- Facial nerve D- Trochlear

13- The lateral rectus muscle of the eye is innervated by the {Final2022}{قسم}

A- optic nerve B- trochlear nerve C- oculomotor nerve

D- facial narva E- Abducent nerve

14- Superior oblique muscle of the eye is innervated by : {Mid2023}{قسم}

A- Abducent nerve B- Trochlear nerve C- Oculomotor nerve

D- Trigeminal nerve E- Chorda tempani nerve

15-The transparent lens in the human eye is held in its place by {Formative2024}

a) ligaments attached to the ciliary body b) smooth muscles attached to the iris

c) smooth muscles attached to the ciliary body d) ligaments attached to the iris

16-levator palpares superioris is innervated by: {Formative2024}

a) Superior division of occulomotor nerve b) Inferior division of occulomotor nerve

c) Abducent nerve d) Trochlear nerve

17- The oculomotor nerve supplies ALL BUT ONE of the following; {قسم}

A) Medial rectus B) Superior oblique C) Inferior rectus

D) Superior rectus E) Inferior oblique

18-levator palpares superioris is innervated by: {قسم}

a) Inferior division of occulomotor nerve b) Superior division of occulomotor nerve

c) Trochlear nerve d) Abducent nerve

19- Which of the following bones form the medial margin of the orbit ? {Final2022}

A- Ethmoid B- Lacrimal C- Nasal D- maxilla

20-The lacrimal gland lies in which anatomical space?

A) Lacrimal fossa B) Orbit apex C) Palpebral conjunctival sac D) Scleral sulcus

21-About the extraocular muscles, one is true

a. all the muscles are supplied by oculomotor nerve

b. the trochlear nerve supplies the lateral rectus muscle

c. superior oblique muscle depresses and abducts the eye ball

d. all the muscles arise from the common tendinous ring

22-Which of the following extraocular muscles is supplied by trochlear nerve

a. Inferior oblique b. Superior oblique c. Lateral rectus

d. Superior rectus e. Medial rectus

23-The palpebral portion of the lacrimal gland lies in:

A) Upper eyelid B) Lacrimal fossa C) Lower eyelid D) Medial canthus

24-Is Not concerning the lacrimal apparatus

a. Lacrimal gland consistes of two parts

b. Lacrimal gland is supplied by parasympathetic fibers from lacrimatory nucleus

c. Nasolacrimal duct open in the inferior meatus

d. The lacrimal sac is located laterally in the orbital cavity

25-Parasympathetic secretomotor fibers to the lacrimal gland originate from:

A) Lacrimal nucleus of the facial nerve B) Trigeminal motor nucleus

C) Oculomotor nucleus D) Hypoglossal nucleus

26-The conjunctival sac is the space between:

A) Cornea and sclera B) Palpebral and bulbar conjunctiva
C) Eyelid skin and tarsal plate D) Orbicularis oculi and tarsal plate

27-The lacrimal gland is located in the:

A) Inferomedial part of orbit B) Medial canthus

C) Posterior orbit D) Superolateral part of orbit

28-The external surface of the eyelids is covered by:

A) Thick skin B) Thin skin C) Palpebral conjunctiva D) Mucous membrane

29-The internal surface of the eyelids is lined by:

A) Epidermis B) Palpebral conjunctiva C) Cornea D) Sclera

30-The lacrimal gland is primarily responsible for:

A) Aqueous component of tears B) Mucous production in tears

C) Lipid secretion D) Producing vitreous humor

31-The lacrimal gland receives its arterial blood supply from:

A) Facial artery B) Ophthalmic artery

C) Maxillary artery D) Superficial temporal artery

32-The lacrimal sac is continuous with the:

A) Lacrimal gland B) Nasolacrimal duct C) Conjunctival sac D) Inferior meatus

33-The lacrimal gland ducts allow tears to flow into the:

A) Nasolacrimal duct B) Orbital septum C) Corneal epithelium D) Conjunctival sac

34-The sensory innervation of the lacrimal gland is provided by:

A) Facial nerve B) Lacrimal nerve C) Maxillary nerve D) Zygomatic nerve

35-Postganglionic parasympathetic fibers reach the lacrimal gland through:

A) Directly from facial nerve

B) Maxillary nerve → zygomatic branch → zygomaticotemporal nerve → lacrimal nerve

C) Ophthalmic nerve → frontal branch → lacrimal nerve

D) Trigeminal nerve → infraorbital nerve → lacrimal nerve

36-The primary action of the levator palpebrae superioris is:

A) Depression of upper eyelid B) Elevation of upper eyelid

C) Retraction of lower eyelid D) Lateral movement of eyelid

37-The nasolacrimal duct opens into the:

A) Superior meatus of the nose B) Middle meatus of the nose

C) Inferior meatus of the nose D) Nasopharynx

38-Sympathetic postganglionic fibers to the lacrimal gland are derived from:

A) plexus around internal carotid artery B) Facial nerve

C) Ophthalmic nerve D) Pterygopalatine ganglion

39-The nasolacrimal duct emerges from the:

A) Lacrimal gland B) Upper end of lacrimal sac

C) Lower end of lacrimal sac D) Conjunctival sac

40-The nasolacrimal duct descends in which direction?

A) Downward, backward, and laterally B) Upward and medially

C) Horizontally forward D) Downward and medially

41-The levator palpebrae superioris is innervated by:

A) Inferior division of oculomotor nerve B) Superior division of oculomotor nerve

C) Facial nerve D) Trochlear nerve

42-The lateral rectus muscle is supplied by:

A) Oculomotor nerve B) Trochlear nerve C) Abducent nerve D) Facial nerve

43-Action of the lateral rectus muscle is:

A) Adduction B) Abduction C) Elevation D) Depression

44-Partial ptosis occurs when:

A) Oculomotor nerve Paralysis B) Sympathetic innervation is paralyzed

C) Facial nerve is paralyzed D) Lacrimal gland secretion increases

45-Complete ptosis occurs due to:

A) Oculomotor nerve paralysis B) Facial nerve paralysis

C) Sympathetic paralysis D) Trigeminal nerve injury

46-The inferior oblique muscle is supplied by:

A) Oculomotor nerve B) Trochlear nerve C) Abducent nerve D) Facial nerve

47-The primary actions of the inferior oblique muscle include:

A) Depression and abduction B) Elevation and abduction

C) Depression and adduction D) Elevation and adduction

48-All of the following are actions of the superior rectus EXCEPT:

A) Elevation B) Adduction C) External rotation D) Internal rotation

49-All of the following are actions of the inferior rectus EXCEPT:

A) Depression B) Adduction C) External rotation D) Internal rotation

50-The smooth (Müller's) part of the levator palpebrae superioris is innervated by:

A) Parasympathetic fibers B) Sympathetic fibers from superior cervical ganglion

C) Facial nerve D) Trigeminal nerve

51-Which part of the fibrous layer covers the anterior part of the eyeball?

A) Sclera B) Cornea C) Choroid D) Retina

52-Which part of the fibrous layer covers the posterior part of the eyeball?

A) Cornea B) Iris C) Choroid D) Sclera

53-All of the following recti muscles are supplied by the oculomotor nerve EXCEPT:

A) Superior rectus B) Medial rectus C) Inferior rectus D) Lateral rectus

54-Action of the medial rectus muscle is:

A) Abduction B) Adduction C) Elevation D) Depression

55-The primary actions of the superior oblique muscle include:

A) Depression and abduction B) Elevation and adduction

C) Elevation and abduction D) Depression and adduction

56-The inferior oblique rotates the cornea:

A) Upward and laterally B) Downward and laterally

C) Upward and medially D) Downward and medially

57-Which nerve supplies most of the recti muscles of the eye?

A) Trochlear B) Oculomotor C) Abducent D) Facial

58-The superior rectus muscle primarily:

A) Elevates, adducts, and internally rotates the eyebal B) Abducts the eyeball

C) Elevates, adducts, and externally rotates the eyeball D) Externally rotates the eyeball

59-The inferior rectus muscle primarily:

A) Elevates the eyeball B) Depresses, adducts, and externally rotates the eyeball

C) Abducts the eyeball D) Depresses, adducts, and internally rotates the eyeball

60-Damage to the abducent nerve will cause:

A) Inability to adduct the eye B) Inability to abduct the eye

C) Complete ptosis D) Loss of pupillary reflex

61-Which of the following muscles elevates the eye and abducts it?

A) Superior rectus B) Inferior rectus C) Superior oblique D) Inferior oblique

62-The superior oblique rotates the cornea:

A) Upward and medially B) Downward and medially

C) Upward and laterally D) Downward and laterally

63-The superior oblique muscle is supplied by:

A) Trochlear nerve B) Oculomotor nerve C) Abducent nerve D) Facial nerve

64-Trochlear nerve palsy results in weakness of:

A) Superior rectus B) Inferior rectus C) Superior oblique D) Lateral rectus

65-Damage to the oculomotor nerve may result in weakness of all of the following EXCEPT:

A) Inferior oblique B) Medial rectus C) Lateral rectus D) Superior rectus

66-The cornea is characterized by all of the following EXCEPT:

A) Covers the anterior eyeball B) Transparent

C) Part of fibrous layer D) Forms posterior 5/6 of the eyeball surface

67-The inner layer of the eyeball is:

A) Cornea B) Sclera C) Choroid D) Retina

68-The outer fibrous layer of the eyeball consists of:

A) Choroid and ciliary body B) Retina C) Sclera and cornea D) Iris and lens

69-The central opening of the iris is called:

A) Pupil B) Cornea C) Lens D) Sclera

70-Contraction of the sphincter pupillae causes:

A) Pupil dilation B) Pupil constriction C) Eyelid elevation D) Eyeball abduction

71-The sphincter pupillae muscle is arranged in:

A) Longitudinal fibers B) Circular fibers C) Oblique fibers D) Radial fibers

72-The middle vascular layer of the eyeball consists of:

A) Sclera and cornea B) Choroid, ciliary body, and iris C) Retina D) Lens and vitreous body

73-The iris represents which part of the eye?

A) Transparent anterior coat B) Colored part of the eye C) Posterior layer D) Optic disc

74-The fovea centralis is the area of:

A) Lowest visual sensitivity B) Pupillary constriction

C) Lens attachment D) Maximum visual sensitivity

75-The sphincter pupillae is innervated by:

A) Sympathetic fibers B) Abducent nerve

C) Trochlear nerve D) Parasympathetic fibers via oculomotor nerve (III)

76-The dilator pupillae is innervated by:

A) Sympathetic fibers from superior cervical ganglion B) Parasympathetic fibers

C) Oculomotor nerve D) Trochlear nerve

77-The central depression in the macula lutea is called:

A) Optic disc B) Fovea centralis C) Blind spot D) Corneal apex

78- The optic disc is the point where:

A) The retina begins B) The optic nerve leaves the retina

C) The cornea meets the sclera D) The lens attaches

79-The anterior chamber of the eye is located:

A) Behind the lens B) Behind the iris and in front of the lens

C) Behind the cornea and anterior to the iris D) Between retina and lens

80-Both anterior and posterior chambers are filled with:

A) Vitreous humor B) Aqueous humor C) Blood D) Lens fibers

81-The vitreous chamber is located:

A) Between cornea and iris B) Between iris and lens

C) Between lens and retina D) Inside the lens

82-Vitreous humor is:

A) Transparent gelatinous substance B) Opaque and fibrous

C) A liquid similar to blood D) Composed of cartilage

1. D	2. D	3. B	4. D	5. B
6. A	7. A	8. A	9. A	10.D
11.D	12.B	13.E	14.B	15.A
16.A	17.B	18.B	19.D	20.A
21.C	22.B	23.A	24.D	25.A
26.B	27.D	28.B	29.B	30.A
31.B	32.B	33.B	34.B	35.B
36.B	37.C	38.A	39.C	40.A
41.B	42.C	43.B	44.B	45.A
46.A	47.B	48.C	49.D	50.B
51.B	52.D	53.D	54.B	55.A
56.A	57.B	58.A	59.B	60.B
61.D	62.D	63.A	64.C	65.C
66.D	67.D	68.C	69.A	70.B
71.B	72.B	73.B	74.D	75.D
76.A	77.B	78.B	79.C	80.B

81.C	82.A	83.	84.	85.

The Ear

1- It relates to the floor of middle ear cavity: {Mid2025}{Summar2025}

a) The internal carotid artery b) The pyramid

c) Promontory d) The bulb of the internal jugular vein

2-Tensor tympani muscle is inserted at: {Formative2025}

a) Pyramid b) Neck of stapes c) Handle of malleus

d) Cartialegonus part of auditory tube

3-The tympanic membrane is at the ……..wall of the middle ear: {Formative2024}

a) Medial. b) Anterior c) Inferior. d) Lateral.

4- The pharyngotympanic tube lies at the .....wall of the middle ear: {Mid2024}

a) Medial. b) Lateral. c) Inferior. d) Anterior.

5-The stapedius muscle pulls the stapes ……. {Formative2024}

a) Superiorly. b) Medially. c) Laterally. d) Posteriorly

6- The facial nerve canal is located in the {قسم}

a) temporal bone. b) greater wing of the sphenoid bone. c) occipital bone

d) mastoid process. e) lacrimal bone.

7- Which of the following is in lateral wall of the middle ear ? {Final2022}

A- ICA B- Tympanic membrane C- Outer wall on inner ear D- IJV

8-The middle ear lies in

a. Posterior cranial fossa b. Cocchlea

c. Petrous part of temporal bone d. Roof of the skull

9-The internal jugular vein is related to which of the following walls of the middle ear cavity?

a. Posterior wall b. Medial wall c. Floor d. Anterior wall

10-Which of the following structures is a part of the bony labyrinth

a. Saccule b. Cochlear duct c. Vestibule d. Utricle e. Semicircular duct

11-The endolymph is present at which of the following sites

a. Bony labyrinth b. Vestibule c. Cochlear duct

d. Cochlea e. Semicirciular canals

12- The middle ear lies in:

A. Basilar part of sphenoid bone. B. Petrous part of temporal bone.

C. Cocchlea. D. Posterior cranial fossa

13- The endolymph is present at which of the following sites?

A. Bony labyrinth. B. Cochlear duct. C. Cochlea. D. Semicircular canals

14-The external auditory meatus is a passage between:

A) Auricle and cochlea B) Auricle and tympanic membrane

C) Tympanic membrane and inner ear D) External ear and middle ear

15-The approximate length of the external auditory meatus is:

A) 1 inch B) 1 cm C) 2 inches D) 2 cm

16-All of the following nerves contribute to the sensory innervation of the external auditory meatus EXCEPT:

A) Auriculotemporal nerve B) Auricular branch of vagus nerve C) Great auricular nerve

D) Lesser occipital nerve E) Facial nerve

17-The external auditory meatus ends at:

A) Tympanic membrane B) Cochlea C) Eustachian tube D) Oval window

18-The tympanic membrane is convex towards:

A) External auditory canal B) Middle ear cavity C) Cochlea D) Auricle

19-The external auditory meatus is composed of:

A) Lateral 1/3 cartilaginous, medial 2/3 bony B) Entirely cartilaginous

C) Entirely bony D) Lateral 2/3 bony, medial 1/3 cartilaginous

20-The tympanic membrane separates:

A) Middle ear and inner ear B) External auditory meatus and middle ear

C) Cochlea and vestibule D) Eustachian tube and middle ear

21-The floor of the tympanic cavity separates it from:

A) Internal carotid artery B) Internal jugular vein C) Cochlea D) Mastoid antrum

22-The upper part of the posterior wall has an opening called:

A) Fenestra vestibule B) Aditus to the mastoid antrum C) Pyramid D) Promontory

23-The anterior wall of the middle ear separates it from:

A) Internal carotid artery B) Internal jugular vein C) Cochlea D) Mastoid antrum

24-The middle ear cavity communicates anteriorly with:

A) External auditory canal B) Mastoid air cells

C) Nasopharynx through the Eustachian tube D) Cochlea

25-The anterior wall of the middle ear contains two canals. The lower and larger canal opens into:

A) Tensor tympani muscle B) Auditory (Eustachian) tube

C) Oval window D) Round window

26-The fenestra vestibuli (oval window) is closed by:

A) Tensor tympani B) Base of stapes C) Tympanic membrane D) Pyramid

27-The pyramid is the origin of which muscle?

A) Stapedius B) Tensor tympani C) Levator veli palatine D) Orbicularis oculi

28-The stapedius muscle is attached to:

A) Malleus B) Incus C) Stapes D) Tympanic membrane

29-The tensor tympani muscle is attached to:

A) Malleus B) Incus C) Stapes D) Tympanic membrane

30-The lateral wall of the middle ear is largely formed by:

A) Mastoid air cells B) Tympanic membrane C) Cochlea D) Auditory tube

31-The fenestra cochleae separates the middle ear from:

A) Cochlear perilymph B) Endolymph C) Auditory tube D) External auditory canal

32-The medial wall of the middle ear is formed by:

A) Lateral wall of inner ear B) Lateral wall of mastoid air cells

C) Lateral wall of pharynx D) Tympanic membrane

33-The middle ear cavity contains:

A) Muscles B) Auditory ossicles C) Nerves D) All of the above

34-The muscles in the middle ear are:

A) Tensor tympani and stapedius B) Levator veli palatini and tensor veli palatini

C) Orbicularis oculi and stapedius D) Auricular muscles

35-The auditory ossicles include:

A) Malleus, incus, stapes B) Malleus, stapes, cochlea

C) Incus, stapes, cochlear duct D) Malleus, incus, tensor tympani

36-Which part of the bony labyrinth houses the oval window?

A) Cochlea B) Vestibule C) Utricle D) Saccule

37-The nerve supplying the tensor tympani is a branch of:

A) Facial nerve B) Glossopharyngeal nerve C) Mandibular nerve D) Vagus nerve

38-Stapedius is innervated by which nerve?

A) Facial nerve B) Glossopharyngeal nerve C) Mandibular nerve D) Vagus nerve

39-The bony labyrinth consists of all the following parts EXCEPT:

A) Vestibule B) Semicircular canals C) Cochlea D) Saccule

40-The fluid filling the bony labyrinth is:

A) Endolymph B) Perilymph C) CSF D) Lymph

41-The membranous labyrinth is filled with:

A) Endolymph B) Perilymph C) CSF D) Lymph

42-The bony labyrinth is lined internally by:

A) Epithelium B) Perichondrium C) Endosteum D) Mucosa

43-Which of the following is NOT part of the membranous labyrinth?

A) Utricle B) Saccule C) Semicircular ducts D) Semicircular canals

44-The cochlear duct is part of the:

A) Middle ear B) External ear C) Bony labyrinth D) Membranous labyrinth

45-The cochlear duct contains the sensory organ for hearing called the:

A) Crista ampullaris B) Macula C) Organ of Corti D) Cupula

46-The cochlear duct is filled with:

A) Perilymph B) Endolymph C) CSF D) Blood plasma

1. D	2. C	3. D	4. D	5. D
6. A	7. B	8. C	9. C	10.C
11.C	12.B	13.B	14.B	15.A
16.E	17.A	18.B	19.A	20.B
21.B	22.B	23.A	24.C	25.B
26.B	27.A	28.C	29.A	30.B
31.A	32.A	33.D	34.A	35.A
36.B	37.C	38.A	39.D	40.B
41.A	42.C	43.D	44.D	45.C
46.B	47.	48.	49.	50.

Meninges

1- Which of the following sinuses is a paired sinus? {Summar2025}

a-superior sagittal sinus b-inferior sagittal sinus c- transverse sinus d-straight sinus

2-Which of the following meningeal fold separate between two cerebral hemisphere: {Formative2025}

a) falx cerebelli b) tentorium cerebelli c) falx cerebri d) diaphragma sellae

3-Which of the following is not a paired sinus of the brain: {Formative2025}

a) transverse b) cavernous c) inferior petrosal d) occipital

4-Venous sinus lies at the junction of the falx cerebri with the tentorium cerebelli: {Formative2024}

a) Straight sinus b) Sigmoid sinus c) Inferior sagittal sinus d) Superior sagittal sinus

5-CSF circulate in which of the following sites? {Formative2024}

a) between pia mater and brain b) subdural space

c) Between dura and bone d) subarachnoid space

6- The Occipital sinus {Final2024}

A- drain into transverse sinus B- run on attached border of falx cerebelli

7-Inferior petrosal sinus drained into {Final2023}

a) Basilar plexus b) Superior sagittal sinus c) Inferior sagittal sinus d) Cavernous sinus

8- Which is true about inferior sagital sinus {Final2024}

a. Runs on the free border of falx cerebri b. Drains into transverse sinus

c. drains into internal jaguar vein d. Takes drainage from superficial middle cerebral vein

9- Dural fold over the pituitary gland is {Final2021}

A- Falx cerebri B- Falx cerebelli C- Tentorium cerebelli D- Diaphragma sellae

10- Which statement about the cerebral meninges Is most accurate? {Final2021}

A- Dura mater has 2 distinct layers surrounding brain

B- The major venous sinuses lie in the subdural space

C- Rupture of middle meningeal artery causes a subarachnoid hemorrhage

D- The pia mater is a thick layer that adherent to brain surfaces

11- Arachnoid granulations is a prolongation from {Final2021}

A- Pia mater B- Dura mater C- Arachnoid mater D- Sub-arachnoid space

12-Which dural venous sinus directly drains into internal jugular vein

a. transverse sinus b. straight sinus c. superior petrosal sinus d. sigmoid sinus

13-Which of the following neurotransmitters is used by the axons of locus ceruleus neurons that distribute throughout much of the brain?

a. Dopamine b. Norepinephrine c. Acetylcholine d. GABA

14-The cavernous sinus communicates with the facial vein through which structure?

A) Inferior ophthalmic vein B) Superior ophthalmic vein

C) Central retinal vein D) Middle cerebral vein

15-Which of the following is a single (unpaired) sinus?

A) Cavernous sinus B) Superior sagittal sinus C) Transverse sinus D) Sigmoid sinus

16-Which of the following is a paired sinus?

A) Inferior sagittal sinus B) Straight sinus C) Sphenoparietal sinus D) Occipital sinus

17-The falx cerebri lies between the:

A) Cerebellar hemispheres B) Cerebral hemispheres

C) Frontal lobes only D) Occipital lobes

18-The following structure limits rotatory movements of the brain within the skull

a. Tentorium cerebelli b. Falx cerebri

c. Diaphragma sellae d. Squamous part of the temporal bone

19-The following nerves are sensory to the dura mater

a. Sixth cervical spinal nerve b. Oculomotor nerve

c. Trochlear nerve d. Trigeminal nerve

20-The falx cerebri is a fold of which meningeal layer?

A) Endosteal dura B) Arachnoid C) Pia D) Meningeal dura

21-The arachnoid mater lies between which two layers?

A) Dura mater externally and pia mater internally B) Skull and dura mater

C) Pia mater and brain tissue only D) Dura mater and bone

22-The diaphragma sellae forms the roof of the:

A) Cavernous sinus B) Sella turcica C) Tentorium cerebelli D) Foramen magnum

23-The endosteal layer of the dura mater lines:

A) The brain surface B) The outer surface of the skull

C) The inner surface of the skull bones D) The spinal cord

24-The meningeal layer of dura is continuous with the dura of the spinal cord through:

A) Foramen ovale B) Foramen magnum C) Jugular foramen D) Optic canal

25-The space between the arachnoid mater and pia mater is:

A) Subarachnoid space B) Subdural space C) Epidural space D) Cavernous sinus

26-The layer of dura that forms tubular sheaths around cranial nerves is the:

A) Endosteal layer B) Meningeal layer C) Arachnoid mater D) Pia mater

27-The primary function of dural folds is to:

A) Produce CSF B) Lubricate the brain

C) Restrict rotatory displacement of the brain D) Protect spinal nerves

28-The diaphragma sellae has a small central opening for the passage of the:

A) Optic nerve B) Pituitary stalk (infundibulum)

C) Internal carotid artery D) Trochlear nerve

29-The most common artery injured in extradural hemorrhage is the:

A) Anterior cerebral artery B) Posterior cerebral artery

C) Middle meningeal artery D) Internal carotid artery

30-Subdural hemorrhage is usually caused by:

A) Tearing of superior cerebral veins B) Injury to middle meningeal artery

C) Fracture of temporal bone D) Occlusion of carotid artery

31-Which of the following nerves is NOT located in the lateral wall of the cavernous sinus?

A) Oculomotor nerve B) Trochlear nerve C) Ophthalmic nerve D) Abducent nerve

32-Which of the following structures passes THROUGH the cavernous sinus?

A) Oculomotor nerve B) Internal carotid artery C) Trochlear nerve D) Ophthalmic nerve

33-All of the following are single venous sinuses EXCEPT:

A) Superior sagittal sinus B) Inferior sagittal sinus

C) Straight sinus D) Cavernous sinus

34-All of the following are paired venous sinuses EXCEPT:

A) Sphenoparietal sinus B) Cavernous sinus C) Transverse sinus D) Occipital sinus

35-The sinus that runs in the free edge of falx cerebri is:

A) Superior sagittal sinus B) Inferior sagittal sinus C) Straight sinus D) Transverse sinus

36-Which cranial nerve passes THROUGH the cavernous sinus alongside the internal carotid artery?

A) Oculomotor nerve B) Abducent nerve C) Maxillary nerve D) Trochlear nerve

37-Arachnoid villi are:

A) Projections of pia mater into the ventricles

B) Projections of arachnoid mater into venous sinuses

C) Projections of dura mater into the brain

D) Projections of arachnoid mater into subdural space

38-Aggregations of arachnoid villi are called:

A) Choroid plexuses B) Pia granulations C) Arachnoid granulations D) Venous granules

39-Arachnoid villi are most numerous along which sinus?

A) Cavernous sinus B) Superior sagittal sinus C) Transverse sinus D) Sigmoid sinus

40-Subarachnoid hemorrhage most commonly results from:

A) Rupture of middle meningeal artery B) Tearing of bridging veins

C) Rupture of a congenital aneurysm on the Circle of Willis D) Trauma to the skull

1. C	2. C	3. D	4. A	5. D
6. B	7. D	8. A	9. D	10.A
11.C	12.D	13.B	14.B	15.B
16.C	17.B	18.B	19.D	20.D
21.A	22.B	23.C	24.B	25.A
26.B	27.C	28.B	29.C	30.A
31.D	32.B	33.D	34.D	35.B
36.B	37.B	38.C	39.B	40.C

Cerebrum

1-primary visual area present in: {Formative2026}

a) temporal lobe b) frontal lobe c) partial lobe d) occipital lobe

2- The visual area is present at the: {Final2023}{Final2022}{Summar2025}

a-frontal lobe b-temporal lobe c-occipital lobe d-partial lobe

3-Which lobe processes visual information? {Formative2025}

a) Frontal b) Occipital c) Temporal d) Parietal

4-Which of the following constitutes the lentiform nucleus? {Formative2024}

a) Putamen and globus pallidus b) Putamen and substantia nigra

c) Caudate and pallidum d) Caudate and putamen

5-The primary auditory area (areas 41 and 42) is situated in: {Formative2024}

a) Parietal lobe b) Frontal lobe c) Occipital lobe d) Temporal lobe

6- Which of the following lies between Globus pallidus and external capsule? {Final2024}

a- Claustrum b- Putamen

7-Corpus callosum is an example of: {Formative2024}

a) Synapses b) Projectional fibers c) Associational fibers d) Commissural fibers

8-Internal capsule is a white matter separates: {Formative2024}

a) Thalmus and lentiform nucleus b) Lentiform nucleus and calstrum

c) Thamus and insula d) Cudate nucleous and lentiform nucleus

9-Mammillary bodies are part of: {Formative2024}

a) Sub thalums b) Epithalmus c) Hypothalamus d) Meta thalamus

10-Concerning cerebral hemisphere, which of the following is TRUE? {Final2023}

a) cingulate gyrus is present in medial surface of parietal lobe

b) precuneus present in parietal lobe c) hippocampus gyrus in temporal lobe

11-Arterial supply of thalamus {Final2023}

a) Anterior cerebral b) Middle cerebral c) Posterior cerebral

12- Thalamus is supplied by the ..... cerebral artery {Final2022}

A- anterior B- middle C- posterior D- Both A & B

13- Example of commissural fibers are the followings EXCEPT {Final2021}

A- Corpus callosum B- Long &Short Association fibers

C- Amygdaloidal nucleus D- Fornix

14- The posterior limb of internal capsule is between {Final2022}

A- Lentiform nucleus & head of caudate B- Lentiform nucleus & body of caudate

C- Lentiform nucleus & thalamus D- Head of caudate & thalamus

15- The followings are inter lobar sulci EXCEPT {Final2021}

A- Central sulcus B- lateral sulcus C- Parito-occipital sulcus D- Inter frontal sulci

16- Broca area is present in {Final2021}

A- Temporal lobe B- Occiptal lobe C- Frontal lobe D- Parital lobe

17- Which one of the following structures is not considered to be part of the basal ganglia? {Final2021}

A- Putamen B- Caudate nucleus C- Dentate nucleus D- Lentiform nucleus

18- ……. area is on the lateral surface of the frontal lobe {Final2022}

A- Wernicke's B- Broca's C- Auditory association D- Both A & B

19- Concerning the central sulcus of the cerebral hemisphere, select the INCORRECT statement ? {Final2022}

A- It is one of the two maior sulci found on the lateral surface

B- it is also called the fissure of Sylvius

C- It separated the motor from the sensory area

D- It runs from the centre of the upper border to a point above the lateral fissure

E- It usually extends for some distance on the medial surface

20-Wernicke's area is present in the:

a) Parietal lobe b) Frontal lobe c) Occipital lobe d) Temporal lobe

21-Broca's area is present in the:

a) Parietal lobe b) Frontal lobe c) Occipital lobe d) Temporal lobe

22-One of the following cortical areas of the brain is present in the parietal lobe:

A.Primary motor area(area 4). B.Primary auditory area(area 41, 42).

C.Primary visual area(area 17). D.General sensory area(area 3,1,2).

23-The two cerebral hemispheres are separated by the:

A. Transverse fissure B. Lateral sulcus C. Longitudinal fissure D. Central sulcus

24-The two cerebral hemispheres are connected by the:

A. Internal capsule B. Corpus callosum C. Thalamus D. Fornix

25-The cerebral cortex is composed of:

A. White matter B. Gray matter

C. Both gray and white matter equally D. Only axons

26-The outer layer of the cerebrum is called the:

A. Subcortical nuclei B. Cerebral cortex C. Corpus callosum D. Internal capsule

27-The largest commissural fiber connecting the two cerebral hemispheres is the:

A. Anterior commissure B. Posterior commissure

C. Internal capsule D. Corpus callosum

28-The folds of the cerebral cortex are called:

A. Sulci B. Gyri C. Fissures D. Lobules

29-The central sulcus separates which lobes of the brain?

A. Frontal and temporal B. Temporal and occipital

C. Parietal and occipital D. Frontal and parietal

30-The gyrus located anterior to the central sulcus is the:

A. Postcentral gyrus B. Precentral gyrus

C. Superior frontal gyrus D. Inferior frontal gyrus

31-The central sulcus is also known as:

A. Fissure of Rolando B. Sylvian fissure

C. Calcarine sulcus D. Longitudinal fissure

32-The lateral sulcus is also called the:

A. Central sulcus B. Rolandic sulcus C. Sylvian fissure D. Parieto-occipital sulcus

33-The lateral sulcus separates which lobes of the brain?

A. Frontal and parietal lobes from temporal lobe B. Frontal from parietal lobe

C. Parietal from occipital lobe D. Temporal from occipital lobe

34- The precentral gyrus is located:

A) Posterior to the central sulcus B) Anterior to the central sulcus

C) Inferior to the lateral sulcus D) Within the occipital lobe

35-The longitudinal fissure separates:

A) Two cerebral hemispheres B) Frontal and parietal lobes

C) Occipital and temporal lobes D) Cerebrum from cerebellum

36-The calcarine sulcus is found in:

A) Frontal lobe B) Parietal lobe C) Occipital lobe D) Temporal lobe

37-Which gyrus is primarily involved in motor speech (Broca's area)?

A) Precentral gyrus B) Superior frontal gyrus

C) Inferior frontal gyrus D) Postcentral gyrus

38-The superior temporal gyrus contains:

A) Primary auditory cortex B) Primary visual cortex

C) Primary motor cortex D) Primary somatosensory cortex

39-The primary visual cortex is located in:

A) Precentral gyrus B) Postcentral gyrus

C) Occipital lobe along calcarine sulcus D) Temporal lobe

40-The insula is hidden within:

A) Central sulcus B) Lateral sulcus

C) Longitudinal fissure D) Parieto-occipital sulcus

41- The calcarine sulcus contains:

A) Primary auditory cortex B) Primary visual cortex

C) Primary somatosensory cortex D) Primary motor cortex

42- The superior temporal gyrus is involved in:

A) Primary motor function B) Auditory processing

C) Visual processing D) Somatosensory processing

1. D	2. C	3. B	4. A	5. D
6. B	7. D	8. A,D	9. C	10.B
11.C	12.C	13.C	14.C	15.D
16.C	17.C	18.B	19.B	20.D
21.B	22.D	23.C	24.B	25.B
26.B	27.D	28.B	29.D	30.B
31.A	32.C	33.A	34.B	35.A
36.C	37.C	38.A	39.C	40.B
41.B	42.B	43.	44.	45.

Brain stem

1-The brain stem does NOT include the: {Formative2026}

a) pons b) medulla oblongata c) midbrain d) Diencephalons

2- The anterior surface of mid brain shows {Final2024}

A- Tectum B- tegmentum C- reticular formation D- Cerebral peduncle

3-Basilar groove present at the anterior surface of: {Formative2024}

a) Spinal cord b) Pons c) Mid brain d) Medulla oblongata

4-Rootlets of glossopharyngeal nerve emerge from the anterior surface of: {Formative2024}

a) Spinal cord b) Mid brain c) Pons d) Medulla oblongata

5- Facial colliculus is formed by facial nerve around: {Final2024}

a- Ocular nucleus b- Abducent nucleus c- Trochlear nucleus

6- Which cranial nerve lies in the junction between pons and medulla? {Formative2023}

a) abducent nerve (VI) b) vestibulocochlear nerve (VIII) and facial nerve(vII)

c) vagus nerve (X) d) glossopharyngeal nerve (IX)

7-The brain stem consists of all EXCEPT:

A. Midbrain B. Pons C. Medulla oblongata D. Thalamus

8- Concerning the medulla oblongata the followings are TRUE EXCEPT {Final2021}

A- It is divided into upper closed & lower open medulla,

B- it contains the nuclei of lower 4 cranial nerves

C- It shares in formation of the floor of 4th ventricle

D- It continues below the foramen magnum as the spinal cord

9- Deep nuclei of the middle four cranial nerves lie in {Final2021}

A- Midbrain B- Pons C- Medulla oblongata D- Spinal cord

10- Solitary Nucleus lies in {Final2021}

A- Midbrain B- Pons C- Medulla oblongata D- Spinal cord

11 - Which of the following does not originate from brain Stem: {Formative2022}

A –CNI B- CNIII C-CNII D- A +C

12-The midbrain, which of the following is correct? {Formative2024}

a) Oculomotor nerve appears medial to cerebral peduncle

b) It lies between pons and upper spinal cord

c) Is supplied by the anterior inferior cerebellar artery

d) It lies in the anterior cranial fossa

13- Which of the following is true about medulla oblengata {Final2024}

a. Pyramid is lateral to olive b. Passs through foramen magnum

c. Between pons and midbrain

14-Medulla oblongata……. {Final2023}

a) pass through foramen magnum & continue as spinal cord b) pyramid lateral to olive

15-The trochlear nerve emerges from the:

A. Ventral pons B. Ventral midbrain C. Dorsal midbrain D. Medulla

16-Regarding brain stem, one is wrong:

A.Medulla oblongata. B.Midbrain. C.Pons. D.Cerebellum

17-The cerebral aqueduct passes through the:

A. Pons B. Medulla C. Midbrain D. Cerebellum

18-Cranial nerves IX, X, XI emerge from the:

A. Pons B. Midbrain C. Medulla D. Cerebellum

19-The tectum of the midbrain contains:

A. Cerebral peduncles B. Red nucleus

C. Superior and inferior colliculi D. Substantia nigra

20-The pons is located between:

A. Midbrain and cerebellum B. Midbrain and medulla

C. Medulla and spinal cord D. Cerebellum and spinal cord

21-The anterolateral sulcus is found on the:

A. Midbrain B. Pons C. Medulla oblongata D. Spinal cord

22-Superior colliculi are concerned with:

A. Hearing reflexes B. Visual reflexes C. Pain sensation D. Voluntary movement

23-Inferior colliculi are concerned with:

A. Vision B. Balance C. Hearing D. Taste

24-The basilar artery lies in the:

A. Midbrain B. Medulla C. Cerebellum D. Pons

25-The medulla oblongata is continuous with the:

A. Pons B. Cerebellum C. Midbrain D. Spinal cord

26-The pyramids of the medulla contain:

A. Ascending sensory fibers B. Corticospinal tracts

C. Spinothalamic tracts D. Cerebellar fibers

27-Inferior cerebellar peduncle connects the cerebellum to the:

A. Midbrain B. Pons C. Medulla D. Thalamus

28-Which cranial nerves emerge from the posterolateral sulcus?

A. IX, X, XI B. V, VII, VIII C. III, IV D. XII

29-Glossopharyngeal nerve emerges from the:

A. Anterolateral sulcus B. Posterolateral sulcus

C. Pontomedullary junction D. Cerebellopontine angle

30-The main arterial supply of the midbrain is from the:

A. Internal carotid artery B. Vertebral artery

C. Basilar artery D. Middle cerebral artery

31-The abducens nerve emerges:

A. From pons B. From medulla

C. At the pontomedullary junction D. From cerebellopontine angle

32-All of the following arteries supply the medulla oblongata EXCEPT:

A. Vertebral artery B. Anterior spinal artery

C. Posterior inferior cerebellar artery D. Superior cerebellar artery

1. D	2. D	3. B	4. D	5. B
6. A	7. D	8. A	9. B	10.C
11.D	12.A	13.B	14.A	15.C
16.D	17.C	18.C	19.C	20.B
21.C	22.B	23.C	24.D	25.D
26.B	27.C	28.A	29.B	30.C
31.C	32.D	33.	34.	35.

Histology

Histological structure of the neurons and neuroglia cells

1-Each nerve is made of fasciculi which are enclosed by the connective tissue sheath called {Formative2026}

a) Perineurium b) Epimysium c) Endoneurium d) Epineurium

2-Which type of cells produces myelin sheeth? {Formative2026}

a) Neurons b) Schwann cells only

c) Oligodendrocytes only d) Both oligodendrocytes and Schwann cells

3- Which of the following forms myelin in the central nervous system? {Summar2025}

a. Astrocytes b. Ependymal cells c. Oligodendrocytes d. Schwann cells

4- Which of the following cells share in the blood brain barrier ? {Mid2025}

a. Schwann cell b. Astrocyte c. Oligodendrocyte

5-Regarding the nerve cell: {Summar2025}

a. It has heterochromatic nucleus b. It contains Nissl granules

c. It is a steroid secreting cell d. It is always rounded in shape

6-The cell responsible for myelination in CNS is: {Formative2025}

a) Astrocyte b) Ependymal cell c) Oligodendroglia d) Schwann cell

7-We can demonstrate neurofibrils by: {Formative2025}

a) Orcein stain b) Toludine blue stain c) PAS d) Silver stain

8-What is the phagocytic cell in the CNS? {Formative2025}

a) Satellite cell b) Schwan cell c) Microglia d) Oligodendroglia

9-Perikaryon refers to: {Formative2024}

a) The axon of nerve cell. b) Cell body of the nerve cell.

c) Cell body of the muscle cell. d) The dendrites of the nerve cell.

10-Which of the following cell types forms myelin in the central nervous system? {Formative2024}

a) Oligodendroglia b) Astrocytes c) Schwann cells d) Microglia

11-Oligodendrocytes functions is: {Formative2024}

a) Phagocytosis. b) Synthesis of myelin c) Synthesis of CSF. d) Secretion of mucin.

12-The histological characteristics of neurons include: {Formative2024}{سؤال في المحاضرة}

a) There is no nucleolus b) heterochromatic nucleus

c) Abundant Nissl bodies d) Sparse rough endoplasmic reticulum

13-Nodes of Ranvier are present in: {Formative2024}

a) Spleen. b) Heart muscle. c) Myelinated nerve fibers. d) Unmyelinated nerve fibers.

14- Concerning the ependymal cells one statement is correct {Mid2024}

a) they are lined the cavities in the peripheral nervous system

b) there are no junctions between adjacent cells

c) the apical surfaces contain microvilli

d) they are neuroepithelial cells

15- Active cells during inflammation of CNS is {Final2021}{Final2022}

A- Astrocyte cells B- Satellite cells C- Microglia

D- Oligodendrocytes E- Both a & c

16-With regard to the perineurium: {سؤال في المحاضرة}

a. It surrounds a nerve fiber b. It is connective tissue

c. It is continuous with the endomysium d. It is continuous with the epimysium

17-What are the supporting cells in the central nervous system called {سؤال في المحاضرة}

a. Schwann cells b. Basket cells c. Ganglion d. Satellite cells

18-Which cell is a macrophage found in the central nervous system? {سؤال في المحاضرة}

A. Kupffer cells b. Microglia c. Oligodendrocyte d. Langerhans cell

19- astrocyte : {Mid2023}

A- Has round cell body B- Present in brain and spinal cord only

C- Protoplasmic type has more glial fibers than fibrous type D- Proliferate after injury

20-Which is TRUE about Neurofibril? {Final2023}

a) Golgi in peri nuclear area only b) More SER

c) More lipofuscin in young cells d) Appear brown by normal stain

21-Ependymal cells

a. Lines the ventricles only b. Do not have cilia or microvilli

c. Squamous cells d. Lines the f the brain and the central canal of the spinal cord

22-The whole nerve (e.g. sciatic nerve) is enclosed by the connective tissue sheath called

a) Perineurium b) Epimysium c) Endoneurium d) Epineurium

23-Each fasciculus is made of many nerve fibers which are enclosed by the connective tissue sheath called

a) Perineurium b) Epimysium c) Endoneurium d) Epineurium

24-Which organelle is absent in dendrites?

A) Mitochondria B) Ribosomes C) RER D) Golgi apparatus

25-The main parts of a neuron include:

A) Cell body, axon, dendrites B) Cell body, myelin sheath, synapse

C) Axon, dendrites, Schwann cells D) Nucleus, mitochondria, ribosomes

26-The cell body of a neuron is also called:

A) Soma B) Axon hillock C) Perikaryon D) Dendrite

27-The main components of a nerve cell include:

A) Nucleus and neuroplasm B) Axon and dendrite only

C) Schwann cells and myelin D) Synapse and mitochondria

28-The cytoplasm of a neuron is called:

A) Axoplasm B) Neuroplasm C) Soma D) Perikaryon

29-The neuron's nucleus is located in the:

A) Axon terminal B) Dendrites C) Cell body (soma) D) Myelin sheath

30-In light microscopy, osmic acid is used to stain:

A) Nissl granules B) Neurofilaments C) Myelin sheath D) Axoplasm

31-Schwann cells are responsible for:

A) Phagocytosis in CNS B) Blood–brain barrier formation

C) Myelination of peripheral nerve fibers D) CSF production

32-The perikaryon of a neuron consists of:

A) Nucleus and neuroplasm B) Axon and dendrites

C) Myelin sheath and nodes of Ranvier D) Synaptic cleft and vesicles

33-The whole nerve (e.g., sciatic nerve) is enclosed by:

A) Perineurium B) Endoneurium C) Epineurium D) Fasciculus

34-The cytoplasm of a nerve cell is described as:

A) Eosinophilic with no granules B) Pale acidophilic with deep basophilic Nissl bodies

C) Basophilic with no organelles D) Acidophilic with no Nissl bodies

35-Nissl granules (bodies) are:

A) Aggregates of RER and ribosomes B) Aggregates of RER and lysosomes

C) Golgi apparatus D) Neurofilaments

36-Silver impregnation of the nerve cell body primarily stains:

A) Nissl granules B) Neurofibrils C) Mitochondria D) Golgi apparatus

37-The color of neurofibrils after silver staining is:

A) Blue B) Red C) Brown or black D) Green

38-The cytoplasm of dendrites contains:

A) Mitochondria, ribosomes, rough endoplasmic reticulum, and Golgi apparatus

B) Mitochondria, ribosomes, rough endoplasmic reticulum, and neurofilaments

C) Mitochondria, ribosomes, , Golgi apparatus and neurofilaments

D) Golgi apparatus, ribosomes, rough endoplasmic reticulum, and neurofilaments

39-The axoplasm contains all of the following EXCEPT:

A) Mitochondria B) Neurofilaments C) Microtubules D) Nissl granules

40-Lipofuscin pigment in neurons:

A) Is abundant in young individuals B) Is a byproduct of mitochondrial activity

C) Accumulates more in older individuals D) Represents active protein synthesis

41-Which staining method is used to visualize neurofibrils and detailed neuron morphology?

A) Hematoxylin and eosin B) Silver impregnation

C) Masson's trichrome D) Periodic acid–Schiff

42-The neuron is primarily a:

A) Lipid-storing cell B) Contractile cell

C) Carbohydrate-storing cell D) Protein-producing cell

43-Each nerve is composed of bundles called:

A) Axons B) Fasciculi C) Dendrites D) Neurofilaments

44-The correct hierarchical order of nerve connective tissue sheaths from outermost to innermost is:

A) Endoneurium → Perineurium → Epineurium B) Epineurium → Perineurium → Endoneurium

C) Perineurium → Epineurium → Endoneurium D) Epineurium → Endoneurium → Perineurium

45-The neuronal nucleus is described as:

A) Small and heterochromatic B) Large, euchromatic with a distinct nucleolus

C) Fragmented and acidophilic D) Absent

46-Nissl granules indicate:

A) High lipid content B) Active protein synthesis

C) Mitochondrial density D) Lysosomal activity

47-Each neuron has:

A) Multiple axons B) Several axons and dendrites

C) No axons, only dendrites D) Only one axon

48-The distal end of an axon branches into:

A) Dendritic spines B) Telodendria C) Nissl bodies D) Neurofibrils

49-The point of origin of the axon from the cell body is called:

A) Dendritic spine B) Nissl body C) Axon hillock D) Telodendria

50-The terminal enlargement of a telodendrium is called:

A) Axon hillock B) Dendritic spine C) Axon terminal D) Nissl granule

51-Multipolar neurons can have different cell body shapes. Which of the following is star-shaped and represents the majority of multipolar neurons?

A) Purkinje cells B) Stellate neurons C) Pyramidal cells D) Bipolar neurons

52-Astrocytes are classified into:

A) Microglia and oligodendroglia B) Fibrous and protoplasmic

C) Schwann and satellite cells D) Ependymal and Purkinje cells

53-Neuroglia in the central nervous system (CNS) include:

A) Schwann cells and satellite cells

B) Astrocytes, oligodendroglia, microglia, and ependymal cells

C) Only astrocytes and microglia D) Only oligodendroglia and Schwann cells

54-Each fasciculus is surrounded by which connective tissue sheath?

A) Epineurium B) Perineurium C) Endoneurium D) Myelin

55-Individual nerve fibers within a fasciculus are surrounded by:

A) Epineurium B) Perineurium C) Endoneurium D) Sarcolemma

56-Neurons can be classified according to the number of dendrites into:

A) Sensory, motor, and interneurons B) Pseudo-unipolar, bipolar, and multipolar

C) Myelinated and unmyelinated D) Upper and lower motor neurons

57-Which type of multipolar neuron has a flask-shaped or fusiform cell body and is found in the cerebellum?

A) Stellate neurons B) Purkinje cells C) Pyramidal cells D) Unipolar neurons

58-Golgi type I neurons are characterized by:

A) Short axons B) No axons C) Long axons D) Only dendritic processes

59-Cytoplasm of astrocytes contains:

A) Golgi apparatus only B) Mitochondria only

C) bundles of neurofilaments D) Nissl granules

60-After CNS injury or inflammation, astrocytes:

A) Undergo apoptosis B) Proliferate and form fibrous scar (gliosis)

C) Transform into oligodendrocytes D) None

61-Neuroglia in the peripheral nervous system (PNS) include:

A) Astrocytes and oligodendroglia B) Microglia and ependymal cells

C) Schwann cells and satellite cells D) Only neurons

62-Star-shaped cells with long branching processes in the CNS are:

A) Oligodendrocytes B) Microglia C) Astrocytes D) Ependymal cells

63-Astrocytic processes form end feet on:

A) Neuronal nuclei B) Myelin sheaths C) Walls of capillaries D) Synaptic vesicles

64-Astrocytes contribute to:

A) Production of cerebrospinal fluid B) Blood-brain barrier formation

C) Myelination in CNS D) Phagocytosis

65-Schwann cells are found in the:

A) Central nervous system B) Peripheral nervous system

C) Brain ventricles D) Spinal cord gray matter

1. A	2. D	3. C	4. B	5. B
6. C	7. D	8. C	9. B	10.A
11.B	12.C	13.C	14.C	15.E
16.B	17.B	18.B	19.D	20.A
21.D	22.D	23.C	24.D	25.B
26.A,C	27.A	28.B	29.C	30.C
31.C	32.A	33.C	34.B	35.A
36.B	37.C	38.B	39.D	40.C
41.B	42.D	43.B	44.B	45.B
46.B	47.D	48.B	49.C	50.C
51.B	52.B	53.B	54.B	55.C
56.B	57.B	58.C	59.C	60.B

61.C	62.C	63.C	64.B	65.B

Histological structure of meninges, choroid plexus and different types of brain barriers

1- What is the dura matter composed of {Summar2025}

a. Loose irregular connective tissue b. Dense irregular connective tissue

c. Simple squamous epithelium d. Nervous tissue

2- All of the following are true about subarachnoid matter except {Mid2025}

a. It is a space between the dura covering in the spinal cord and the vertebra

b. Its contains CSF fluid protects from trauma

c. It is connected with the brain ventricles

3- Choroid plexus {Mid2024}

a) is formed of pia mater folds b) is formed of dura mater folds

c) is formed of dense connective tissue d) is covered by stratified columnar epithelium

4-Regarding arachnoid mater which is WRONG: {Formative2024}

a) Avascular connective tissue b) the outer layer of meninges

c) connected to pia via trabeculae d) avascular connective tissue

5-Concerning dura matter which of the following is true? {Formative2024}

a) loose connective tissue b) contain venous sinuses

c) dense connective tissue d) both b& c

6-Regarding BBB, which of the following is false? {Formative2024}

a) perivascular feet rest on capillary from Schwan cell

b) perivascular feet rest on capillary from astrocyte

c) endothelium rests on basement membrane

d) capillary endothelial cells have tight junctions

7- Choroid plexus is : {Mid2023}

A- Pia matter fold B- Dura matter fold C- Has stratified squamous epithelium

8- Regarding arachnoid mater which is WRONG {Final2021}

A- Avascular connective tissue B- vascular connective tissue

C- connected to pia via trabeculae D- the outer layer of meninges E- both b&d

9- Regarding choroid plexus {Mid2021}

A- It is an invagination of dura mater B- It has continuous capillaries

C- It elaborates CSF D- It is lined with cuboidal to columnar cells E- C&D

10- Concerning blood brain barrier, which of the following is TRUE ? {Mid2021}

A- Capillary endothelial cells have tight junctions

B- Perivascular feet rest on capillary from oligodendroglia

C- Capillary endothelium is fenestrated

D- Perivascular feet rest on capillary from astrocyte

E- A&D

11- The axon hillock contains {Mid2021}

A- Ribosomes B- Microtubules C- Golgi body D- Synaptic vesicles E- B&D

12- Rexed lamination describes which of the following structures : {Mid2021}

A- Nerve cells in the grey matter B- Nerve fibers in the white matter

C- Neuroglia in the white matter D- A&B E- None of the above

13- Which of the following structures is not found in grey matter ? {Mid2021}

A- Cell bodies B- Dendrites C- Neuroglia

D- Unmyelinated axons E- Myelinated axons

14- Nissl substance in nerve cell bodies is associated with {Mid2021}

A- Spread of stimulus B- Synaptic junctions C- Axons

D- Protein synthesis E- Myelin formation

15-What is the avascular layer of meninges? {سؤال في المحاضرة}

A. dura B. arachnoid C. pia mater D. both dura and arachnoid

16-The innermost meningeal layer that closely adheres to the brain surface is:

A) Dura mater B) Arachnoid mater C) Pia mater D) Perineurium

17-Dura mater consists of

a. White matter b. Loose connective tissue with CSF as ground substance

c. Gray matter d. Dense fibrous connective tissue

18-Arachnoid granulations perforate the

a. Dura of the sagittal sinus b. Dura of the tentorium c. Choroid plexus d. Pia mater

19-The coverings of the brain and spinal cord are called:

A) Neuroglia B) Ventricles C) Meninges D) Fascia

20-Cerebrospinal fluid (CSF) is found in the:

A) Subarachnoid space B) Subdural space C) Epidural space D) Intracellular space

21-The primary function of arachnoid villi is to:

A) Produce cerebrospinal fluid B) Protect the brain mechanically

C) Reabsorb cerebrospinal fluid into venous blood D) Form myelin sheath

22-The meninges are arranged from outer to inner as:

A) Pia → Arachnoid → Dura B) Arachnoid → Dura → Pia

C) Dura → Arachnoid → Pia D) Dura → Pia → Arachnoi

23-Arachnoid villi terminate in:

A) Arteries B) Capillary beds C) Lymphatic vessels D) Venous sinuses

24-The outermost meningeal layer is:

A) Pia mater B) Arachnoid mater C) Dura mater D) Endoneurium

25-Which meningeal layer is thin, delicate, and avascular?

A) Dura mater B) Arachnoid mater C) Pia mater D) Endoneurium

26-The surface of the arachnoid mater is covered by:

A) Stratified squamous epithelium B) Simple cuboidal epithelium

C) Simple squamous epithelium D) Pseudostratified epithelium

27-The pia mater is composed mainly of:

A) Dense connective tissue with few vessels

B) Loose connective tissue containing many blood vessels

C) Fibrous connective tissue devoid of vessels D) Nervous tissue

28-The tunnel through which blood vessels penetrate the CNS along with pia mater is called:

A) Subarachnoid space B) Arachnoid villus

C) Virchow–Robin canal D) Perivascular space

29-Arachnoid villi are formed when the arachnoid mater:

A) Invaginates into the pia mater B) Covers cerebral gyri

C) Perforates the dura mater D) Forms the blood–brain barrier

30-Which meningeal layer closely follows the irregularities of the CNS surface?

A) Dura mater B) Arachnoid mater C) Perineurium D) Pia mater

1. B	2. A	3. A	4. B	5. D
6. A	7. A	8. E	9. E	10.E
11.B	12.A	13.E	14.D	15.B
16.C	17.D	18.A	19.C	20.A
21.C	22.C	23.D	24.C	25.B
26.C	27.B	28.D	29.C	30.D

Myelination in PNS and CNS and Types of nerve fibers and nerve endings

1-Which of the following is a type of non-encapsulated nerve ending found in the epidermis {Formative2026}

a) Pacinian corpuscle b) Ruffini corpuscle

c) Free nerve ending d) Meissner's corpuscle

2- What are the nonmyelinated segments called {Mid2025}

a. Nodes of Ranvier b. Intermodal segments

3- The internodal segments of the peripheral nerve axon at the electron microscope {Mid2024}

a) are formed by one Schwann cell b) contain concentric lamellae of dark and light lines.

c) are free of myelin sheath d) lake of the neurolemmal sheath

4-Ruffini's nerve endings; {Formative2024}

a) The nerve endings not branched b) Non encapsulated nerve endings

c) Spindle-shaped d) A receptor contain mainly elastic fibers.

5-Concerning Pacinian corpuscles {Formative2023}

A- Small, ovoid receptor

B- Myelinated terminals branch within the core

C- Contain numerous concentric ,adherent layers of flattened cells

D- The corpuscle enclosed with thick connective tissue capsule

6- In central nervous system : {Mid2023}

A- Non myelinated fibers must be surrounded by layer of oligodendriocyte

B- Myelinated cell is with neurolimmal sheath

C- node of Ranvier separates myelin sheath of the same cell

7-Concerning Pacinian corpuscles all the following statements are true EXCEPT: { سؤال في المحاضرة}

a-Large, ovoid receptor

b- myelinated nerve terminals branch in the core.

c-Contain numerous concentric, adherent layers of flattened cells.

d-The corpuscle enclosed with delicate connective tissue capsule

e- both b&c

8- Concerning Pacinian corpuscles {Final2021}

A- Small, ovoid receptor

B- Myelinated terminals branch within the core

C- The corpuscle enclosed with thick connective tissue capsule

D- Contain numerous ,adherent layers of flattened cells

E- both b&d

9- Regarding Golgi tendon it ? {Final2021}

A- is not enclosed by connective tissue sheath B- contains core of elastic bundles

C- is supplied by spray nerve endings

D- contains myelinated nerve terminals have dilated ends around the tendon core

E- both b &d

10-Which cell type is responsible for myelin formation in the CNS?

A) Schwann cell B) Oligodendroglia C) Astrocyte D) Microglia

11-Synaptic vesicles are most commonly located

a. Witin the axon initial segment b. within the synaptic cleft

c. within presynaptic axon terminals d. within postsynaptic dendrites

12-Which of the following is TRUE about CNS myelinated fibers?

A) They have a continuous neurolemma like PNS fibers B) They lack a neurolemma

C) They are always unmyelinated D) Each fiber is individually wrapped by Schwann cells

13-Approximately how many times can the Schwann cell membrane wrap around a single axon to form myelin?

A) 5 times B) 20 times C) 50 times D) 100 times

14-How many intrafusal fibers are typically found in a single muscle spindle?

A) 1–3 B) 6–12 C) 15–20 D) 20–30

15-What is formed when the plasmalemma of a Schwann cell encircles an axon?

A) Axon hillock B) Node of Ranvier C) Myelin sheath D) Endoneurium

16-Which type of cell is responsible for forming the myelin sheath in the peripheral nervous system?

A) Oligodendrocyte B) Schwann cell C) Astrocyte D) Microglia

17-How is the myelin sheath formed around a peripheral axon?

A) By layering Schwann cell cytoplasm around the axon

B) By spirally wrapping the Schwann cell membrane around the axon

C) By oligodendrocytes forming multiple layers in the CNS

D) By fibroblasts secreting collagen

18-Among the following, which type of nerve fiber conducts impulses the fastest?

A) Small-diameter unmyelinated fiber B) Small-diameter myelinated fiber

C) Large-diameter myelinated fiber D) Large-diameter unmyelinated fiber

19-Which of the following is present in the axon terminal of a motor neuron?

A) Nucleus of the neuron B) Mitochondria C) Myelin sheath D) Node of Ranvier

20-The nodes of Ranvier are:

A) Gaps in the myelin sheath B) Regions with extra myelin layers

C) The cell body of the neuron D) Axon terminals

21-The connective tissue surrounding the muscle spindle is continuous with:

A) Epimysium B) Endomysium C) Perimysium D) Sarcolemma

22-Which type of intrafusal fiber contains a cluster of nuclei at the center?

A) Nuclear chain fibers B) Nuclear bag fibers C) Extrafusal fibers D) Golgi tendon fibers

23-What is the term for the segment of an axon between two nodes of Ranvier?

A) Axon hillock B) Dendrite C) Internodal segment D) Synaptic terminal

24-The speed of nerve conduction is directly proportional to:

A) Axon diameter and myelin thickness B) Length of dendrites only

C) Number of synapses D) Presence of neuroglia

25-What is the neurolemma?

A) Axonal cytoplasm B) Nucleus of Schwann cell

C) Cell membrane of Schwann cell surrounding myelin D) A type of glial cell

26-In H&E-stained slides, how does the myelin sheath typically appear?

A) Dark purple layer surrounding axon

B) Vacuolated or empty space between axolemma and neurolemma

C) Bright red continuous layer D) As granular cytoplasm of Schwann cell

27-In the peripheral nervous system, a single Schwann cell myelinates:

A) Multiple axons simultaneously B) Only one axon segment

C) The entire neuron including dendrites D) No axons, only glial cells

28-Under electron microscopy, the myelin sheath appears as:

A) A homogenous dark layer

B) Many concentric lamellae with alternating light and dark lines

C) Granular cytoplasm surrounding the axon D) A single thick membrane

29-Which of the following are examples of non-encapsulated (free) sensory nerve endings?

A) Free nerve endings and Merkel's discs B) Pacinian corpuscles and Meissner's corpuscles

C) Ruffini endings and Krause end bulbs D) Muscle spindles and Golgi tendon organs

30-The thickened region of the axon terminal membrane where neurotransmitter release occurs is called:

A) Postsynaptic density B) Presynaptic membrane C) Neurolemma D) Axolemma

31-Which type of intrafusal fiber has nuclei arranged in a longitudinal chain?

A) Nuclear chain fibers B) Nuclear bag fibers C) Extrafusal fibers D) Pacinian fibers

1. C	2. A	3. B	4. C	5. C
6. C	7. B	8. D	9. C	10.B
11.C	12.B	13.C	14.B	15.C
16.B	17.B	18.C	19.B	20.A
21.C	22.B	23.C	24.A	25.C
26.B	27.B	28.B	29.A	30.B
31.A	32.	33.	34.	35.

Physiology

BBB and CSF

1-The cerebrospinal fluid is produced by the {Formative2026}

a) Arachnoid villi b) Choroid plexus c) Dura matter d) Pia matter

2- Which of the following has a much lower concentration in the cerebrospinal fluid (CF) than in cerebral capillary blood? {Final2021}{Mid2023}{Formative2026}{د}

A- Na+ B- K+ C- Osmolality D- Protein E- Mg2+

3- Glucose is the main fuel of brain, the amount of glucose in CSF is.....in plasma {Mid2025}

A) the Same as B) more than C) less than D) absent

4- Na concentration in the CSF is……. its concentration in the plasma. {Summar2025}

a) more than b) less than c) same as d) absent

5-Which structrure of the ventricular system synthesizes cerebrospinal fluid (CSF)? {Formative2022}

a-Cerebral aqueduct b-Arachnoid granulations

c-Choroid plexus d-Subarachnoid cisterns

6 - The blood–brain barrier {Formative2022}

A. Slows equilibration of solutes between blood and brain tissue fluids.

B. Is a more effective barrier for fat-soluble substances than water-soluble substances.

C. Is a more effective barrier in infants than in adults.

D. Is a more effective barrier for CO2 than for O2

7- Cerebrospinal fluid {Final2022}{Formative2024}

A- Is formed in the arachnoid granulations

B- Provides the brain with most of its nutrition

C- Protects the brain from injury when the head is moved

D- Has a lower pressure than that in the cerebral venous sinuses

8- The cerebrospinal fluid is present in {Final2022}

A- Subarachnoid and subdural space

B- Subarachnoid space and ventricles of the brain

C- Subdural space but not the subarachnoid space

D- Subdural space and ventricles of the brain E- None of the above

9- Which of the following best describes the total volume of cerebrospinal fluid in an adult human nervous system (in milliliters)? {Mid2024}

a) 150 b) 500 c) 50 d) 300

10-CSF circulate in which of the following sites? {Formative2024}

a) between pia mater and brain b) subdural space

c) Between dura and bone d) subarachnoid space

11-Functions of CSF {Final2023}

a) Protection of brain when the head is moved b) Provide most of brain nutrients

c) Its pressure is less than that of cerebral venous blood

12- Which of the following statements about the cerebrospinal fluid (CSF) is TRUE? {Final2021}{Final2022}

A- It is absorbed by the choroid plexus

B- Its absorption is dependent on CF pressure and venous sinuses pressure

C- It circulates in the epidural space

D- It has a higher protein concentration than plasma

13-The blood brain barrier is impermeable to ----------

a. plasma proteins. b. CO_2 and O_2 c. Lipid-soluble substances. d. Water

14-Blood brain barrier permits the entry of

a. Most large molecules and proteins b. Low lipid (fat) soluble molecules

c. Toxins and foreign substances d. Lipid soluble molecules

15-Cerebral blood flow tends to increase when

a. Extravascular H concentration increases b. Systemic venous pressure rises

c. Cerebral activity decreases d. Intracranial pressure increases

e. Arterial pressure falls below 60mmhg

16-Which of the following ions is present in similar concentration in CSF and blood?

A. K^+ B. Ca^{2+} C. Na^+ D. Mg^{2+}

17-Which ion is lower in CSF than in blood?

A. Na^+ B. K^+ C. Mg^{2+} D. Cl^-

18-Which ion is higher in CSF than in blood?

A. Na^+ B. K^+ C. Mg^{2+} D. Ca^{2+}

19-Which substance is higher in CSF than in blood?

A. K^+ B. Ca^{2+} C. Protein D. Creatinine

20-CSF is primarily secreted by:

A. Arachnoid villi B. Ependymal lining C. Choroid plexuses D. Pia mater

21-What happens if CSF pressure falls below 68 mm H_2O?

A. CSF production ceases B. CSF flow reverses

C. CSF becomes turbid D. CSF absorption stops

22-Which of the following substances cannot cross the adult BBB under normal conditions?

A. Oxygen B. Carbon dioxide C. Plasma proteins D. Alcohol

23-Which type of drug crosses the BBB most readily?

A. Ionized water-soluble drugs B. Large proteins

C. Non-ionized, lipid-soluble drugs D. Charged molecules

24-Lipid-soluble substances that can cross the BBB include:

A. Cholesterol and bile pigments B. Plasma proteins

C. Alcohol and anesthetics D. Albumin

25-The rate of CSF secretion is approximately:

A. 0.1 ml/min B. 0.5 ml/min C. 1 ml/min D. 2 ml/min

26-The adult BBB is essentially impermeable to:

A. CO_2 B. O_2 C. Cholesterol D. Water

27-Inflammation, irradiation, or tumors can:

A. Strengthen the BBB B. Reduce drug entry into the brain

C. Disrupt the BBB, allowing entry of substances normally excluded

D. Increase neurotransmitter escape

28-The CSF turns over how many times per day?

A. 1–2 times B. 2–3 times C. 5–6 times D. 3.7 times

29-CSF primarily fills:

A. Only the spinal cord B. Only the subarachnoid space

C. The ventricles and the subarachnoid space D. Only the cisterns

30-The BBB prevents:

A. Escape of neurotransmitters from CNS into general circulation

B. Glucose transport into neurons

C. Oxygen from entering the CNS D. Lipid-soluble drugs from entering the brain

31-Which of the following requires a carrier to cross the BBB?

A. Oxygen B. Carbon dioxide C. Glucose D. Alcohol

32-CSF absorption occurs mainly by:

A. Osmosis B. Diffusion C. Bulk flow via arachnoid villi D. Active transport in ventricles

33-Which of the following spaces is filled with CSF?

A. Epidural space B. Subdural space C. Perineurial space D. Subarachnoid space

34-The rate of CSF production per day is approximately:

A. 250 ml/day B. 550 ml/day C. 750 ml/day D. 1000 ml/day

35-At normal CSF pressure (~112 mm H_2O):

A. Filtration exceeds absorption B. Absorption exceeds filtration

C. Filtration and absorption are equal D. CSF flow stops

36-Normal lumbar CSF pressure ranges from:

A. 50–100 mm H_2O B. 50–100 mmg C. 80–180 mmHg D. 80–180 mm H_2O

37-At approximately what CSF pressure does absorption stop?

A. 80 mm H_2O B. 100 mm H_2O C. 68 mm H_2O D. 120 mm H_2O

1. B	2. D	3. C	4. C	5. C
6. A	7. C	8. B	9. A	10.D
11.A	12.B	13.A	14.D	15.A
16.C	17.B	18.C	19.D	20.C
21.D	22.C	23.C	24.C	25.B
26.C	27.C	28.D	29.C	30.A
31.C	32.C	33.D	34.B	35.C
36.D	37.C	38.	39.	40.

The synapses

1- Renshaw cell in the spinal cord acts by releasing: {Summar2025}

a) Acetylcholine b) Glycine c) Glutamate d) All of the above

2- The most common type of synapses is {Summar2025}

a) Axodendritic b) Axoaxonic c) Axosomatic d) Dendrite somatic

3- Neurotransmitter is released into the synapse and makes its way to receptors on the……. of the target neuron. {Summar2025}

a. postsynaptic axon. b. presynaptic cell soma.

c. postsynaptic dendrite. d. presynaptic dendrite.

4- Which of the following is true about Axo-Axonic synapse {Mid2025}

a. Highest excitability due to opening of Na channels

b. Highest excitability due to opening of L type channels

c. Highest excitability due to opening of T type channels

d. Highest excitability due to opening of aquaporin II type channels

5-Post synaptic membrane contains: {Formative2025}

a) Receptor of acetyl choline b) Acetyl choline esterase

c) Adrenalin d) Acetyl choline

6- The most active part in triggering action is {Final2025}

a. Neurotransmitter- receptor complex

7-it takes 2 msec for transmission of a signal from certain dorsal root ganglia to a certain A.H.C, How many interneurons are present? {Formative2026}

a) 5 b) 3 c) 4 d) 0.5

8- It takes 2 msec for transmission of a signal from certain dorsal root ganglia to a certain A.H.C. How many synapses are present? {Summar2025}

a) 4 b) 3 c) 5 d) 0.5

9- Synaptic conduction is mostly orthodromic (one way conduction) because: {Formative2022}

A. Dendrites cannot be depolarized.

B. Once repolarized, an area cannot be depolarized.

C. The strength of antidromic impulses is less.

D. The neurotransmitter is in the presynaptic terminal.

10- IPSP differs from EPSP in ? {Mid2021}

A- Being of shorter duration B- Being unable to summate spatially

C- Moving the membrane potential away from threshold

D- Depending upon opening of voltage K + channels

11-Concerning synapsis which of the following is CORRECT? {Mid2021}{Formative2022}

A- The electrical synapses are more common in CNS

B- Amphetamine decreases synaptic transmission

C- Acidosis increases synaptic transmission

D- Hypoxia decreases synaptic transmission

12-Excitatory post synaptic potentials can be caused by {Final2023}

a) Cl influx b) Na influx c) K outflux d) Ca outflux

13- Opening of Cl channels leads to {Final2023}

a) inhibition of post synaptic neuron b) stimulation of post synaptic neuron

14- Opening of ligand-gated Cl-channels causes? {Mid2021}{Final2022}{Formative2024} {Mid2024}

A- Inhibition of the postsynaptic neuron B- Depolarization of the postsynaptic neuron

C- Initiation of an action potential D- Block of ligand-gated cation channels

15- Temporal summation ? { Mid2021}{Final2022}{Mid2024}

A- Occurs when several presynaptic fibers are stimulated simultaneously

B- Occurs when presynaptic fiber is repeatedly stimulated

C- Is due to overlap of the discharging zone

D- Stimulus occurs after depolarization produced by the preceding stimulus

16- Presynaptic inhibition ? {Mid2021}{Final2022}

A- Is produced as a result of GABA release

B- Leads to decreased release of excitatory transmitter

C- Is involved in gate theory of pain control

D- All of the above

17-About the action of a transmitter in a synapse, the most active part is: {Formative2025}

a) The neurotransmitter b) The receptor

c) The receptor neurotransmitter complex d) The vesicle with V snrf and T snrf

18- If transmission takes 2 mms then how many interneurons {Mid2025}

a. 3 b. 5 c. 4

19- Ranshow cell is activated by {Mid2025}

a. glutamate b. Glysine c. Ach d. All of them

20- At chemical synapses, excitatory postsynaptic potentials (EPSPs) can be produced by {Final2021}{Final2024}{Formative2024}

A- Opening of Na+ channels B- Opening of K+ channels

C- Opening of Cl- channels D- Closure of Ca+2 channels

21- Which ONE of the following neurotransmitters would you expect to find in the synapse during fast Inhibitory synaptic transmission? {Final2021}{Final2022}{Formative2024}

A- GABA B- Acetylcholine C- Noradrenaline D- Glutamate

22- In Case of alkalosis in epileptic patient, the protective method to prevent excitation : {Mid2023}{Final2024}

A- Increase hydrogen ions in blood B- Decrease arousal neurotransmitters

C- Increase sleep neurotransmitters D- fatigue in synaptic transmission

23- Concerning synapsis which of the following is correct? {Mid2024}

a) The electrical synapses are more common in CNS.

b) Amphetamine decreases synaptic transmission

c) Acidosis increases synaptic transmission

d) Hypoxia decreases synaptic transmission.

24-Which electrical event is characteristic of inhibitory synaptic interactions? {Guyton}

A) A neurotransmitter agent that selectively opens ligand-gated chloride channels is the basis for an inhibitory postsynaptic potential

B) Because the Nernst potential for chloride is about −70 mV, chloride ions tend to move out of the cell along its electrochemical gradient

C) A neurotransmitter that selectively opens potassium channels will allow potassium to move into the cell

D) An increase in the extracellular sodium concentration usually leads directly to an inhibitory postsynaptic potential

25-The release of neurotransmitter at a chemical synapse in the central nervous system is dependent upon which of the following? {Guyton}

A) Synthesis of acetylcholinesterase

B) Hyperpolarization of the synaptic terminal

C) Opening of ligand-gated ion calcium channels

D) Influx of calcium into the presynaptic terminal

26-Prolonged changes in neuronal activity are usually achieved through the activation of which of the following? {Guyton}

A) Voltage-gated chloride channels

B) Transmitter-gated sodium channels

C) G-protein–coupled channels

D) Voltage-gated potassium channels

27-Which of the following is characteristic of the events occurring at an excitatory synapse? {Guyton}

A) There is a massive efflux of calcium from the presynaptic terminal

B) Synaptic vesicles bind to the postsynaptic membrane

C) Voltage-gated potassium channels are closed

D) Ligand-gated channels are opened to allow sodium entry into the postsynaptic neuron

28-A transmitter substance released from a presynaptic neuron activates a second messenger G-protein system in the postsynaptic neuron. Which one of the following postsynaptic responses to the transmitter substance is NOT a possible outcome? {Guyton}

A) Activation of cyclic adenosine monophosphate (cAMP)

B) Activation of cyclic guanosine monophosphate (cGMP)

C) Activation of gene transcription

D) Closing an ion channel E) Opening an ion channel

29- A pool of presynaptic neurons innervate the dendrites of a postsynaptic neuron. Electrical signals are transferred from the dendrites to the soma of the postsynaptic neuron by which process? {Guyton}

A) Action potential B) Active transport C) Capacitive discharge

D) Diffusion E) Electrotonic conduction

30-Concerning synapsis which of the following is CORRECT

A- The electrical synapses are more common in CNS

B- Amphetamine decreases synaptic transmission

C- Alkalosis increases synaptic transmission

D- Hypoxia increases synaptic transmission

31- Synaptic transmission is characterized by all of the following, EXCEPT: {د}

a- impulses are transmitted from the pre to the postsynaptic neurons

b- alkalosis depresses synaptic transmission.

c- fatigue is due to exhaustion of neurotransmitter stores.

d- repetitive stimulation leads to synaptic potentiation.

32- Synaptic plasticity shows all of the following, EXCEPT: {د}

a- ability to change the function of the synapse according to the demand.

b- play an important role in memory.

c- may lead only to long term potentiation. d- explain the process of habituation.

33- Regarding synapses: {د}

a- presynaptic knobs contain vesicles which have t-snare in their membranes

b- v-snare of the vesicles has high affinity to t-snare in the active zone of the presynaptic membrane

c- Ca2+ voltage gated channels of the pre-synaptic knob open during repolarization

d- synaptic cleft contain high concentration of K+.

34-Concerning synapsis which of the following is CORRECT

A- The electrical synapses are more common in CNS

B- Amphetamine decreases synaptic transmission

C- Acidosis increases synaptic transmission

D- Hypoglycemia decreases synaptic transmission

35- Regarding presynaptic inhibition: {د}

a- the postsynaptic neuron membrane is hyperpolarized.

b- there is decrease release of excitatory chemical transmitters from the presynaptic neuron.

c- there is increased Ca2+ influx into presynaptic neuron.

d- there is closure of K+ channels of the presynaptic neuron.

36- Glycine: {د}

a- acts mainly as inhibitory chemical transmitter.

b- action is potentiated by strychnine.

c- causes depolarization of the postsynaptic membrane.

d- is responsible for the process of sensitization

37-The most excitable part of a neuron is:

a) Dendrites b) Soma c) Axon hillock d) Node of Ranvier

38-The synaptic cleft mainly contains:

a) Intracellular fluid b) Extracellular fluid c) Synaptic vesicles d) Mitochondria

39-Excitation or inhibition of the postsynaptic neuron, depending on ……

a- Type of transmitter mainly b- Type of receptor mainly

40-Renshaw cell is stimulated or excited by:

a) Acetylcholine b) Glycine c) GABA d) Noradrenaline

41-When Renshaw cell is stimulated , it release …….

a) Acetylcholine b) Glycine c) GABA d) Noradrenaline

42-Synaptic delay is about……..

a. 0.5msec b. 0.5sec c. 0.05msec d. 0.05sec

43-if synaptic delay about 7 msec , there is …….. interneurons

a. 14 b. 15 c. 16

44-if synaptic delay about 7 msec , there is …….. synapses

a. 14 b. 15 c. 16

45-The ability to change the function of a synapse according to demand is called:

a) Synaptic fatigue b) Synaptic plasticity

c) Synaptic delay d) Synaptic inhibition

46- Concerning organization of synapses: {د}

a- axo-axonic is the most excitable. b- axo-dentritic is the most excitable

c- axo-somatic is the least excitable. d- axo-dendritic is the least in number.

47-Synaptic transmission can be inhibited by:

a) hypercalcemia. b) alkalosis. c) acidosis. d) thyroid hormones

48-Synaptic transmission:

a) is inhibited by hypercalcemia. b) is stimulated by alkalosis.

c) does not show fatigue. d) is stimulated by acidosis.

e) shows delay about 15 m.sec

49-When an excitatory neurotransmitter is released it may lead to:

a) potassium entry. b) calcium entry. c) hyperpolarization.

d) chloride entry. e) EPSP

50-Synaptic transmission may involve:

a) release of neurotransmitter from post synaptic neuron.

b) entry of Ca++ to postsynaptic neuron.

c) occurrence of EPSP in presynaptic neuron.

d) outflow of K+ from presynaptic neuron.

e) Na+ entry to postsynaptic neuron

51-Presynaptic inhibition leads

a. Decrease in neurotransmitter release at the synapse

b. Decrease in membrane conductance of the pre-synaptic nerve ending

c. Action potential of post-synaptic neurons

d. Excitation of post-synaptic membrane

52-In presynaptic inhibition:

a) the release of the excitatory neurotransmitter from the presynaptic knob is inhibited.

b) the postsynaptic membrane is hyperpolarized.

c) the presynaptic knobs release glycine.

d) the presynaptic knobs release GABA.

53-Presynaptic inhibition is characterized by:

a) an increase in calcium entry to presynaptic neuron.

b) inhibition of release of neurotransmitter from presynaptic neuron.

c) depolarization of postsynaptic neuron.

d) inhibitory neuron terminating on postsynaptic neuron.

e) an increase in sodium influx to post synaptic neuron

54- Inhibitory post synaptic potential: {د}

a- involves increased K+ efflux. b- involves increased Na+ efflux.

c- is a state of partial depolarization. d- increases excitability of the postsynaptic membrane.

55- Inhibitory presynaptic potential: {د}

a- needs inhibitory third neuron. b- needs excitatory third neuron.

c- leads to depolarization of the presynaptic neurone.

d- directly inhibits the postsynaptic neurone,

56-regarding summation:

a) Spatial summation is more common than temporal summation.

b) Temporal summation requires discharge from multiple presynaptic neurons.

c) Spatial summation requires repetitive discharge from single presynaptic neuron

d) summation is rarely required in synapses.

57-Temporal summation of EPSP results from:

a) simultaneous stimulation of multiple knobs. b) repetitive stimulation of a single knob.

c) convergence. d) divergence

58-Spatial summation of EPSP is helped by:

a) parallel circuits. b) divergence. c) convergence. d) lateral inhibition.

59-Convergence in the neural pool leads to:

a) spread of stimulus. b) spatial summation.

c) temporal summation. d) after discharge

60-Which of the following is characteristic of the events occurring at an excitatory synapse?

a. Chloride channels are opened in the post-synaptic membrane

b. Synaptic vesicles bind to the postsynaptic membrane

c. Voltage-gated potassium channels are closed

d. Ligand-gated channels are opened to allow sodium entry into the postsynaptic neuron

61-The minimum time for transmission across one synapse is

a. 0.5 msec b.0.5 sec c. 1 msec d. 1.5 msec d.1 sec

62-The membrane potential will depolarize by the greatest amount if the membrane permeability increases for which of the following?

a. Potassium and chloride b. Chloride c. Sodium d. Sodium and potassium

63-Excitatory post-synaptic potentials (EPSP) are produced by

a. Opening of chloride channels in post-synaptic membrane

b. Transient inflow of positive charges to the postsynaptic neurons

c. Closure of calcium channels in the pre-synaptic membrane

d. Hyperpolarization of post-synaptic membrane

64-Synaptic transmission depends upon

a. Presence of voltage-gated Ca ++ channels in membrane of synaptic knobs

b. Diffusion of neurotransmitters from synaptic knobs into the soma and dendrites of postsynaptic neurons

c. Direct transmission of impulses from the presynaptic neuron to the postsynaptic neuron

d. Presence of voltage-gated chloride - channels in the subsynaptic membrane

65-IPSP could result from

a. Opening of ligand-gated cation channels b. Closure of potential-gated Ca ++ channels

c. Closure of Cl-channels d. Opening of K + channels

66- Excitatory post synaptic potentials

a. Is due to opening of Na voltage-gated channels in postsynaptic membrane

b. Obeys all or none rule

c. Of the dendrites can reach an action potential

d. None of the above

67-Which of the following is an inhibitory neurotransmitter:

a) histamine. b) glutamate. c) GABA. d) acetyl choline. e) noradrenaline

68-which of the following is an inhibitory neurotransmitter:

a. Glutamate b. Glycine. c. Aspartate. d. Adrenaline.

69- Excitatory postsynaptic potential: {د}

a. can not be summated b. is a state of local depolarization.

c. is due to opening of voltage gated Na+ channels.

d. is due to opening of chemical gated K+ channels.

70- Excitatory postsynaptic potential: {د}

a- is due to increase permeability of the postsynaptic membrane to Na+

b- is a state of complete depolarization.

c- can produce an action potential without summation.

d- of dendrites can reach to an action potential.

71- Excitatory postsynaptic potential: {د}

a- is not propagated. b- shows reversal of polarity.

c- lasts about 2 seconds. d- may reach action potential at the soma.

72- EPSPs : {د}

a- result from release of GABA from presynaptic neuron.

b- can be temporally summated.

c- result from increase permeability of the postsynaptic membrane to K+

d- obeys the all or none rule.

73- Regarding synaptic potentials: {د}

a. presynaptic inhibition causes hyperpolarization of post-synaptic neuron.

b. during IPSP a neuron can not discharge an action potential.

c. presynaptic facilitation is due to increased secretion of chemical transmitter due to stimulation by a 3rd excitatory presynaptic neuron.

d. EPSP can be caused by glycine.

74- Regarding synaptic potentials: {د}

a presynaptic inhibition causes hyperpolarization of post-synaptic neuron.

b. during IPSP a neuron can not discharge an action potential

c. Post-tetanic facilitation is due to increased secretion of chemical transmitter due to stimulation by a 3 excitatory presynaptic neuron.

d. EPSP can be caused by glycine.

75- An inhibitory neurotransmitter increases the permeability of the postsynaptic neurone to: {د}

a- K+ b- Na+ c- Ca2+ d- HCO3

76- Inhibitory post synaptic potentials: {د}

a- is due to presynaptic inhibition. b- causes a state of hyperpolarization.

c- is associated with influx of Na+ ions. d- can not be summated.

77- IPSP: {د}

a- is a state of depolarization of the postsynaptic membrane.

b- is produced by binding of acetylcholine to its receptor.

c- results from inflow of Cl- ions and outflow of K+ ions from the postsynaptic membrane.

d- results from the release of catecholamines from the presynaptic knob.

78- IPSPs: {د}

a- are a consequence of presynaptic inhibition.

b- interact with other fast and slow potentials to move the membrane potential of the postsynaptic neuron toward or away

c- can be produced by an increase in Na+ conductance from the firing level.

d- can be produce by an increase in Ca2+ conductance.

79- Grand post-synaptic potential, may cause the postsynaptic membrane to: {د}

a- be more excitable if EPSP > IPSP. b- be less excitable if IPSP > EPSP.

c- reach firing if there is summation of EPSPs d- all of the above

80- Regarding synaptic transmission: {د}

a- is depressed by alkalosis.

b- is increased by acidosis.

c- repetitive stimulation of a single presynaptic neurone leads to development of an action potential in the postsynaptic neurone

d- is increased by hypoxia.

81- Synaptic transmission: {د}

a- is increased by alkalosis. b- shows a delay of about 0.05 ms.

c- is increased by acidosis. d- decreased during tetanus disease

82- Synaptic transmission is facilitated by all of the followings, EXCEPT: {د}

a- coffee and tea. b- alkalosis. c- strychnine. d- hypoxia.

83- Regarding synaptic transmission, it is decreased by: {د}

a- alkalosis. b- hyperventilation. c- increased secretion of GABA d- theobromine.

84- If the central delay in one reflex is 4 msec., then the: {د}

a- number of synapses in this reflex is two. b- number of synapses in this reflex is four.

c- number of synapses in this reflex is eight. d- it is a monosynaptic reflex.

85- All of the following about synaptic transmission is true EXCEPT: {د}

a- It is unidirectional from presynaptic to postsynaptic neuron.

b- It undergoes fatigue by repeated stimulation.

c- It shows a delay by about 5 msec at each synapse.

d- It is depressed by anesthetics and hypnotics.

86-Lateral inhibition: {د}

a- Needs only inhibitory neurons

b- Helps sharpening of sensation.

c- Means that inhibited central neurons are surrounded by excitatory neurons.

d- Needs only one type of chemical transmitter.

87-The ability to detect two points simultaneously applied to the skin is based on which physiologic mechanism? {Guyton}

A) Presynaptic inhibition B) Lateral inhibition

C) Medial inhibition D) Feed-forward inhibition

88-Electrical synapse differ from chemical synapse in that

a. They are slow in conduction b. Common in central nervous system

c. There are gap junctions in pre and post synaptic memberane

d. Unidirectional e. Always inhibitory

89-Excitatory post synaptic potentials (EPSP) are produced by

a. Increased calcium outflux b. Increased sodium influx

c. Increased chloride influx d. Increased potassium outflux

90-In a given neuronal circuit

a. Facilitation may be produced by overlap of two maximal inputs

b. Occlusion may be produced by overlap of submaximal input

c. Reverberation allows maintained discharge despite stopping of original stimulus

d. Convergence helps relay of one input signal to multiple neurons

91-After discharge is caused by:

a) projection mechanism. b) fasciculation along the pathway.

c) lateral inhibition. d) reverberating circuits

92-After discharge:

a) can be caused by lateral inhibition. b) can be caused by parallel circuits.

c) is due to presence of Renshow cells. d) leads to spread of the stimulus

93-Parallel circuit is a mechanism for

a) pain inhibition. b) hyperalgesia. c) referred pain. d) after discharge

94- Release of chemical transmitter at synapse is dependent on

a. Synthesis of acetylcholine esterase

b. Hyperpolarisation of synaptic terminal

c. Fusion of synaptic vesicles with postsynaptic membrane

d. Influx of calcium into synaptic terminal

95-Regarding the mechanism of synaptic transmission: {د}

a- Ca ions are released from the presynaptic knob.

b- Presynaptic knobs release only an excitatory transmitter.

c- Chemical transmitter causes development of only excitatory post-synaptic potentials.

d- Excitatory transmitters open Na+ channels.

96- concerning synapses: {د}

a- electrical synapses are more common in CNS than chemical synapses

b- chemical synapses act in both directions.

c- Chemical synapses are characterized by presence of gap junctions between pre- and postsynaptic neurons.

d- Chemical synapses show synaptic delay.

97-Chemical synapses in the nervous system :

a- allow diffusion of chemical substances form the presynaptic neuron into the postsynaptic neuron

b- allow transmission of potential changes in direction only; from presynaptic to the postsynaptic neurons

c- have potential-gated ionic channels

d- are more numerous in peripheral nervous system than the central nervous system

98-Synaptic transmission is terminated by:

a- block of presynaptic receptors

b- elevation of Ca ++ concentration in synaptic cleft

c- reuptake of neurotransmitters by postsynaptic neurons

d- degradation of neurotransmitters by specific enzymes

99- Released transmitter is inactivated by all of the following except: {-}

a- active reuptake. b- inactivation by enzymes.

c- diffuse away in ECF. d- phagocytosed by glial cells.

100-Excitatory or inhibitory action of neurotransmitter is determined by

a. Function on its postsynaptic receptor b. Its molecular composition

c. Shape of synaptic vesicles d. Distance between pre- and post- synaptic membranes

1. B	2. A	3. C	4. A	5. A
6. A	7. A	8. A	9. D	10.C
11.D	12.B	13.A	14.A	15.B
16.D	17.C	18.A	19.C	20.A
21.A	22.D	23.D	24.A	25.D
26.C	27.D	28.D	29.E	30.C
31.B	32.C	33.B	34.D	35.B
36.A	37.C	38.B	39.B	40.A
41.B	42.A	43.B	44.A	45.B
46.A	47.C	48.B	49.E	50.E
51.A	52.A	53.B	54.A	55.A
56.A	57.B	58.C	59.B	60.D
61.A	62.C	63.B	64.A	65.D
66.A	67.C	68.B	69.B	70.A
71.A	72.B	73.C	74.C	75.A
76.B	77.C	78.B	79.D	80.C
81.A	82.D	83.C	84.C	85.C
86.B	87.B	88.C	89.B	90.C

91.D	92.B	93.D	94.D	95.D
96.D	97.B	98.D	99.D	100. A

Somatosensory function

1-The primary sensory cortex is concerned for coding of the stimulus detected by the sensory receptors as regard: {Formative2026}

a) modality, location, intensity, and duration.

b) specific energy, adequate threshold, sensation, and duration.

c) adequate threshold, receptive field, adaptation, and projection.

d) modality, adequate threshold, sensitivity, and location.

2-About touch receptors all are true except: {Formative2026}

a) Adapt rapidly b) Superficial receptors

c) Respond to mechanical changes d) Fatigue rapidly

3- The sensory unit is: {Summar2025}

a) Single afferent neuron with all its receptor endings

b) All neurons in the same area of the CNS

c) All receptors in the same area of the skin

d) All nerve fibers with all its motor end plates

4- When receptors causes large change in intensity of the stimulus that leads it all small change in frequency this demonstrate Which function of the receptor ? {Mid2025}

a. Compressor, weber fechner law b. Compressor, law of projection

c. Generator molars law d. Transducer Weber fishner

5-Gradual decrease in the rate of discharge from the receptors under constant stimulation is called: {Formative2024}

a) adequate threshold b) receptor potential c) adaptation d) fatigue

6-. Which column of grey matter receives the sensory input for the body ? {Formative2022}

A. posterior column B. anterior column C. lateral column D. intermediate column

7- A single sensory axon and all of its peripheral branches constitute {Formative2022}

A. receptive field B. sensory unit C. dermatome D. sensory nerve

8-A single sensory axon transmitting painful stimulus and all of its peripheral branches constitute {Formative2024}

a) Receptive field b) Sensory nerve c) Dermatome d) Sensory unit

9- The four basic parameters of a stimulus which are detected by the sensory receptors and are encoded by somatosensory areas are ? {Mid2021}{Final2024}{Formative2024}

A- Modality, adequate threshold, sensitivity, and location

B- Adequate threshold, receptive field, adaptation, and projection

C- Sensitization, discrimination, energy, and projection

D- Modality, location, intensity, and duration

10- Sensory coding include : {Mid2023}

A- modality, adequate threshold, sensitivity, and location

B- adequate threshold, receptive field, adaptation, and projection.

C- modality, location, intensity, and duration

D- sensitization, discrimination, energy, and projection

11-Proprioceptive information reaches consciousness through the: {Formative2024}

a) olivocerebellar pathway b) spinocerebellar pathway

c) dorsal column -medial lemniscal pathway d) spinothalamic system

12-Which of the following is not dorsal column sensation? {Final2023}

a) Sense of position b) Tactile localization

c) Crude touch d) Sense of muscle movement

13-Which of the following best describes the concept of specificity in sensory nerve fibers that transmit only one modality of sensation? {Mid2024}{Guyton}

A) Frequency coding principle B) Concept of specific nerve energy

C) Singularity principle D) Labeled line principle

14 - In sensory receptors one of the following is correct : {Formative2022}

A. Stimulus energy is converted into a local depolarization.

B. The generator potential is graded and self-propagating.

C. A generator potential can be produced by only one form of energy.

D. The frequency of action potentials generated doubles when the strength of the stimulus doubles.

15-In sensory receptors one of the following is correct : {Formative2024}

a) Stimulus energy is converted into a local depolarization.

b) A single receptor can be stimulated many forms of energy

c) The frequency of action potentials generated doubles when the strength of the stimulus doubles.

d) The generator potential follows the all or none rule.

16- Discrimination of the intensity of stimuli results from the: {د}

a- presence of adequate stimulus for each receptor.

b- presence of specific pathway for each sensation.

c- specific centre for each sensation.

d- Weber-Fechner law or Power low.

17-Rods and cones of the retina are a type of: {د}

a- Mechanoreceptors. b- Thermoreceptors. c- Nociceptors. d- Photoreceptors.

18-Touch receptors are a type of: {د}

a- Mechanoreceptors. b- Thermoreceptors. c- Nociceptors. d- Photoreceptors.

19- Taste buds contain receptor cells that considered a type of: {د}

a- Mechanoreceptors. b- Thermoreceptors. c- Nociceptors. d- chemoreceptors.

20- Which of the following receptors are NOT found in the skin: {د}

a. Mechanoreceptors. b. Thermoreceptors. c. Nociceptors. d. Photoreceptors.

21- Specificity: {د}

a- explains the process of adaptation.

b- means that each receptor is most sensitive to a specific form of energy called the adequate stimulus.

c- is a state of partial depolarization. d- is due to high intensity of the stimulus.

22- The receptor potential: {د}

a- is conducted actively to the 1st node of Ranvier.

b- can be summated.

c- is accompanied with absolute refractory period.

d- has the same duration like that of the action potential.

23- Receptor potential: {د}

a- is an action potential.

b- is due to non-specific opening of Na+ channels.

c- duration is shorter than that of the action potential.

d- brings the membrane potential away from the firing level.

24- Pacinian corpuscles are: {د}

a- pain receptors. b- rapidly adapting touch receptors.

c- slowly adapting touch receptors. d- innervated by C nerve fibers.

25- Fast adaptation receptors include: {د}

a- pain receptors. b- auditory receptors.

c- paccinian corpuscles. d- arterial barorecptors.

26-Depolarization of a receptor:

a) is called action potential. b) obeys all or none law.

c) is graded. d) is caused by Ca++ inflow.

27-Receptor potential:

a) is a state of partial depolarization of the receptor. b) is not graded. c) is not summated.

d) is caused by increased permeability to K+. e) its duration = 2 msec.

28-Receptor potential:

a) does not obey all or none law. b) can not be summated.

c) has a duration greater than 2 sec

d) results from an increase in the membrane permeability of the receptor to Ca++

e) its amplitude is inversely proportional to the intensity of the stimulus

29- Renshaw cell: {د}

a- Shows the phenomenon of negative feedback inhibition. b- Is activated by glycine.

c- Is present in dorsal horn of the spinal cord. d- Are excitatory neurons

30- Renshaw cell: {د}

a-shows the phenomenon of lateral inhibition b-is activated by glycine.

c-is present in dorsal horn of the spinal cord. d- are excitatory neurons.

31-Sensory receptor potentials {د}

a- are action potentials

b- always bring membrane potential of a receptor cell toward threshold

c- always bring membrane potential of a receptor cell away from threshold

d- are graded in size, depending on stimulus intensity

32-Receptor potential {د}

a. Is actively conducted to the 1 node of Ranvier

b. Has a duration shorter than that of the action potential

c. Amplitude is proportional to the intensity of the stimulus

d. Is due to specific increase of the membrane permeability to Na ions

33- Generator potential: {د}

a- is a state of partial depolarization. b- duration is shorter than that of the action potential.

c- is accompanied with absolute refractory period. d- can not be summated.

34-Adaptation is rapid in:

a) pain receptors. b) touch receptors. c) cold receptors.

d) hot receptors. e) muscle spindle

35-Adaptation of receptors:

a) is a decrease in response in spite of constant stimulation.

b) is a decrease in response when the stimulus decreased.

c) is rapid in pain receptors.

d) is slow in temperature receptors.

e) is moderate in touch receptors

36-Stimulation of receptors leads to:

a) action potential . b) partial depolarization. c) complete depolarization.

d) non specific response. e) all or non response

37-Slowly adapting (tonic) receptors include all the following except:

a. Baroreceptors b. Auditory receptors c. Pacinian corpuscles d. Nociceptors

38-Rapidly adapting (phasic) receptors include:

a. Muscle spindle b. Taste receptors c.Touch receptors d. Nociceptors

39-Moderately adapting receptors include all the following except:

a. Taste receptors b. Smell receptors c. Temperatue receptors d. Nociceptors

40- Posterior column damage in the spinal cord may impair all of the following EXCEPT {Mid2021}

A- Vibration sense B- Pain sensation

C- Touch sensation D- The ability to stand steadily with the eyes closed

41- In sensory receptors all of the following is CORRECT EXCEPT {Mid2021}

A- Stimulus energy is converted into a local depolarization

B- The generator potential is graded and self-propagating

C- A generator potential can be produced by only one form of energy

D- The frequency of action potentials generated doubles when the strength of the stimulus doubles

42- Impaired stereognosis in the absence of a detectable defect in touch and pressure sensation points to a lesion in (the): {Formative2022}

a) dorsal column b) medial lemniscus

c) parietal lobe posterior to the postcentral gyrus d) any of the above locations

1. A	2. D	3. A	4. A	5. C
6. A	7. B	8. D	9. D	10.C
11.C	12.C	13.D	14.A	15.A
16.D	17.D	18.A	19.D	20.D
21.B	22.B	23.B	24.B	25.C
26.C	27.A	28.A	29.A	30.A
31.D	32.C	33.A	34.B	35.A
36.B	37.C	38.C	39.D	40.B
41.C,D	42.C	43.	44.	45.

Physiology of pain

1-The pain produced by ischemia is poorly localized and throbbing while pain from a needle stick is well localized and sharp. Which of the following comparisons of ischemic and needle stick pain is correct? {Final2021}{Final2022}{Final2024}{Formative2026}

a) Ischemic pain sensory fibers are classified as A delta (AΔ) sensory fibers

b) Ischemic pain sensory fibers terminate within the substantia gelatinosa of the spinal cord

c) Ischemic pain is transmitted to the brain through the neospinothalamic tract

d) Ischemic pain is produced by overstimulating somatic touch receptors

2- About referred pain, all the following are true except: {Mid2025}{Summar2025}

a) It occurs as a result of migration of organs during development

b) It is explained by the convergence projection theory

c) It always accompanies cutaneous pain

d) It is a major manifestation of visceral pain

3- Characters of primary hyperalgesia include: {Summar2025}

a) It occurs in the injured skin area and the surrounding area of flare

b) The pain threshold is not lowered

c) Touch in this area is not painful d) It occurs in the healthy skin area

4- Concerning pain sensation, the following is true: {Summar2025}

a) Pain receptors in viscera are the same as in the skin

b) Pain receptors are rapidly adapting receptors

c) Slow pain is conducted by type A nerve fibers

d) Pain sensation is relayed in thalamus and sensory cortex

5- The following theories can explain pain analgesia except the: {Summar2025}

a) Gate theory b) Descending analgesic tracts

c) Presence of neurotransmitters such as enkephalins in the analgesic pathway

d) The convergence projection theory

6- pain receptors {Mid2025}

a. Are rapidly adapting receptors b. are free nerve endings c. Aren't widely distributed

7-Pain receptors in the skin are typically classified as which of the following? {Guyton}

A) Encapsulated nerve endings B) A single class of morphologically specialized receptors

C) The same type of receptor that detects position sense D) Free nerve endings

8- Pain receptors are {Final2021}

A- Similar in structure to Pacinian corpuscles B- Quick to adapt to a constant stimulus

C- More easily stimulated in injured tissue D- Stimulated by cutting the wall of the gut

9-All are TRUE about referred pain, EXCEPT…….. {Final2023}

a) pain felt away from its original site b) renal pain is referred at the testicles

c) somatic superficial pain is usually referred

d) occur in areas of the same dermatological origin

10-Which of the following is true about gate control theory in pain analgesia system {Final2025}

a. Touch receptors send inhibitory collaterals to DRG to inhibit release of substance P

b. Touch receptors send inhibitory collaterals to substantia gelatinosa to inhibit release of substance P

11- The sensation of pain from left leg is {Final2024}

A- carried by the same tract that carry temperature from the left leg

B- relayed by synapsing across the posterior root ganglion

12- Cutaneous pain: {Formative2023}

A.Could be caused by overstimulation of touch receptors.

B.Could be referred to other organs of the same dermatomal nerve supply.

C. Could be associated with hyperalgesia.

D. Receptors adapt to stimulation more quickly than touch receptors.

13-exaggerated pain sensation is called {Formative2024}

a) hyperalgesia b) analgesia c) hyperthesia d) red flare

14- Regarding hyperalgesia one of the following is CORRECT ? {Final2021}

A- It is the hyposensitivity to pain sensation B- It is divided into 3 types

C- The primary hyperalgesia associated with redness

D- The secondary hyperalgesia very near to the site of injury

15- All Of the following is TRUE about cutaneous pain EXCEPT : {Mid2023}

A- evokes spasm of nearby muscle B- Is always sharp

C- By A delta D- Always with hyperalgesia

16- Cutaneous hyperalgesia means {Mid2023}

A- painless stimulus become painful

17- Interneurons that use the neurotransmitter enkephalin to inhibit afferent pain signals are most likely found in which region of the central nervous system? {Mid2024}{Guyton}

a) Dorsal horn of spinal cord b) Postcentral gyrus c) δ-type A d) Type C fiber

18- Pain receptors in the gut and urinary tract may be stimulated by all of the following EXCEPT {Mid2021}

A- Cutting through their wall with a sharp scalpel B- Distention

C- Excessive acid fluid D- Vigorous rhythmic contractions behind an obstruction

19- Cutaneous pain {Mid2021}

A- Can be caused by overstimulation of touch receptors

B- Can be referred to other organs of the same dermatomal nerve supply

C- Can be associated with hyperalgesia

D- Receptors adapt to stimulation more quickly than touch receptors

20- A 32-year-old woman experienced the sudden onset of a severe cramping pain in the abdominal region. She also became nauseated. List some of the common features of visceral pain? {Mid2021}{Final2022}{Mid2024}{Formative2024}

A- It results from activation of nociceptors in the viscera that are innervated by the same fibers as innervate skin, induces rapid shard pain, causes spasms of the visceral muscle, and shows relatively rapid adaptation

B- It is mediated by Aδ and C fibers in the ventral roots of spinal nerves, radiates to a nearby or distant somatic structure, is accompanied by sweating, and is relayed to the cortex by the spinothalamic tract

C- It is poorly localized, is accompanied by sweating, radiates to a somatic structure that may be some distance away, and is relayed to the somatosensory cortex via the spinothalamic tract

D- It is well localized, is accompanied by sweating, radiates to a somatic structure that May be some distance away, and causes spasms of visceral muscle

21-About visceral pain , one is TRUE ? {Final2023}

a) usually accompanied with tachycardia b) show somatic reflex contraction of nearby muscles

c) usually localized d) carried on A delta

22-Pain from the stomach is referred to which area of the body? {Guyton}

A) Upper right shoulder area B) Abdominal area above the umbilicus

C) Proximal area of the anterior and inner thigh D) Abdominal area below the umbilicus

23-Stimulation of which brain area can modulate the sensation of pain? {Guyton}

A) Superior olivary complex B) Locus ceruleus C) Periaqueductal gray area D) Amygdala

24- Which of the following is a group of neurons in the pain suppression pathway that uses enkephalin as a neurotransmitter? {Guyton}

A) Postcentral gyrus B) Nucleus raphe magnus

C) Periaqueductal gray area D) Type AB sensory fibers

25-Which of the following is the basis for referred pain? {Guyton}

A) Visceral pain signals and pain signals from the skin synapse with separate populations of neurons in the dorsal horn

B) Visceral pain transmission and pain transmission from the skin are received by a common set of neurons in the thalamus

C) Visceral pain signals are rarely of sufficient magnitude to exceed the threshold of activation of dorsal horn neurons

D) Some visceral pain signals and pain signals from the skin provide convergent input to a common set of neurons in the dorsal horn

26-Which statement concerning visceral pain signals is correct? {Guyton}

A) They are transmitted along sensory fibers that course mainly with sympathetic nerves in the abdomen and thorax

B) They are not stimulated by ischemia in visceral organs

C) They are transmitted only by the lightly myelinated Б-type A sensory fibers

D) They are typically well localized

27-The pathway of which system crosses in the ventral white commissure of the spinal cord within a few segments of entry and then courses to the thalamus contralateral to the side of the body from which the signal originated? {Guyton}

A) Anterolateral system B) Dorsal column–medial lemniscal system

C) Corticospinal system D) Spinocerebellar system

28-Which of the following is a property of C fibers? {د}

A) Have the slowest conduction velocity of any nerve fiber type

B) Have the largest diameter of any nerve fiber type

C) Are afferent nerves from muscle spindles

D) Are afferent nerves from Golgi tendon organs

29- The fast pain sensation that accompanies a painful trauma is carried to C.N.S. by fibres that belong to: {د}

a- A alpha fibres. b- A delta fibres. c- C fibres. d- B fibres.

30- Pain: {د}

a- receptors are stimulated by a rise in bradykinin concentration.

b- threshold differs from one person to another.

c- sensation is enhanced by endorphins.

d- sensation is enhanced by GABA released by A fibres.

31- The pricking pain: {د}

a- terminate in the non-specific thalamic nuclei.

b- is lost by compression of the nerve.

c- is blocked by local anesthetics better than slow pain.

d- is a severe dull aching pain.

32- Slow pain: {د}

a- is carried along type A delta fibres. b- pathway activates the non-specific thalamic nuclei.

c- is perceived mainly in sensory area I. d- lasts for a short time.

33- Gallbladder pain is referred to the : {د}

a- umbilicus. b- tip of the right scapula. c- left shoulder. d- left lumbar region.

34- Cardiac pain is referred to the: {د}

a- umbilicus. b- tip of the right scapula c- left shoulder. d- left lumbar region.

35- Deep pain: {د}

a- is due to stimulation of pain receptors in the skin. b- is pricking in character.

c- is poorly localized. d- is carried along a delta fibres.

36- Concerning visceral pain, the followings are true statements, EXCEPT: {د}

a- may cause reflex vomiting.

b- may cause contraction of overlying skeletal muscle.

c- is well localized.

d- may cause change in arterial blood pressure.

37- All the following may give rise to visceral pain, EXCEPT: {د}

a- distension of hallow viscus. b- ischemia.

c- spasm of hallow viscus. d- incision with scalpel in the gut wall.

38- Visceral pain: {د}

a- is a localized pain. b- is accompanied by spasm of its overlying muscle.

c- can be elicited by sharp cutting of viscera. d- is carried along garcail and cunate tracts.

39- Cutaneous pain: {د}

a- receptors adapt to stimulation more rapidly than touch receptors.

b- can be caused by excitation of receptors by K+ ions released in injured tissues.

c- can be caused by over-stimulation of touch receptors.

d- is carried into the spinal cord by a long A beta fibers.

40- Cutaneous pain : {د}

a- threshold is the same for males and females.

b- is carried only on myelinated nerve fibres.

c- is carried on the spinothalamic tract of the same side.

d- may refer to the viscera

41- The analgesia system includes: {د}

a- Raphe magnus nucleus in the post horn of the spinal cord.

b- the periaquiductal area in the mid-brain.

c- the sensory area I. d- supra-optic nuclei of the hypothalamus

42- Serotinin is used as neurotransmitter in pain suppression by neuron in the following region : {د}

a- raphe magnus nucleus . b- periaqueductual gray matter .

c- post-central gyrus . d- nucleus cuneatus .

43-An interneuron in which region uses enkephalin to inhibit pain transmission ? {د}

a- post-central gyrus . b- raphe magnus nucleus .

c- dorsal horn of spinal cord . d- periaqueductal gray matter .

44- Regarding secondary hyperalgesia: {د}

a- pain threshold is low. b- may be due to sun burn.

c- skin is reddish in colour. d- may be caused by thalamic syndrome.

45- Concerning primary hyperalgesia: {د}

a- pain threshold is high. b- may be due to sun burn.

c- skin is normal. d- may be caused by thalamic syndrome.

46-Convergence projection mechanism explains:

a) after discharge. b) primary hyperalgesia.

c) referred pain. d) secondary hyperalgesia.

47-Regarding referred pain:

a) it is a character of deep pain. b) can be explained by reverberating circuit.

c) occurs to right shoulder in cardiac pain. d) is a character of cutaneous pain.

e) can be explained by convergence projection mechanism

48-Visceral pain is:

a) carried by A-delta afferent. b) caused mainly by ischemia.

c) mainly pricking in character. d) carried in ventral spinothalamic tract.

49-Deep pain is:

a) carried by A-delta afferents. b) well localized. c) carried by C fibers.

d) associated with increased heart rate. e) associated with increased blood pressure

50-Slow pain is:

a) carried by A-delta fibers. b) carried by C fibers. c) well localized than fast pain.

d) carried by A-beta fibers. e) reaches sensory cortex

51-Fast pricking type of cutaneous pain is:

a) carried by C fibers. b) carried by A-beta fibers.

c) poorly localized. d) carried by A-delta fibers

52-Fast pricking type of pain:

a) is poorly localized. b) is transmitted along C fibers.

c) has rapid onset and short duration. d) its chemical transmitter is substance P.

e) terminates in reticular formation of the brain stem

53-It is correct to say that:

a) primary hyperalgesia is due to facilitation along the pathway.

b) fast cutaneous pain is carried by C-fibers.

c) slow cutaneous pain terminates in sensory cortex.

d) slow cutaneous pain terminates in thalamus

54-Primary hyperalgesia is due to:

a) convergence. b) facilitation along the pathway. c) reverberating circuits.

d) convergence projection mechanism. e) hypersensitivity of receptors

55-Regarding thermal receptors:

a) the range of discharge of warm receptors is between 15 °C and 35 °C.

b) the maximum rate of discharge of hot pain receptors occurs at 40 °C.

c) warm receptors are more numerous than cold receptors.

d) the maximum rate of discharge of cold receptors is at 25 °C

56-Pain receptors are also called:

A) Mechanoreceptors B) Nociceptors C) Thermoreceptors D) Proprioceptors

57-Pain receptors are absent in:

A) Periosteum of bone B) Brain parenchyma C) Dura mater D) Pleura

58-Which of the following tissues is most pain-sensitive?

A) Liver parenchyma B) Lung parenchyma C) Periosteum D) Brain parenchyma

59-Pain from lung disease is mainly due to:

A) Damage to alveoli B) Stretch or irritation of pleura

C) Airway obstruction D) Pulmonary capillary rupture

60-Pain caused by compression or injury of nerve vessels is called:

A) Nociceptive pain B) Neuropathic pain C) Referred pain D) Visceral pain

61-The lateral spinothalamic tract primarily carries:

A) Crude touch B) Pain & temperature C) Vibration & proprioception D) Fine touch

62-The cell body of the first-order neuron in the pain pathway is located in:

A) Posterior horn of spinal cord B) Cerebral cortex

C) Thalamus D) Dorsal root ganglion

63-Second-order neurons in the pain pathway are located in:

A) Thalamus B) Posterior horn of spinal cord C) Dorsal root ganglion D) Medulla

64-Which tract carries pain and temperature sensations?

A) Dorsal column-medial lemniscus B) Corticospinal tract

C) Lateral spinothalamic tract D) Vestibulospinal tract

65-The third-order neuron in the pain pathway projects from the:

A) Thalamus → somatosensory cortex B) Posterior horn → cortex

C) Dorsal root ganglion → thalamus D) Brainstem → cerebellum

66-The anterior (ventral) spinothalamic tract primarily carries:

A) Pain & temperature B) Proprioception

C) Fine touch & vibration D) Crude touch & pressure

67-Pain due to decreased blood supply to muscles that occurs during exertion but not at rest is called:

A) Neuropathic pain B) Nociceptive pain C) Ischemic pain D) Referred pain

68-Which of the following features is characteristic of ischemic pain in muscles?

A) Sharp, stabbing pain at rest B) Dull, aching pain relieved by activity

C) Cramping pain during exertion relieved by rest D) Constant burning pain

69-Which of the following types of pain is usually not referred?

A) Cutaneous pain B) Visceral pain C) Cardiac pain D) Renal colic

70-Mild visceral pain is mainly transmitted via:

A) Somatic sensory fibers B) Sympathetic fibers

C) Parasympathetic fibers D) Motor fibers

71-Severe visceral pain may be mediated by:

A) Somatic fibers only B) Parasympathetic fibers

C) Sympathetic fibers D) Corticospinal tract

1. B	2. C	3. A	4. D	5. D
6. B	7. D	8. C	9. C	10.B
11.A	12.C	13.A	14.C	15.A
16.A	17.A	18.A	19.C	20.C
21.B	22.B	23.C	24.C	25.D

26.A	27.B	28.A	29.B	30.A
31.B	32.B	33.B	34.C	35.C
36.C	37.D	38.B	39.B	40.A
41.B	42.A	43.C	44.D	45.B
46.C	47.E	48.B	49.C	50.B
51.D	52.C	53.D	54.E	55.D
56.B	57.B	58.C	59.B	60.B
61.B	62.D	63.B	64.C	65.A
66.D	67.C	68.C	69.A	70.B
71.B	72.	73.	74.	75.

Stretch reflex

1-The muscle spindle remains capable of responding to stretch even the muscle is contracting. This is because of: {Formative2026}

a) autogenic inhibition b) reciprocal innervation

c) co-activation of α and ý motor neurons d) presence of stretch receptors in the tendon

2-Which reflex does not show the property of reciprocal inhibition? {Formative2026}

a) Flexion withdrawal reflex b) Abdominal muscle reflex

c) Plantar flexion reflex d) Positive supporting reaction

3- Muscle tone is reduced with the following diseases except {Mid2025}{Summar2025}

a. upper motor neuron lesion b. Lower motor neuron lesion

c. Cerebellar lesions d. toxicity with curare like drugs

4- The withdrawal reflex: {Summar2025}

a) is monosynaptic. b) is polysynaptic. c) is more rapid than the stretch reflex.

d) disappears in complete transection of the spinal cord below level of the lesion.

5- The following is true regarding dynamic stretch reflex: {Mid2024}

a) Its receptor is slowly adapting. b) Its afferent is secondary fibers.

c) It is important for rapid correction of position. d) It is related to muscle pump.

6- All are true regarding the stretch reflex, EXCEPT: {د}{Summar2025}

a) can be elicited by a passive stretch of the muscle.

b) depends on the muscle spindle which is in parallel with the extrafusal muscle fibers.

c) it is a polysynaptic reflex.

d) stretch of a muscle is followed by its contraction.

7- The primary endings innervating the muscle spindle: {د}{Summar2025}

a) only innervates the nuclear chain fibres.

b) innervate both nuclear chain and nuclear bag fibres.

c) are a slowly conducting nerves having a diameter of 4 microns.

d) discharge during the static response only.

8- When dynamic γ -motor neurons are activated at the same time as α -motor neurons to muscle? {Mid2021}

A- Rapid inhibition of discharge in spindle la afferents takes place

B- The muscle will not contract

C- The number of impulses in spindle 1a afferents is smaller than when a discharge alone is increased

D- The number of impulses in spindle 1a afferents is greater than when a discharge alone is increased

9-When dynamic ý motor neurons are activated at the same time as α motor neurons to muscle, {Formative2026}

A) the number of impulses in spindle Ia afferents is greater than when α discharge alone is increased.

B) the number of impulses in spindle Ia afferents is smaller than when α discharge alone is increased.

C) clonus is likely to occur.

D) prompt inhibition of discharge in spindle Ia afferents takes place.

10- The inverse stretch reflex {Mid2021}

A- Occurs when 1a spindle afferents are inhibited

B- Is a monosynaptic reflex initiated by activation of the Golgi tendon organ

C- Uses type II afferent fibers from the Golgi tendon organ

D- Is a disynaptic reflex with an excitatory interneuron inserted between the afferent and efferent limbs

11- The inverse stretch reflex: {Summar2025}{Mid2025}

a) depends on a receptor situated in the middle of the muscle.

b) is elicited when the muscle is subjected to excessive stretch.

c) leads to increased force of muscle contraction.

d) is a monosynaptic reflex.

12- Which of the following doesn't obey reciprocal innervation {Final2025}

a. Positive supporting reaction

13-Which of the following reflexes don't have possibility of reciprocal inhibition? {Final2023}

a) Flexor reflex b) Abdominal reflex c) Positive supporting reaction

14-γ efferent fiber supply ……. {Final2025}

a. Intrafusal muscle fibers

15-Regarding stretch reflex which of the following is False: {Formative2024}

a) After discharge is not from the properties of this reflex.

b) It is a monosynaptic reflex.

c) Gamma efferent neuron is considered as the efferent of stretch reflex.

d) Nuclear bag fiber is responsible for tendon jerk reflex.

16-Concerning muscle spindles each of the following statements is true EXCEPT: {Formative2024}

a) Flower spray and annulospiral afferent nerve fibers carry sensation from intrafusal muscle fibers.

b) Contain nuclear bag and nuclear chain intrafusal muscle fibers.

c) Spindle shaped and found parallel to long axis of muscle.

d) There is no motor supply

17-Which of the following best describes the sensory capabilities of the muscle spindle under normal physiological conditions? {Formative2024}{Final2024}

a) Rate of change in length only b) Length only

c) Tension and rate of change in length d) Length and rate of change in length

18- Regarding the role of muscle spindles and Golgi tendon organs. {Formative2024}

a) Muscle spindles measure the tension of skeletal muscle.

b) Muscle spindles are only of importance in the regulation of posture.

c) Muscle spindles receive both afferent and efferent nerve fibers.

d) Golgi tendon organs measure the length of skeletal muscles.

19-which neurons of the following supply the skeletal muscle cells (intrafusal fiberswithin the muscle spindles? {Formative2024}

a) Granule cells b) Gamma motor neurons

c) Pyramidal neurons d) Alpha motor neurons

20- Withdrawal reflexes are {Mid2021}

A- Initiated by nocuous stimulation of the skin B- Can lead to the appearance of clonus

C- An example of a stretch reflex

D- Accompanied by the same response on both sides of the body

21- During a voluntary movement, the Golgi tendon organ provides the central nervous system with information about {Mid2021}

A- The length of the muscle being moved B- The velocity of the movement

C- The change in joint angle produced by the movement

D- The tension developed by the muscle being moved

22- What are the elements of the deep tendon reflex ? {Mid2021}

A- Golgi tendon organ, group 1b afferent fibers, spinal inhibitory interneuron, contralateral A - motor neuron, and extrafusal fibers of the skeletal muscle

B- Golgi tendon organ, group II afferent fibers, spinal inhibitory interneuron, ipsilateral a- motor neuron, and intrafusal fibers of the skeletal muscle

C- Group la sensory fibers originating in the contractile portion of the intrafusal fibers of the muscle spindle, spinal dorsal horn neuron, ipsilateral a-motor neuron, and extrafusal fibers of the skeletal muscle

D- Group la sensory fibers originating in the central portion of the intrafusal fibers of the muscle spindle, ipsilateral a-motor neuron, and extrafusal fibers of the skeletal muscle

23- Ankle joint jerk reflex lies at the level of {Final2021}

A- L2-4 B- S1 C- C5,6 D- S4,5

24-Which reflex is responsible for polysynaptic excitation of contralateral extensors? {د}

A) Stretch reflex (myotatic) B) Golgi tendon reflex (inverse myotatic)

C) Flexor withdrawal reflex D) Subliminal occlusion reflex

25- Physiologically, an intact innervated muscle can normally be made to contract by stimulation of {Formative2022}

A. α-motor neurons B. γ-motor neurons C. Either one of the above

26- A 3-month-old baby is brought to her pediatrician for a checkup. Stroking the plantar surface of the foot produces a reflex extension of the large toe rather than the expected flexion. The Babinski sign elicited by the physician indicates damage to which of the following? {Formative2024}

a) Basal ganglia b) Cerebellum c) Brainstem d) Pyramidal tracts

27- Complete transection of the spinal cord at the level of T1 would most likely result in {Final2021}

A- temporary loss of stretch reflexes below the lesion

B- temporary loss of conscious proprioception below the lesion

C- permanent loss of voluntary control of movement above the lesion

D- permanent loss of consciousness above the lesion

28- Which of the following is a characteristic of nuclear bag fibers? {Final2021}{د}

A- They are one type of extrafusal muscle fiber

B- They detect dynamic changes in muscle length

C- They give rise to group Ib afferents

D- They are innervated by alpha-moto neurons

29- Muscle stretch leads to a direct increase in firing rate of which type of nerve? {Final2021}{د}

A- α-Motoneurons B- y-Motoneurons C- Group Ia fibers D- Group Ib fibers

30- Gamma motor neuron is : {Mid2023}

A- represent pathway for facilitation of stretch reflex by higher centers

B- Efferent of stretch reflex

C- Supply extrafusal fibers D- Supply central part

31- Which of the following best describes the reflex arc and muscle response of the Golgi tendon organ? {Mid2024}

a) Monosynaptic reflex arc; relaxation of muscle

b) Disynaptic reflex arc; relaxation of muscle

c) Monosynaptic reflex arc; contraction of muscle

d) polysynaptic reflex arc; relaxation of muscle

32- Which reflex is responsible for monosynaptic excitation of ipsilateral homonymous muscle? {Final2021}

A- Stretch reflex (myotatic) B- Golgi tendon reflex (inverse myotatic)

C- Flexor withdrawal reflex D- Subliminal occlusion reflex

33- Which reflex cause polysynaptic excitation of the contralateral extensor muscle : {Mid2023}

A- Stretch reflex B- Inverse stretch reflex

C- Sunliminal occlusion reflex D- Flexor withdrawal reflex

34- Which reflex is responsible for polysynaptic excitation of contralateral extensors? {Final2021}

A- Stretch reflex (myotatic) B- Golgi tendon reflex

C- Flexor withdrawal reflex D- Subliminal occlusion reflex

35- A weight Lifter Is attempting to set a new personal best in competition As the lifter attempts to raise the weight to his chest by flexing his forearms at the elbow, he suddenly drops the weight to the floor .Which neurons were excessively stimulated and caused the lifter to drop the weight? {Final2021}

A- Renshaw cells B- Lateral reticulospinal tract neurons

C- Gamma motor neurons D- Class Ib afferent neurons

36-A weight lifter is in a competition and he lifts a heavy weight then he drops it. What happen in case of stimulation of gamma and alpha motor neurons at the same time? {Final2023}

a) rate of discharge in Ia fibers is less than activation of alpha alone

b) rate of discharge in Ia fibers is more than activation of alpha alone

c) no change in rate of discharge in Ia fibers

37- The most Important role of the gamma motoneurons is to {Final2021}{Final2024}

A- Stimulate skeletal muscle fibers to contract

B- Maintain Ia afferent activity during contraction of muscle

C- Generate activity in Ib afferent fibers

D- Detect the length of resting skeletal muscle

E- Prevent muscles from producing too much force

38- The excessive muscle tone produced in decerebrate rigidity (lesion which leads to interruption of inhibitory descending motor tracts) can be reversed by {Final2021}

A- stimulation of group Ia afferents

B- cutting the dorsal roots

C- transection of cerebellar connections to the lateral vestibular nucleus

D- stimulation of a-motoneurons

39.The stretch reflex: {د}

a) is a polysynaptic reflex. b) is a monosynaptic reflex.

c) receptor is the Pacinian corpuscle. d) has a long central delay.

40-Which reflex is responsible for monosynaptic excitation of ipsilateral homonymous muscle? {د}

A) Stretch reflex (myotatic) B) Golgi tendon reflex (inverse myotatic)

C) Flexor withdrawal reflex D) Subliminal occlusion reflex

41- Regarding the muscle spindle: {د}

a) the primary afferent fibres respond to the static length of muscle as well as to its rate of change in length.

b) stretching the extrafusal fibres causes the intrafusal muscle fibers to be stretched.

c) they are excited during overstretch of the muscle.

d) they are in series with the extrafusal muscle fibers.

42- The gamma efferent system: {د}

a) Modifies the muscle spindle sensitivity.

b) supplies the central zone of intrafusal muscle fibres.

c) is inhibited by the vestibulospinal tract.

d) Supplies the extrafusal muscle fibers.

43- The nuclear bag intrafusal fibers that signal the rate of changes in muscle length is innervated by: {د}

a) corticospinal neurons. b) alpha motor neurons.

c) dynamic gamma motor neurons. d) static gamma motor neurons.

44- Gamma motor nerve fibers: {د}

a) cause the contraction of the peripheral part of the intrafusal fibers.

b) cause contraction of the middle portions of the muscle spindle.

c) cause relaxation of the intrafusal fibres.

d) Thickness is about 16 µ.

45- The secondary endings innervating the muscle spindle: {د}

a. only innervates the nuclear chain fibers. b. only innervates the nuclear bag fibers.

c. are a rapidly conducting nerves having a diameter of 16 microns.

d. discharge during the dynamic response only.

46- Muscle tone, all are true, EXCEPT: {د}

a) is defined as resistance offered by muscle to passive stretch.

b) is maintained by gamma efferent discharges.

c) maintains body posture against effect of gravity.

d) is decreased in UMNL.

47- Regarding the muscle tone, all are true, EXCEPT: {د}

a) the primary and secondary afferents are both stimulated when the spindle is stretched.

b) the gamma efferent discharges are firing independently of the descending tracts from areas in the brain.

c) hypertonia may occur during hyperthyroidism.

d) hypotonia commonly occurs as a result of spinal motor neuron damage.

48-All of the followings are inhibitory supraspinal centers to the stretch reflex, EXCEPT: {د}

a premotor area 6. b. Neocerebellum.

c. lentiform nucleus of the basal ganglia d. Red nucleus.

49- All of the followings are facilitatory supraspinal centers to the stretch reflex, EXCEPT: {د}

a Pontine reticular formation. b. Neocerebellum.

c. vestibular nucleus d. premotor area 6

50- Hyporeflexia is present in: {د}

a) Parkinsonism. b) LMNL. c) hyperthyroidism. d) UMNL

51- Tendon jerks are exaggerated in: {د}

a) anterior horn cell disease. b) dorsal root lesions.

c) Chorea. d) internal capsule lesion.

52- Golgi tendon organ: {د}

a) has a threshold of stimulation equal to the muscle spindle.

b) lies in parallel with the extra-fusal muscle fibers.

c) when stimulated reflexely cause the contraction of the muscle in the tendon of which it is situated.

d) when stimulated, reflexely relaxes the muscle in the tendon of which it is situated.

53-The only reflex which doesn't obey the phenomenon of reciprocal innervation is: {د}

a) Stretch reflex b) Inverse stretch reflex.

c) Flexion withdrawal reflex. d) Positive supporting reaction.

54- The ankle jerk is exaggerated: {د}

a) LMNL b) in Parkinsonism.

c) in neocerebellar disease. d) on the right side after damage in the left internal capsule.

55- Muscle tone is reduced in the following conditions, EXCEPT: {د}

a) lower motor neuron lesion. b) upper motor neuron lesion.

c) neocerebellar lesions. d) toxicity with curare-like drugs.

56- The inverse stretch reflex: {د}

a) depends on a receptor which is situated in the middle of the muscle.

b) is elicited when the muscle is subjected to excessive stretch.

c) leads to increased force of muscle contraction.

d) is a monosynaptic reflex.

57- The withdrawal reflex: {د}

a) is monosynaptic. b) is polysynaptic.

c) is more rapid than the stretch reflex. d) disappears in UMNL.

58-the receptor of the inverse stretch reflex is: {د}

a. muscle spindle b. Golgi tendon organ

c. Pacinian corpuscle. d. Free nerve endings

59-The extrafusal muscle fibers are innervated by: {د}

a. lateral spinothalamic neurons. b. alpha motor neurons.

c. dynamic gamma motor neurons. d. static gamma motor neurons.

60- The nuclear chain intrafusal fibers are innervated by: {د}

a. corticospinal neurons. b. alpha motor neurons.

c. dynamic gamma motor neurons. d. static gamma motor neurons.

61- Motor neurons in the spinal cord may be the site of convergence of signals arising from all the following sites, EXCEPT: {د}

a) cerebral cortex. b) cerebellum.

c) reticular formation. d) cells in the dorsal root ganglia.

62- Regarding the crossed extensor reflex, all are true, EXCEPT: {د}

a) is an antigravity reflex. b) is a polysynaptic spinal reflex.

c) has a shorter latency than the stretch reflex. d) never appear after spinal cord transection

63- Concerning the flexor withdrawal reflexes: {د}

a) they involve reciprocal innervation. b) their central delay equals 0.5 msec.

c) it is a monosynaptic reflex. d) it provides an example of a deep reflex

64- Regarding the muscle spindle: {د}

a) Has an afferent connection belonging to la fibers.

b) Has an efferent connection belonging to type II fibers.

c) Contains the extrafusal muscle fibers.

d) Responds when the muscle is overstretched

65-The afferent from the Golgi tendon organs belongs to: {د}

a) Group Ia fibers b) group Ib fibers c) group II fibers d) group Ill fibers

66- Which of the following receptors receives an efferent: {د}

a) Pacinian corpuscle b) Muscle spindle c) Golgi tendon organ d) Pain receptors

67-Skeletal muscle tone:

a) is a voluntary act. b) keeps posture against gravity.

c) increases if the muscle shortens. d) all of the above

68-The dynamic stretch reflex:

a) is the basis of muscle tone. b) its receptor is nuclear chain.

c) is mediated through stimulation of the secondary nerve endings.

d) shows rapid contraction of the muscle followed by rapid relaxation

69-Dynamic stretch reflex is:

a) slower than withdrawal reflex. b) the mechanism of muscle tone.

c) polysynaptic reflex. d) increased in UMNL.

70-Static stretch reflex is:

a) the base of tendon jerk. b) stimulated by red nucleus. c) a polysynaptic reflex.

d) inhibited by vestibular nucleus. e) the base of muscle tone

71-Static stretch reflex is:

a) responsible for tendon jerk. b) transmitted through primary ending only.

c) initiated at nuclear chain intrafusal fibers. d) initiated at nuclear bag intrafusal fibers

72-Primary ending from muscle spindle:

a) discharge when muscle stretch is maintained. b) explain alpha – gamma coactivation.

c) is the afferent for dynamic stretch reflex. d) is the afferent for static stretch reflex

73-Dynamic stretch reflex:

a) has secondary ending as an afferent. b) has alpha neuron as an afferent.

c) is the mechanism of muscle tone. d) has gamma motor neuron as an efferent

74-Muscle tone:

a) is less in antigravity muscles. b) decreases venous return.

c) is based on the static stretch reflex. d) is characterized by fatigue.

e) is based on the dynamic stretch reflex

75-Muscle spindle:

a) is present in muscle tendon. b) has afferent A-delta fibers. c) has afferent C fibers.

d) has afferent A-alpha fibers. e) is facilitated by motor area 4

76-Nuclear bag fibers in muscle spindle are:

a) slowly responding. b) slowly adapting. c) rapidly responding.

d) connected to secondary ending. e) responsible for static stretch reflex

77-The crossed extensor reflex:

a) has a central delay of 5 m.sec. b) shows the phenomenon of reciprocal innervations.

c) does not show after discharge. d) is a monosynaptic spinal reflex

78-The excessive muscle tone produced in decerebrate rigidity can be reversed by {د}

A) stimulation of group Ia afferents B) cutting the dorsal roots

C) transection of cerebellar connections to the lateral vestibular nucleus

D) stimulation of α-motoneurons E) stimulation of γ-motoneurons

79-Complete transection of the spinal cord at the level of T1 would most likely result in

A) temporary loss of stretch reflexes below the lesion

B) temporary loss of conscious proprioception below the lesion

C) permanent loss of voluntary control of movement above the lesion

D) permanent loss of consciousness above the lesion

1. C	2. D	3. A	4. B	5. C
6. C	7. B	8. D	9. A	10.C
11.B	12.A	13.C	14.A	15.C
16.D	17.D	18.C	19.B	20.A
21.D	22.D	23.B	24.C	25.C
26.D	27.A	28.B	29.C	30.A
31.B	32.A	33.D	34.C	35.D
36.B	37.B	38.B	39.B	40.A
41.B	42.A	43.C	44.A	45.A
46.D	47.B	48.B	49.D	50.B
51.D	52.D	53.D	54.D	55.B
56.B	57.B	58.B	59.B	60.D
61.B	62.C	63.A	64.A	65.B
66.B	67.B	68.D	69.D	70.E
71.C	72.C	73.B	74.C	75.E
76.C	77.B	78.B	79.A	80.

The motor cortex

1-Lesion in area 4 is characterized by all of the following except: {Formative2026}

a) Plantar flexion response of all toes

b) Flaccid paralysis of the affected muscles

c) Lesion is contralateral

d) Absent superficial reflexes

2- Lesion in area 4 in the cerebral cortex leeds to all of the following except {Mid2025}

a. Fanning of all toes

b. Flaccid paralysis

c. Loss of superficial reflexs

3-Motor area 4 of cerebral cortex:

a) inhibits stretch reflex.

b) controls fine skilled movements.

c) is less excitable than area 6.

d) is located in post central gyrus

4- Primary motor area is present in {Final2024}

a- Precentral gyrus

5- Which of the following parts of the body has cortical motoneurons with the largest representation on the primary motor cortex (area 4)? {Final2021}{د}

A- Shoulder

B- Ankle

C- Fingers

D- Elbow

6-Which of the following features is characteristic of the supplementary motor cortex? {Formative2024}

a) Stimulation of the supplementary motor cortex leads to bilateral movements, typically involving both hands.

b) It is located just anterior to the premotor cortex on the lateral surface of the hemisphere.

c) The supplementary cortex functions totally independent of the premotor and primary motor cortex

d) Like the premotor cortex, stimulation in the supplementary motor cortex leads to discrete movement of individual muscles.

7-Which body part is represented most laterally and inferiorly within the primary motor cortex? {Formative2024}

a) Abdomen b) Hand c) Face d) Lower limb

8- The precentral gyrus and corticospinal tract are essential for {Mid2021}{Formative2024}{Final2024}

A- Vision B- Auditory identification C- Kinesthesia D- Voluntary movement

9- Motor aphasia results from lesions in {Mid2021}{Final2022}

A- Visual association area B- Auditory area

C- Exner area D-Somatic sensory area

10-The patient understands the meaning of words but finds it difficult to talk : {Final2023}

a) motor apraxia b) visual aphasia c) motor aphasia d) sensory aphasia

11- Patient will understand spoken but find it difficult to speech {Final2022}{Formative2022} {Final2024}

A- sensory visual aphasia B- Sensory auditory aphasia

C- Global aphasia D- Motor aphasia

12-Which of the following is not a feature of corticospinal tract lesion? {Formative2025}

a) Cogwheel rigidity b) Exaggerated deep tendon reflexes

c) Plantar extensor response d) Spasticity

13- Which of the following has the greatest area of representation in the primary motor cortex? {Mid2024}

a) Knee b) Thumb c) Toes d) Trunk.

14- Which of the following tracts is responsible for gross movement ? {Final2025}

a. Corticospinal tract b. Extra pyramidal Tract

15-Which extra-pyramidal tract mediate movement of head & neck due to visual & auditory stimuli : {Final2023}

a) vestibulospinal b) rubrospinal c) tectospinal d) reticulospinal

16-initiation of fine movements is the function of: {د}

a. Motor area 4 b. supplementary motor area

C. premotor area D. superior temporal area.

17- A lesion of Broca's area produces: {د}

a) apraxia b) motor aphasia. c) agraphia. d) word deafness.

18- A patient who suffers from hemiplegia is able to write his thought to his family but is unable to express these ideas as spoken words is affected by: {د}

a) visual agnosia. b) sensory aphasia. c) motor aphasia. d) apraxia.

19-Motor apraxia: {د}

a) is due to lesion in primary motor area 4. b) is due to lesion in sensory area.

c) is accompanied with muscle paralysis.

d) is characterized by loss of memory of voluntary movements.

20-Which body part is represented superiorly and medially within the postcentral gyrus? {Guyton}

A) Upper limb B) Lower limb C) Abdomen D) Genitalia

21- The larger area of representation in the motor cortex includes muscles: {د}

a) that extend the shoulder. b) that flex the knee.

c) of the trunk. d) of vocalization.

22- Regarding the primary motor cortex (area 4), all are true, EXCEPT: {د}

a) is responsible for fine delicate movements.

b) parts of the body are represented in an area which is directly proportional in size to the complexity of movement carried by that part.

c) sends facilitatory impulses to gamma motor neurons.

d) is inhibitory to muscle spindles.

23- The cortical motor area 4 on the left side of the body: {د}

a) there is a map representing body parts on the right side.

b) the body is represented in an erect (upright) pattern.

c) controls motor activity of the left side of the body.

d) the area representing a particular body part is directly proportional in size to the size of that part or organ.

24-Corticospinal tract:

a) originates only from primary motor area.

b) terminate mainly on the anterior horn cells of the opposite side.

c) is inhibitory to stretch reflex.

d) passes through the anterior limb of internal capsule

25- The pyramidal tract: {د}

a) arises only from area 4. b) supplies only the same side of the body.

c) most of the fibres do not cross. d) responsible for fine movements.

26- Most of fibres of the pyramidal tract have direct synapse with: {د}

a) anterior alpha motor neurons. b) gamma motor neurons.

c) interneurons of the spinal cord. d) dorsal horn cells of spinal cord.

27- The pyramidal tract originates from: {د}

a) all areas of cerebral cortex including parietal, frontal, temporal and occipital lobes.

b) motor, premotor areas as well from somatic sensory cortex.

c) area 4 only.

d) motor area 4 and premotor area 6 only.

28- In controlling the fine movement of the hands, the corticospinal tract terminates directly on: {د}

a) Renshaw cells. b) spinal cord interneurons.

c) spinal alpha motor neurons. d) posterior horn cells.

1. A	2. A	3. B	4. A	5. C
6. A	7. C	8. D	9. C	10.C
11.D	12.A	13.B	14.B	15.A
16.A	17.B	18.C	19.D	20.B
21.D	22.D	23.A	24.B	25.D
26.C	27.B	28.C	29.	30.

The higher sensory centers

1- primary sensory cortex {Mid2025}

a. responsible for perception and integration of cutaneous and kinesthetic sensations

b. Sensory memory

c. Orientation of position of each part of the body relative to each other

2-Inability to put right hand over the left hand is called {Formative2024}

A. phantom limb B. Thalamic phantom limb C. Stereognosis D. Tactile localization

3- Ablation of somatosensory area I does not significantly impair ? {Mid2021}

A- Joint position sensation B- Tactile localization

C- Tactile discrimination D- Pain sensation

4- Ablation in somatosensory area 1 doesn't cause significantly impair in : {Mid2023}

A- Tactile discrimination B- tactile localization

C- Pain sensation D- Joint position sensation

5- Which of the following is true regarding thalamus function? {Mid2024}

a) It receives sensation from the same side of the body.

b) Intralaminar nuclei is concerned with the motor function of the thalamus.

c) It is a part of the system concerned with emotional reactions.

d) None of the above.

6-The highest degree of pain localization comes from which of the following? {Guyton}

A) Simultaneous stimulation of free nerve endings and tactile fibers

B) Stimulation of free nerve endings by bradykinin

C) Nerve fibers traveling to the thalamus by way of the paleospinothalamic tract

D) Stimulation of Ϭ-type A fibers

7-Within the primary somatosensory cortex, the various parts of the contralateral body surface are represented in areas of varying size that reflect which of the following? {Guyton}

A) The relative size of the body parts

B) The density of the specialized peripheral receptors

C) The size of the muscles in that body part

D) The conduction velocity of the primary afferent fibers

8- The size of representation of various body parts in the primary somatic sensory cortex is dependent on: {د}

a- relative size of the body parts . b- density of the receptors in that organ.

c- size of the muscles in that body part. d- conduction velocity of the afferent fibers

9-Discrimination of locality of sensation depends on the:

a) rate of discharge along the afferent nerve. b) projection from the cortex.

c) presence of specific pathway for each sensation. d) weber-fescher's law

10-Syringomyelia leads to:

a) loss of all sensations. b) upper motor neuron lesion.

c) loss of deep sensation. d) loss of spinothalamic sensation.

11-Syringomyelia:

a) leads to UMN paralysis. b) leads to loss of fine touch.

c) does not lead to loss of pain sensation.

d) means narrowing of the central canal of the spinal cord.

e) lesion occurs in the lower cervical and upper thoracic regions (jacket distribution).

12-The distributing center that controls activity of the cortex is:

a. Brain stem b. Hypothalamus c. Cingulate gyrus

d. Hippocampus. e. Thalamus

13- Stimulation of the dorsomedial nucleus of the thalamus , leads to {Final2022} {Formative2022}

A- rage B- fear C- placidity D- hyperactivity

1. A	2. B	3. D	4. C	5. C
6. A	7. B	8. B	9. B	10.D
11.E	12.E	13.C	14.	15.

Pharmacology

General and Local Anaesthetics

1- All of the following are true about halothane except: {Summar2025}

a) it produces a dose dependent hypotension

b) it produces fulminant hepatic necrosis

c) it sensitizes myocardium to arrhythmogenic effects of epinephrine

d) it has rapid induction and recovery time

2- All of the following is true about halothane EXCEPT {Mid2025}

a. it causes dose dependant hypotension b. it is hepatotoxic

c. it sensetizes myocardial (anti-arrythmic) cells D- it has fast induction and recovery time

3- Which of the following is barbiturates intravenous anesthetic agents? {Final2025} {Summar2025}

a) Halothane b) Thiopental c) Etomidate d) Ketamine

4- Which of the following inhalation agents is well well-tolerated agent in children? {Mid2024}

a) Enflurane b) Desflurane c) Isoflurane d) Halothane

5- Which of the following has Slow induction? {Final2024}

A- enflurane B- sevoflurane C- isoflurane D- desflurane

6-One of the following drugs sensitize myocardium to arrhythmogenic effects of epinephrine {Formative2025}{سؤال فى المحاضرة}

a) Halothane b) Atropine c) Lidocaine d) Desflurane

7- One of the following drugs induce anesthesia rapidly but irritates tracheobronchial tree? {Final2022}{Final2021}

A- Enflurane B- Desflurane C- Isoflurane D- Lidocaine

8- Malignant hyperthermia syndrome is manifested by the following signs except {Formative2022}

a) Hyperthermia b) Muscular relaxation c) Tachycardia d) Metabolic acidosis

9-Malignant hyperthermia syndrome is treated with: {Formative2024}{سؤال في المحاضرة}

a) Adrenaline b) Ca c) Dantrolene d) Atropine

10- One of the following drug induce dissociative anesthesia ? {Mid2021}{Final2022}

A- Enflurane B- Halothane C- Ketamine D- Lidocaine

11- The primary mechanism of action of local anesthetics is {Mid2021}

A- Activation of ligand-gated potassium channels

B- Blockade of voltage-gated sodium channels

C- Stimulation of voltage-gated N-type calcium channels

D- Blockade of GABA-gated chloride channels

12- One of the following anaesthetic drug increase heart rate and blood pressure {Formative2022}{Final2021}

a) Enflurane b) Halothane c) ketamine d) Lidocaine

13- Ketamine : {Mid2023}

A- Stimulate NMDA receptor B- Contraindicated in hypovolemic shock

C- increase cerebral blood flow and intracranial tension

D- cause rapid induction and rapid recovery

14- Which of the following drugs irritate the tracheobronchial tree ? {Final2022}

A- Isoflurane B- Halothane C- Desflurane D- Sevoflurane

15- The route of local anesthetic administration, which is associated with instillation within subarachnoid space is {Final2021}

A- Topical anesthesia B- Infiltrative anesthesia

C- Regional anesthesia D- Spinal anesthesia

16- Which of the following local aesthetics is a useful antiarrhythmic agent ? {Final2021}{Final2022}

A- Cocaine B- Lidocaine C- Bupivacaine D- Ropivacaine D- biperiden

17- Which one of the following local anesthetics is amide one? {Merit}

A. Lidocaine B. Procaine C. Tetracaine D. Benzocaine

18-Local aesthetics produce: {Formative2024}

a) Used for sedation for the entire body

b) Alleviation of anxiety and pain with an altered level of consciousness

c) Blocking pain sensation without loss of consciousness

d) Analgesia, amnesia, loss of consciousness

19- The most potent local anaesthetic agent in this list is: {Formative2023}

a-Procalne b-Lidocaine c-Cocaine d-Bupivacaine

20- Which of the following local anesthetics is more likely to cause allergic reactions ? {Final2022}{Formative2022}{Mid2023}

A- lidocainne B- bupivacaine C- procaine D- ropivacaine

21-It is an alpha 2 agonist used as a pre-anesthetic medication to produce analgesia & sedation: {Merit}

A. Benzodiazepines B. Scopolamine C. Dexmedetomidine D. Fentanyl

22-Which of the following anesthetics is hepatotoxic and cardiotoxic? {Merit}

A. Halothane B. Isoflurane C. Desflurane D. Sevoflurane

23-Prevent bradycardia and reduce respiratory secretion in preanesthetic state: {Merit}

A. Clonidine B. Scopolamine C. Dexmedetomidine D. Fentanyl

24-The first sensation lost by local anesthetics is: {Merit}

A. Temperature B. Touch C. Motor function D. Pain

25-Which of the following is a gaseous inhalation anesthetic?

A) Halothane B) Isoflurane C) Sevoflurane D) Nitrous oxide

26-Which of the following inhalation anesthetics is associated with the highest risk of hepatotoxicity?

A) Halothane B) Isoflurane C) Sevoflurane D) Desflurane

27-Which of the following anesthetics is a potent analgesic but weak anesthetic when used alone?

A) Halothane B) Isoflurane C) Nitrous oxide D) Sevoflurane

28-Which inhalation anesthetic is preferred in pediatric anesthesia due to its non-irritating smell and rapid induction?

A) Isoflurane B) Sevoflurane C) Desflurane D) Halothane

29-Which of the following intravenous anesthetics is classified as an ultrashort-acting barbiturate?

A) Ketamine B) Propofol C) Thiopental D) Etomidate

30-Which IV anesthetic is unique because it produces dissociative anesthesia?

A) Propofol B) Ketamine C) Thiopental D) Etomidate

31-Which of the following inhalation anesthetics can generate nephrotoxic fluoride ions?

A) Halothane and Isoflurane B) Enflurane and Sevoflurane

C) Desflurane and Nitrous oxide D) Isoflurane and Nitrous oxide

32-Chronic exposure to …… can cause megaloblastic anemia

A) Halothane B) Isoflurane C) Nitrous oxide D) Sevoflurane

33-The drug of choice for treating malignant hyperthermia is:

A) Propofol B) Dantrolene C) Succinylcholine D) Nitroglycerin

34-The drug commonly used to stop postanesthetic shivering is:

A) Naloxone B) Meperidine C) Propofol D) Dantrolene

1. D	2. D	3. B	4. D	5. A
6. A	7. B	8. B	9. C	10.C
11.B	12.C	13.C	14.C	15.D
16.B	17.A	18.C	19.D	20.C
21.C	22.A	23.B	24.D	25.D
26.A	27.C	28.B	29.C	30.B
31.B	32.C	33.B	34.B	35.

Opioid Analgesics

1- Which of the following drugs is a competitive antagonist of opioid receptors? {Summar2025}

a) Meperidine b) Naloxone c) Clonidine d) Tramadol

2- Morphine is contraindicated in one of the following conditions: {Summar2025}

a) Cough b) Cardiac asthma with Pulmonary Edema

c) Postoperative pain d) Head injury

3-Morphine is contraindicated in head injury because: {Mid2024}{Formative2025}

a) It is liable to cause addiction b) It can raise systolic blood pressure

c) It can raise intracranial tension d) It does not relieve the pain of head injury

4-Opioid analgesics produce: {Formative2024}

a) Drug dependence b) Drug tolerance c) Both of the above d) None of the above

5-Opioid receptors mostly involved in pain are: {Formative2024}

a) Sigma receptors b) Delta receptors c) Kappa receptors d) Mu receptors

6-All the following are pharmacological effects of opioid analgesics except: {Formative2024}

a) Mydriasis b) Euphoria c) Constipation d) Emesis

7-Opioid full antagonist...? {Final2023}

a-Naloxone

8- Methadone is often used in drug treatment programs for narcotic (opioid) addiction. What is the classification of this medication ? {Mid2021}

A- Non-steroidal anti-inflammatory B- Opioid analgesic

C- Antimigrainous agent D- Agent for gout

9-Which one of these drugs has euphoric effect: {Formative2024}

a) Morphine b) Aspirin c) Ibuprofen d) Diclofenac

10- The diagnostic triad of opioid overdosage Is {Final2021}

A- Mydriasis, coma and hyperventilation B- Coma, depressed respiration and miosis

C- Mydriasis, chills and abdominal cramps D- Miosis, tremor and vomiting

11- Triad of opioid toxicity : {Mid2023}

A- coma, respiratory depression, miosis B- Coma, hyperventilation, mydriasis

12- Opioids is contraindicated in head injuries …… {Final2025}

a. because it cause elevation in ICP

13- Antidote used in morphine toxicity? {Final2024}

A- naloxone

14- Opiod receptor antagonist {Mid2025}

a. Naloxone b. Methadone c. Meperidine d. Pentazocine

15- The relief produced by intravenous morphine in dyspnea from pulmonary edema is associated with reduced: {Final2021}

A- Perception of shortness of breath B- Patient anxiety

C- Cardiac preload and afterload D- All of the above

16- Postanesthetic shievering is managed by administration of one of the following drug? {Final2021}{Final2022}

A- Adrenaline B- Meperidine C- NSAIDs D- Morphine

17-All the following are central actions of morphine except: {Merit}

A. Miosis B. biliary colic C. Cough suppression D. Nausea and vomiting

18-The following drug used for treatment of morphine addiction: {Merit}

A. Naloxone B. Codeine C. Methadone D. Tramadol

19-Mu (μ) receptors produce all the following except: {Merit}

A. Sedation. B. Analgesia. C. Respiratory depression. D. Hallucinations

20-Which one of the following is a central action of morphine? {Merit}

A. Constipation. B. Biliary pain. C. Urine retention. D. Nausea & vomiting.

21-Morphine tolerance developed to all the following actions except: {Merit}

A. Analgesic effect. B. Emetic effect. C. Euphorigenic effect. D. Constipating effect

22-Which one of the following drugs used in treatment of alcoholism? {Merit}

A. Codeine B. Tramadol C. Naltrexone D. Buprenorphine

23-Which one of the following is not correct regarding naloxone? {Merit}

A. It has short duration B. It is used in treatment of morphine dependence

C. It is given parenteral D. It is used for neonatal asphyxia caused by morphine

24-All the following are morphine precautions except: {Merit}

A. Acute myocardial infarction B. Bronchial asthma C. Addison disease D. Head injuries

25-Acute morphine side effects include all the following except: {Merit}

A. Adrenal insufficiency B. Urine retention. C. Mental clouding. D. Constipation.

26- Which one of the following NOT used for treatment of morphine addiction: {Merit}

A. Methadone B. Naloxone C. Lofexidine D. Buprenorphine

27- Morphine toxicity is best treated with: {Merit}

A. Clonidine B. Methadone C. Codeine D. Naloxone

28-The antidote of choice for morphine poisoning is: {Merit}

A. Nalorphine B. Nalbuphine C. Naltrexone D. Naloxone

29-Tolerance of morphine developed to all the following effects except:

A. Analgesic B. Sedative C. Emetic D. Constipating

30-The treatment of choice for acute opioid overdose is:

A) Methadone B) Naloxone C) Buprenorphine D) Loperamide

31-All of the following are clinical uses of naloxone EXCEPT:

A) Treatment of acute opioid poisoning

B) Counteracting neonatal opioid-induced fetal asphyxia

C) Diagnosis of opioid addiction

D) Chronic management of alcohol dependence

32-The M1 opioid receptor is primarily responsible for:

A) Spinal analgesia B) Supraspinal analgesia

C) Respiratory depression D) Decreased GIT motility

33-M2 opioid receptor mediates:

A) Spinal analgesia B) Supraspinal analgesia C) Only euphoria D) Only miosis

34-Miosis induced by opioids is primarily due to:

A) M1 receptor activation B) M2 receptor activation

C) Histamine release D) Dopamine receptor blockade

35-In pulmonary edema, if respiratory depression is a concern, which drug is preferred?

A) Morphine B) Furosemide C) Codeine D) Tramadol

36-Which of the following is a synthetic opioid used for diarrhea with minimal CNS effects?

A) Diphenoxylate B) Morphine C) Codeine D) Fentanyl

37-One mechanism of action of these drugs is:

A) Closing voltage-gated Ca^{2+} channels on presynaptic nerve terminals

B) Opening voltage-gated Ca^{2+} channels

C) Inhibiting K^{+} channels D) Blocking Na^{+} channels

38-Which drug is superior to morphine in treating myocardial ischemia?

A) Furosemide B) Nitroglycerin C) Codeine D) Tramadol

39-The primary clinical use of opioids is:

A) Treatment of hypertension B) Analgesia for severe pain

C) Anti-inflammatory therapy D) Antidepressant therapy

40-Which of the following opioids is commonly used as an antitussive?

A) Morphine B) Fentanyl C) Codeine D) Methadone

41-Meperidine blocks shivering mainly through:

A) μ-opioid receptor activation B) α2-adrenoceptor subtypes

C) Kappa receptor activation D) Sodium channel blockade

42-……..commonly used opioid for diarrhea

A) Loperamide B) Methadone C) Tramadol D) Buprenorphine

43-Cross-tolerance of opioids is an example of:

A) Pharmacokinetic tolerance B) Pharmacodynamic tolerance

C) Metabolic tolerance D) Behavioral tolerance

44-Which of the following is commonly used orally for morphine addiction treatment?

A) Methadone B) Buprenorphine C) Levo-α-acetylmethadol D) Morphine

45-The primary clinical use of naloxone is:

A) Treatment of diarrhea B) Treatment of acute opioid poisoning

C) Chronic pain management D) Sedation

46-Buprenorphine is:

A) A μ full agonist B) A μ partial agonist
C) An opioid antagonist D) A non-opioid analgesic

47-Levo-α-acetylmethadol is:

A) μ full agonist B) μ partial agonist C) Opioid antagonist D) SSRI

48-The classic triad of acute opioid toxicity includes:

A) Hypertension, mydriasis, tachypnea B) Coma, pinpoint pupils, respiratory depression

C) Seizures, fever, hypotension D) Euphoria, diaphoresis, agitation

49-Naloxone is usually administered by:

A) Oral route B) Injection C) Transdermal patch D) Inhalation

50-Naltrexone is approved for the treatment of:

A) Acute opioid overdose B) Alcoholism C) Chronic pain D) Diarrhea

51-Naltrexone is classified as a:

A) Pure opioid antagonist B) Partial μ-opioid agonist

C) μ-opioid agonist D) Kappa receptor agonist

52-Compared to naloxone, naltrexone is:

A) Less potent and shorter-acting B) More potent and longer-acting

C) Equally potent but shorter-acting D) Only effective intravenously

53-Naltrexone can be used for all of the following EXCEPT:

A) Acute opioid overdose reversal B) Maintenance therapy in ex-opioid abusers

C) Prevention of opioid relapse D) Treatment of alcohol dependence

1. B	2. D	3. C	4. C	5. D
6. A	7. A	8. B	9. A	10.B
11.A	12.A	13.A	14.A	15.D
16.B	17.B	18.C	19.D	20.D
21.D	22.C	23.B	24.A	25.A
26.B	27.D	28.D	29.D	30.B
31.D	32.B	33.A	34.B	35.B
36.A	37.A	38.B	39.B	40.C
41.B	42.A	43.B	44.A	45.B
46.B	47.A	48.B	49.B	50.B
51.A	52.B	53.A	54.	55.

Antidepressant drugs

1-Which neurotransmitter is linked to OCD and depression? {Formative2026}

a) Epinephrine b) Dopamine c) Serotonin d) Both B & C

2-Cheese reaction is a hypertensive crisis occur with concomitant administration of: {Formative2025}

a) Selegiline with fluoxetine b) Imipramine with mirtazapine

c) Imipramine with propranolol d) Selegiline with tyramine rich food

3-Best antidepressants used in treatment of panic disorder: {Formative2024}{Mid2025}

a) Clomipramine b) Selegiline c) Fluoxetine d) Imipramine

4-tricyclic antidepressants are not considered the 1st line antidepressant as they produce the following side effects except: {Formative2024}

a) torsades de piontes b) urinary retention

c) hypothyroidism d) orthostatic hypotension

5- Agomelatine a novel antidepressant acts by the following mechanism: {Mid2024}

a) Block reuptake of serotonin and norepinephrine reuptake

b) Block reuptake of serotonin

c) Block reuptake of norepinephrine and dopamine

d) Melatonin (MT1 and MT2) receptor agonist and 5HT2c antagonist.

6-Which one of the following antidepressant agents can be also prescribed for treatment of neuropathic pain. {Summar2025}

a) Bupropion b) SSRIs as fluoxetine c) SNRIS as duloxetine d) MAOIS as phenelzine

7-At high doses the following drug may cause parkinsonism-like dysfunction. {Formative2025}

a) Amphetamine. b) Reserpine. c) Diazepam. d) propranolol.

8- Which synapses are involved in depression ? {Mid2021}

A- Dopaminergic synapses B- Serotoninergic synapses

C- Cholinergic synapses D- All of the above

9- Which of the following antidepressants is nonselective MAO inhibitors ? {Mid2021}

A- Moclobemide B- Selegiline C- Tranylcypromine D- Fluoxetine

10-Which of the following antidepressants can be used in treatment of depression and neuropathic pain

A. Venlafaxine B. Fluoxetine C. Agomelatine D. Selegilin

11- The following drug at a high dose it may cause parkinsonism-like dysfunction ? {Mid2021}{Final2022}

A- Ethosuximide B- Baclofen C- Diazepam D- Reserpine E- Ketamine

12- Tricyclic antidepressants are {Final2021}

A- Highly selective serotonin reuptake inhibitors B- Monoamine oxidase inhibitors

C- Selective norepinephrine reuptake inhibitors

D- Mixed norepinephrine and serotonin reuptake inhibitors

13- which of the following drugs is selective MAO B inhibitors? {Final2022}

A- Risperidone B- Olanzepine C- selegilline D- Fluxetine

14- Cheese interaction occur in combination of : {Mid2023}

A- selegiline+tyramine rich food

15-Which one of the following could be used for treatment of nocturnal enuresis? {Merit}

a. Imipramine b. Fluoxetine c. Bupropion d. Duloxetine

16-Serotonin syndrome commonly results from the coadministration of: {Merit}

a. SSRIs + duloxetine. b. MAOIs + nefazodone. c. SSRIs + MAOIs. d. TCAs + mirtazapine.

17- High dose of SSRIs in combination with MAOIs leads to: {Merit}

A. Discontinuation syndrome B. Serotonin syndrome

C. Hypertensive crisis D. CNS depression

18-Lithium produces all the following adverse reactions except: {Merit}

A. Hypothyroidism B. Nephrogenic diabetes insipidus

C. Parkinsonism D. Teratogenesis

19-Which one of the following is a side effect of TCAs? {Merit}

A. Serotonin syndrome B. Atropine-like actions

C. Hepatotoxicity D. parkinsonian syndrome

20- It can be used for smoking cessation: {Merit}

A. Fluoxetine B. Imipramine C. Mirtazapine D. Bupropion

21-Which of the following is a reversible selective MAO-A inhibitor?

A) Tranylcypromine B) Phenelzine C) Moclobemide D) Selegiline

22-Which TCA adverse effect is primarily due to alpha-1 adrenergic receptor blockade?

A) Constipation B) Orthostatic hypotension C) Dry mouth D) Seizures

23-Which of the following is a tertiary amine TCA?

A) Desipramine B) Nortriptyline C) Amitriptyline D) Maprotiline

24-Which of the following is a secondary amine TCA?

A) Imipramine B) Doxepin C) Desipramine D) Clomipramine

25-Which of the following drugs is a selective serotonin reuptake inhibitor (SSRI)?

A) Amitriptyline B) Fluoxetine C) Duloxetine D) Phenelzine

26-Which of the following correctly pairs a TCA with its class?

A) Amitriptyline – Secondary amine B) Nortriptyline – Secondary amine

C) Imipramine – Secondary amine D) Desipramine – Tertiary amine

27-According to the monoamine hypothesis, depression is primarily caused by a deficit of:

A) GABA and glutamate B) Dopamine and acetylcholine

C) Norepinephrine and/or serotonin D) Glycine and histamine

28-At therapeutic doses, TCA sodium channel blockade can cause:

A) Antiarrhythmic effects B) Severe seizures

C) Urinary retention D) Orthostatic hypotension

29-Which of the following is a tricyclic antidepressant (TCA)?

A) Fluoxetine B) Amitriptyline C) Duloxetine D) Agomelatine

30-Which of the following is a selective serotonin reuptake inhibitor (SSRI)?

A) Venlafaxine B) Fluoxetine C) Amitriptyline D) Agomelatine

31-Duloxetine belongs to which class of antidepressants?

A) SNRI B) TCA C)SSRI D) MAOI

32-Which antidepressant class is associated with hypertensive crisis if patients consume tyramine-rich foods?

A) TCAs B) MAOIs C) SSRIs D) SNRIs

33-Which of the following is not an SSRI?

A) Paroxetine B) Sertraline C) Citalopram D) Venlafaxine

34-Phenelzine is classified as:

A) Selective MAO-A inhibitor B) Selective MAO-B inhibitor

C) Nonselective MAOI D) Reversible MAOI-A

35-Which of the following is a common side effect of TCAs due to anticholinergic activity?

A) Dry mouth and constipation B) Hyponatremia C) Tremors D) Insomnia

36-Clomipramine is classified as:

A) Secondary amine TCA B) Tertiary amine TCA C) SSRI D) SNRI

37-Venlafaxine is classified as:

A) Tertiary amine TCA B) SSRI C) SNRI D) MAOI

38-The primary mechanism of action of TCAs is:

A) Blocking NMDA receptors B) Inhibiting reuptake of norepinephrine and serotonin

C) Activating GABA receptors D) Inhibiting monoamine oxidase

39-Blurred vision caused by TCAs is due to:

A) Alpha-1 adrenergic blockade B) Histamine H1 blockade

C) Sodium channel blockade D) Muscarinic receptor blockade

40-SNRIs inhibit the reuptake of which neurotransmitters?

A) Dopamine and serotonin B) Norepinephrine and serotonin

C) Norepinephrine only D) Serotonin only

41-Which of the following is an example of an SNRI?

A) Amitriptyline B) Duloxetine C) Fluoxetine D) Clomipramine

42-Which of the following is an atypical antidepressant?

A) Fluoxetine B) Amitriptyline C) Bupropion D) Phenelzine

43-Selegiline at low doses selectively inhibits:

A) MAO-A B) MAO-B C) Both MAO-A and MAO-B D) Neither

44-SNRIs are particularly effective in patients with:

A) Depression and neuropathic pain B) Hypertension C) Epilepsy D) Asthma

45-Which of the following MAOIs is nonselective?

A) Selegiline B) Moclobemide C) Tranylcypromine D) Rasagiline

46-Moclobemide is a:

A) Nonselective MAOI B) Selective MAO-A inhibitor

C) Selective MAO-B inhibitor D) SSRI

47-Nefazodone is classified as:

A) TCA B) Atypical antidepressant C) SNRI D) MAOI

1. D	2. D	3. C	4. C	5. D
6. C	7. B	8. D	9. C	10.A
11.D	12.D	13.C	14.A	15.A
16.C	17.B	18.C	19.B	20.D
21.C	22.B	23.C	24.C	25.B
26.B	27.C	28.A	29.B	30.B
31.A	32.B	33.D	34.C	35.A
36.B	37.C	38.B	39.D	40.B
41.B	42.C	43.B	44.A	45.C
46.B	47.B	48.	49.	50.

Pathology

Basic Pathology of CNS. Pathology of Cerebrovascular Diseases

1-What differentiates a Transient Ischemic Attack (TIA) from a stroke? {Formative2026}

a) Symptoms resolve within 24 hours b) Symptoms persist longer than 24 hours

c) It's due to hemorrhage d) It causes permanent brain damage

2-Which type of cerebral edema results from blood-brain barrier disruption? {Formative2026}

a) Vasogenic b) Interstitial c) Osmotic d) Cytotoxic

3-Which of the following is a cause of increased CSF production? {Formative2026}

a) Arnold-Chiari malformation b) Thrombosis of superior sagittal sinus

c) Choroid plexus papilloma d) Aqueductal stenosis

4-The most common cause of subarachnoid hemorrhage is: {Formative2026}

a) Trauma b) Arteritis

c) Thrombosis of cerebral veins d) Rupture of berry aneurysm

5-The most dangerous venous sinus is: {Formative2026}

a) cavernous sinus b) transverse sinus c) superior sagittal sinus d) sigmoid sinus

6-The most common site of cerebral hemorrhage is in the {Formative2025}

a) Cerebral hemisphere. b) Pons. c) Cerebellum. d) Basal ganglia and thalamus.

7-The most common, non-traumatic cause of cerebral hemorrhage is ..... {Formative2025}

a) Ruptured aneurysm. b) Hypertension c) Hemorrhagic blood disease. d) Vasculitis.

8-The commonest cause of massive cerebral hemorrhage in old age is …… {Formative2025} {Summar2025}

a) Trauma b) Hypertension

c) Rupture of cerebral aneurysm d) Hemorrhagic blood diseases

9- Global ischemia due to {Mid2025}

a. systemic hypotenshion b. thrombosis c. embolism

10- Increase CSF in ……. {Final2022}

A- hydrocephalus B- Cerebral oedema C- Encephalitis D- None of the above

11- In communicating hydrocephalus the following is not correct: {Formative2023}

A. The obstruction is in the subarachnoid space

B. The obstruction is around the mid- brain.

C. CSF can pass from the ventricular system to the subarachenoid space.

D. CSF cannot pass from the ventricular system to the subarachenoid space.

12- all about communication hydrocephalus are TRUE EXCEPT : {Mid2023}

A- CSF can pass from ventricular system to subarachnoid space

B- Obstruction in subarachnoid space

C- obstruction at midbrain

D- CSF can't pass from ventricular system to subarachnoid space

13- Cerebral hemorrhage is characterized by all EXCEPT : {Mid2023}{Formative2023}

A- Mostly traumatic hemorrhage B- accounts for 10%-20% of all strokes

C- Mostly spontaneous(non-traumatic) hemorrhage D- more in males

14- Petecheal cerebral hemorrhage is……. {Formative2022}

a- Induced by ruptured aneurysm b- Associated with systemic hypertention

c- A common complication of brain tumors. d- Capillary hemorrhage

e- All of the above

15-Cerebral edema results in all of the following, EXCEPT {Final2023}

a) Gyri are flattened b) Sulci are narrowed

c) Brain is softer d) Ventricular cavities are dilated

16- One of the following is CORRECT for subdural hemorrhage {Mid2021}

A- The blood is arterial in origin B- The blood is venous in origin

C- The blood is capillary in origin D- Caused mainly by rupture of middle meningeal artery

E- Caused mainly by rupture of berry aneurysm

17- Extradural Hemorrhage due to {Mid2025}

a. Rupture of Middle meningeal artery b. Cerebral Hemorrhage

c. Laceration d. Petechial hemorrhage

18- Hydrocephalus resulting from {Final2021}

A- Failure of the development of the brain vesicle B- Aque ductal stenosis

C- Failure of cleavage of the forebrain D- Failure of development of skull bone

19- Cerebral infarction is characterized by {Mid2021}

A- Liquefaction with cavity formation B- Formation of brain hematoma

C- Dense infiltration by lymphocytes and plasma cells

D- Shrinkage of infarcted area after 2 days E- None of the above

20- One of the following is INCORRECT for global brain ischemia {Mid2021}

A- May have good prognosis B- May induce brain death C- May induce neuronal damage

D- Caused by septic shock E- Caused by cerebral thrombosis

21- Global brain ischemia is caused by……. {Formative2022}

a- Sthersclerotic cerebral aneurysm b- Severe hypotension c- Meningitis

d- Ruptured aneurysm of cerebral artery d- Thrombosis of cerebral artery

22- Petechial cerebral hemorrhage is ……. {Final2022}

A- Induced by ruptured aneurysm B- Associated with systemic hypertension

C- A common complication of brain tumors D- Capillary hemorrhage

E- All of the above

23- Massive cerebral hemorrhage occurs due to the following EXCEPT {Final2022}

A- ruptured aneurysm B- systemic hypertension C- Capillary hemorrhage

D- brain tumors E- All of the above

24- which of the following is the most common cause of Cerebral hemorrhage, in young age ? {Final2022}

A- Tumor B- Hypotension C- Anemia D- Rapture OF Cerebral Aneurysm

25 Hydrocephalus means abnormal dilation of the ventricles due to: {ثناء}

A. Severe inflammation B. Malignant tumors C. Cerebral aneurysm

D. Accumulation of CSF E. Cerebral haemorrhage

26-CSF is normally formed by the: {ثناء}

A. Choroid plexus B. Arachnoid villi C. Pia mater D. Arachnoid mater

27-CSF is normally absorbed by the: {ثناء}

A. Choroid plexus B. Arachnoid villi C. Pia mater D. Arachnoid mater

28-Gliosis means: {ثناء}

A. Regeneration of the brain tissue B. Necrosis of the brain tissue

C. Fibrosis of the brain tissue D. None of the above

29-Congenital cerebral aneurysm affects: {ثناء}

A. Blood vessels of circle of Willis B. Cerebral arteries

C. Meningeal arteries D. Internal carotid artery

30-Mycotic cerebral aneurysm is due to: {ثناء}

A. Congenital weakness of the wall of cerebral vessels

B. Septic emboli from subacute bacterial endocarditis

C. Atherosclerosis of the cerebral vessels

D. Increased intracranial tension

31-Subdural haemorrhage occurs: {ثناء}

A. Between dura and skull bones B. Between dura and arachnoid

C. In subarachnoid space D. Intracerebral

32-Subarachnoid haemorrhage is due to: {ثناء}

A. Trauma of the meningeal artery B. Trauma of the cortical veins

C. Rupture of cerebral aneurysm D. Sudden rise in blood pressure

E. Haemorrhagic blood diseases

33-Hypertensive patients are at high risk of: {ثناء}

A. Extradural haemorrhage B. Subdural haemorrhage C. Subarachnoid haemorrhage

D. Peticheal intracerebral haemorrhage E. Massive intracerebral haemorrhage

34-All of the followings can increase intracranial tension Except

a. Intracranial tumors b. Intracranial inflammation c. Hemorrhage d. Infarction

35-Cytotoxic edema is characterized by:

A) Extracellular fluid accumulation

B) Intracellular fluid accumulation in neurons, glia, or endothelial cells

C) CSF accumulation only D) BBB disruption without cell swelling

36-Vasogenic edema is primarily caused by:

A) Breakdown of the blood-brain barrier B) Increased intracellular osmoles

C) Neuronal apoptosis D) Hypoperfusion only

37-Hydrostatic cerebral edema, a subtype of vasogenic edema, is commonly associated with:

A) Malignant hypertension B) Brain tumors C) Hypoglycemia D) Viral meningitis

38-Osmotic edema occurs due to:

A) Breakdown of the blood-brain barrier

B) Decreased plasma osmolality causing water to move into brain cells

C) Intracellular toxin accumulation D) Mechanical trauma only

39-The primary mechanism of interstitial edema is:

A) Breakdown of the blood-brain barrier B) Rupture of the CSF-brain barrier

C) Cellular toxin accumulation D) Intracranial hemorrhage

40-Which type of intracranial hemorrhage occurs between the skull and the dura mater?

A) Subarachnoid hemorrhage B) Subdural hemorrhage

C) Extradural (epidural) hemorrhage D) Intracerebral hemorrhage

41-A common cause of extradural hemorrhage is:

A) Ruptured berry aneurysm

B) Trauma with skull fracture tearing the middle meningeal artery

C) Hypertensive intracerebral bleed D) Tumor erosion

1. A	2. A	3. C	4. D	5. A
6. D	7. B	8. B	9. A	10.A
11.D	12.D	13.A	14.D	15.D
16.B	17.A	18.B	19.A	20.E
21.B	22.D	23.C	24.D	25.D
26.A	27.B	28.C	29.A	30.B
31.B	32.C	33.E	34.D	35.B
36.A	37.A	38.B	39.B	40.C
41.B	42.	43.	44.	45.

Infections of the CNS

1-All are characters of CSF in septic Meningitis except {Mid2025}{قسم}

a. Elevated protein content b. High sugar content

c. prescence of polymorphs , pus cells d. increased tension

2- CSF in viral meningitis shows : {Mid2023}{Formative2023}

A- Marked protein elevation B- Increased number of lymphocytes

C- Many PNLS and pus cells D- Marked decrease in glucose content

3-Encephalitis is …… {Formative2025}{قسم}

a) inflammation of brain. b) inflammation of the meninges.

c) inflammation of spinal cord. d) inflammation of nerves.

4- What term refers to an inflammation of the brain tissues? {Summar2025}

a. Encephalitis b. Meningitis c. Sinusitis d. Meningoencephalitis

5- Complications of TB meningitis does NOT include {Final2021}

A- Increased intracranial tension B- Compression of the cranial nerves leading to their paralysis

C- Hydrocephalus D- May result in tuberculoma

E- May result in brain abscess

6-Temporal brain abscess is usually complication of {سؤال فى المحاضرة}

a. Otitis media b. Mastoiditis c. Sinusitis, d. lung infection

7- Cerebellar brain abscess is usually a complication of {Final2021}{Final2022}

A- Suppurative otitis media B- Suppurative lung diseases C- Mastoiditis

D- Skull fractures E- Systemic pyemia

8-Cerebellar brain abscess is usually a complication of: {ثناء}

A. Suppurative otitis media B. Suppurative lung diseases C. Sinusitis

D. Skull fractures E. Systemic pyaemia

9- Parietal lobe abscessis usually a complication of

A- Suppurative otitis media B- Suppurative lung diseases

C- Mastoiditis D- Skull fractures

10- Web of fibrin is in which of the following ? {Final2022}

A- T.b meningitis (tuberculosis) B- Viral meningitis

C- Fungal meningitis D- All the above

11-CSF shows cob web coagulum of fibrin on standing in case of: {ثناء}

A. Acute pyogenic meningitis B. Chronic meningitis C. Brain abscess

D. Encephalitis E. Cerebral infarction

12-Modes of infection in pyogenic meningitis include all EXCEPT: {ثناء}

A. Blood spread B. Blood transfusion C. Local spread from paranasal sinuses

D. Via skin E. Direct implantation in case of skull fracture

13-The following may be complicated by brain abscess EXCEPT: {ثناء}

A. Suppurative otitis media. B. Bacterial endocanditis. C. Suppurative lung disease.

D. Supurative peritonitis. E. Suppurative mastoiditis.

14-Causes of non- suppurative meningitis include all except……. {قسم}

a. Tuberculosis b. Viral c. Fungal d. Meningococci

15-…… is the commonest causative organisms of septic meningitis. . {قسم}

a.Tuberculosis b. Entero-viruses c. Meningococci d. staphylococci

16-Meningococcal infection is transmitted commonly by ……. {قسم}

a. droplet infection b. blood transfusion c. swallowing d. skin inoculation

17-All are characters of Viral meningitis except. {قسم}

a. Less fulminate clinical course than septic meningitis. b. Self-limited.

c. Commonly due to entero-virus infection.

d. There are no distinctive macroscopic findings

e. CSF shows increased number of neutrophils, marked protein elevation and decreased glucose content

18-In septic meningitis, the CSF pressure is typically:

A) Normal B) Decreased C) Increased D) Variable

19-What happens to CSF protein content in septic meningitis?

A) Decreases B) Remains normal C) Increases D) Fluctuates randomly

20-Which type of cells are predominantly seen in the CSF during bacterial meningitis?

A) Lymphocytes B) Plasma cells C) PNLs D) Eosinophils

21-In viral (aseptic) meningitis, the CSF cell count is typically:

A) Predominantly neutrophils B) Predominantly lymphocytes

C) Predominantly eosinophils D) Acellular

22-Compared to bacterial meningitis, viral (aseptic) meningitis is usually:

A) More severe and fulminant B) Less severe and self-limiting

C) Always fatal D) Associated with high CSF neutrophils

23-How is CSF glucose typically affected in bacterial meningitis?

A) Increased B) Decreased C) Unchanged D) Variable depending on age

24-Which symptom reflects CNS involvement in meningitis?

A) Photophobia B) Irritability C) Disturbed conscious level D) All of the above

25-The appearance of CSF in tuberculous meningitis is usually:

A) Purulent and cloudy B) Clear with a fine network of fibrin on standing

C) Bloody D) Milky

1. B	2. B	3. A	4. A	5. E
6. A	7. C	8.	9. A	10.A
11.B	12.B	13.D	14.D	15.C
16.A	17.E	18.C	19.C	20.C
21.B	22.B	23.B	24.D	25.B

Microbiology

Encephalitis

1-Immune response in case of infection with prions there is :- {Formative2025}

a) No inflammatory or immune response to these diseases as they are normal human proteins.

b) Activation of cytotoxic T cell.

c) Activation of TH17 cell.

d) Exaggerate immune response as they act as a super antigens.

2-Prions transmitted by: {Formative2024}

a) Arthropod born vector b) Inhalation.

c) Blood transfusion. d) Ingestion of infected tissues mainly the brain

3- Arbovirus encephalitis the west nile fever is transmitted by …… {Final2025}

a. Mosquitos b. Flies

4- Encephalitis can be caused by the following viruses EXCEPT {Mid2021}

A- Rabies B- Herpes simplex C- Measles

D- Respiratory syncytial virus E- Arboviruses

5- Encephalitis is most commonly {Mid2021}

A- Secondary to meningitis B- Viral C- Fungal

D- Autoimmune E- None of the above

6- All the following viruses can cause encephalitis except: {Formative2022}

a- Arboviruses b- Arenaviruses C- Hepatitis viruses D- Herpes viruses

7- All the following bacteria can cause encephalitis except: {Formative2022}

a- Rickettsia b- Mycoplasma c- Bordetella d- Mycobacterium tuberculosis

8- Approximately 15% of encephalitis cases caused by HSV1 are {Final2024}

A- following activation of latent infection in the trigeminal ganglia

B- following activation of latenet infection in the sacral ganglia

9- Kuru and Creutzfeldt-Jakob disease (CJD) are thought to be caused by which of the following? {Formative2022}

a. Slow viruses b. Prions c. Cell wall deficient bacteria d. Environmental toxins

10- What is the other name for Bovine spongiform encephalopathy? {Final2021}

A- Kuru disease B- Scrapie disease C- Sheep's disease

D- Mad dog disease E- Mad cow disease

11- The followings are characters of prion-mediated diseases except: {قسم}

a. They are confined to CNS causing neurodegeneration (amyloid) and spongiform changes due to neuronal vacuolation and neuronal loss.

b. They have a long incubation period and a chronic progressive course with dementia and end fatally.

c. There is immune response to these diseases

d. They are transmissible by ingestion of infected tissues mainly the brain, transplanted tissues and contaminated surgical instruments.

12- The followings are human prion-mediated diseases except: {قسم}

a. Creutzfeldt-Jakob disease (CJD) b. Gerstmann-Sträussler-Scheinker syndrome

C. Scrapie d. kuru.

13-The most common cause of infectious encephalitis..... {قسم}

a. herpes simplex type I, b. herpes simplex type II,

c. varicella-zoster virus (VZV), d. all of the above

14- HSV 2 is transmitted: {قسم}

a. sexually or to newborn during birth b. ingestion

c. inhalation d blood transfusion.

15- HSV 1 is transmitted by: {قسم}

a. droplets b. contact with secretions

c. sexually or to newborn during birth. d. A and b

16-Viruses establish latent infection: {قسم}

a. HSV-1 b. HSV2 c. Varicella zoster virus d. All of the above

17- Arboviruses means: {قسم}

a. Many viral diseases are transmitted by the bite of an arthropod vector.

b. Many viral diseases are transmitted by the bite of an animal vector.

c. Many viral diseases are transmitted by ingestion.

d. Non of the above

18-The most common cause of encephalitis is …….

A. HSV 1 B .HSV2 C. Bacterial cause D. CMV

19-Approximately 15% of viral encephalitis cases are caused by:

A. HSV-2 B. Varicella-zoster virus C. HSV-1 D. Rabies virus

20-Latent HSV-1 infection is most commonly found in the:

A. Dorsal root ganglion B. Autonomic ganglia C. Olfactory bulb D. Trigeminal ganglion

21-HSV-1 reaches the temporal lobe primarily by:

A. Hematogenous spread B. Cerebrospinal fluid circulation

C. Travel down sensory neurons D. Direct invasion from nasal mucosa

22-Rabies presenting with hyperactivity, agitation, delirium, hydrophobia, and seizures is called:

A. Paralytic rabies B. Latent rabies C. Furious rabies D. Chronic rabies

23-HSV-1 most commonly affects which part of the brain?

A. Frontal lobe B. Parietal lobe C. Occipital lobe D. Temporal lobe

24-Which of the following is a common neurological feature of encephalitis?

A. Bradycardia B. Seizures C. Jaundice D. Polyuria

25-Cytomegalovirus (CMV) encephalitis occurs primarily in:

A. Healthy adults B. Neonates only C. Immunocompromised D. Elderly patients

26-Arbovirus encephalitis is transmitted primarily by:

A. Respiratory droplets B. Direct contact

C. Bite of infective mosquitoes D. Fecal–oral route

27-Typical CSF findings in viral encephalitis include:

A. Neutrophilic pleocytosis, low glucose

B. Lymphocytic pleocytosis, mildly elevated protein, normal glucose

C. Eosinophilic pleocytosis, low protein D. High red blood cells, low protein

28-Rabies encephalitis has two clinical forms. The most common form (≈80% of cases) is:

A. dumb rabies B. Furious rabies C. Chronic rabies D. Latent rabies

29-The glucose level in the CSF of most viral encephalitis cases is:

A. Low B. High C. Normal D. Variable

30-All of the followings can cause brain abscess except

A. Bacteria B. Viruses C. Protozoa D. Fungi

31-Otitis media can cause abscess in:

A) Temporal lobe B) Frontal lobe C) Parietal lobe D) Occipital lobe

32-Sinusitis can cause abscess in:

A) Temporal lobe B) Frontal lobe C) Parietal lobe D) Occipital lobe

33-Prion-mediated diseases are unique because:

A. They elicit a strong immune response

B. They cause massive inflammation in the CNS

C. They cause neurodegeneration without any inflammatory or immune response

D. They trigger antibody-mediated immunity

34-Pre-exposure prophylaxis for rabies involves:

A. Live attenuated vaccine B. Killed (inactivated) rabies vaccine

C. Oral polio vaccine D. MMR vaccine

35-Post-exposure prophylaxis for rabies includes:

A. Killed vaccine only B. Killed vaccine + rabies hyperimmune globulin

C. Hyperimmune globulin only D. Antibiotics

36-The most common symptom of a brain abscess is:

A) Headache B) Seizures C) Fever D) Vomiting

37-Prion diseases affect:

A. Humans only B. Animals only

C. Both humans and animals D. Peripheral nerves only

38-Diseases caused by prions are also called:

A. Retroviral encephalitis B. Transmissible spongiform encephalopathies

C. Demyelinating disorders D. Autoimmune encephalitis

39-The hallmark pathological feature of prion diseases is:

A. Demyelination of neurons B. Vascular infarcts

C. Spongy, “Swiss cheese-like” holes in the brain parenchyma D. Neuronal hyperplasia

1. A	2. D	3. A	4. D	5. B
6. C	7. C	8. A	9. B	10.E
11.C	12.C	13.D	14.A	15.D
16.D	17.A	18.A	19.C	20.D
21.C	22.C	23.D	24.B	25.C
26.C	27.B	28.B	29.C	30.B
31.A	32.B	33.C	34.B	35.B
36.A	37.C	38.B	39.C	40.

Meningitis

1- Which of the following is the most common cause of neonatal meningitis? {Summar2025}

a. Haemophilus influenzae b b. Streptococcus agalactiae

c. Neisseria meningitides d. Streptococcus pneumoniae

2-Which pathogen is a common bacterial cause of meningitis? {Summar2025}

a. Pseudomonas aeruginosa b. Staphylococcus epidermidis

c. Enterobacter aerogenes d. Streptococcus pneumoniae

3-The most common cause of hospital acquired meningitis is …. {Final2025}

a. Pseudomonus aeruginosa b. Streptococcus pneumonia

c. Streptococcus agalactiae d. Haemophilus influenzae b

4-The classical triad of meningitis is …. {Final2025}

a. Headache, fever , Neck stiffness

5-Causes of non- suppurative meningitis include all, except: {Mid2024}{Formative2025} {Summar2025}

a) Tuberculosis b) Meningococci c) Fungal d) Viral

6-The following is incorrect as regards to CSF changes in bacterial meningitis. {Formative2025}

a) Decrease sugar content. b) Increase protein content

c) Decrease protein content d) Presence of neutrophils.

7-The most common cause of community acquired meningitis is ….

a. Pseudomonus aeruginosa b. Pneumococci

c. Streptococcus agalactiae d. Haemophilus influenzae b

8-The most common cause of meningitis in young children is ….

a. Pseudomonus aeruginosa b. Pneumococci

c. Streptococcus agalactiae d. Haemophilus influenzae b

9-All Neisseria species give which of the following biochemical reactions?

A. Catalase negative B. Oxidase positive C. Urease positive D. Coagulase positive

10-- Which of the following organisms causes epidemic meningitis cases at college campuses? {Mid2021}{Formative2023}

A- Hemophilus influenzae type b B- Neisseria meningitides C- Streptococcus pneumoniae

D- Listeria monocytogenes E- M. Tuberculosis

11-Fungal meningitis is caused caused by which of the followings? {Mid2023}{Formative2023}

a.Cryptococcus neoformans. b. lesteria monocytogenes

c. Aspergillus flavus d. pneumococci

12-CAMP test is used for identification of: {Formative2024}

a) H. influenzae b) E. coli c) Pseudomonas d) Str. agalactia

13-……is the commonest causative organisms of septic meningitis. {سؤال في المحاضرة}

a. Tuberculosis b. Entero-viruses c. Meningococci d. staphylococci

14-All are characters of CSF in septic meningitis except : {سؤال في المحاضرة}

a.Increased tension b. Presence of many PNLS, pus cells and bacteria.

c. Elevated protein content. d. increased sugar content.

15-CSF in acute septic meningitis does NOT show: {ثناء}

A. Increased amount B. Increased protein content C. Increased sugar content

D. Increased polymorphonuclear cells E. Organisms

16-All are characters of Viral meningitis except {سؤال في المحاضرة}

a. Less fulminate clinical course than septic meningitis. b. Self-limited.

c. Commonly due to entero-virus infection. d. There are no distinctive macroscopic findings

e. CSF shows increased number of neutrophils, marked protein elevation and decreased glucose content

17- Mode of transmission of meningitis {Final2024}

A) air droplets

18- Which of the following Causes aseptic meningitis ? {Final2024}

A- mycobacterium Tuberculosis B- H. influenza C- N. meningitides D- Pneumococci

19- Choose the CORRECT answer; encapsulated yeast transmitted by inhalation of soil contaminated with bird excreta and results in meningitis ? {Mid2021}

A- Cryptococcus B- Blastomyces C- Candida

D- Aspergillus E- None of the above

20- Which procedure is most likely used to test for meningitis ? {Mid2021}

A- Throat culture B- Lumber puncture C- Tuberculosis skin test

D- None of the above E- CAMP test

21- meningitis can be caused by: {Formative2022}

a. bacteria b. fungus c. virus d. all of the above

22-all of the followings are examples of bacterial meningitis except: {Formative2022}

a. Streptococcus pneumonia b. Neisseria meningitides

c. Listeria monocytogenes d.cryptococcus

23- Acute bacterial meningitis is transmitted by : {Mid2023}

A- Blood transfusion B- aerosolization of respiratory tract secretion

24-Causes of septic meningitis include all of the followings except:- {Formative2025}

a) Cryptococcus neoformans b) H. influenzae c) N. meningitidis d) Pneumococci

25- Which Of The Following Indicate Septic Meningitis ?

A- Increase in number of neutrophils (PML)

26- .....is the commonest causative organism of septic meningitis. {Mid2024}

a) Tuberculosis b) Entero-viruses c) Meningococci d) Staphylococci

27- Meningococcal infection is transmitted commonly by {Mid2024}

a) Droplet infection b) Blood transfusion c) Swallowing d)Skin inoculation

28-Serogroup A meningococcus is most commonly associated with:

A. Sporadic meningitis in children B. Epidemic meningitis worldwide

C. Neonatal sepsis D. Urinary tract infections

29- Meningitis is described as septic or pyogenic when the CSF contains mainly: {قسم}

a. Polymorphonuclear leucocytes as inbacterial meningitis.

b. Lymphocytes as in viralmeningitis.

c. Polymorphonuclear leucocytes as infungal meningitis.

d. Lymphocytes as inbacterial meningitis.

30- N. meningitidisis one of the most common causes of bacterial meningitis and appears microscopically as: {قسم}

a. Gram positive lancet-shaped cocci (i.c. oval with pointed ends) arranged in pairs or short chains.

b. Gram negative cocci arranged in pairs with the adjacent sides flattened; "kidney shaped appearance".

c. Gram positive cocci arranged in grape-like clusters, non motile and non spore forming.

d. Gram positive cocci arranged in chains. Some strains are capsulated.

31- Satellitism can occur with: {قسم}

a. Listeria monocytogenes b. M. tuberculosis c. H. influenza d. Str. Pneumoniae

32-……. appears as yeast cells with gelatinous capsule and causes meningitis. {قسم}

a. Candida albicans b. Cryptococcus neoformans

c. Aspergillus Niger d. Aspergillus flavus

33- A diagnosis of meningitis is made through the analysis of CSF. Which of the following results would be expected with a diagnosis of bacterial meningitis? {قسم}

a) Cloudy CSF, elevated WBC, elevated protein and decreased glucose

b) Clear CSF, decreased WBC, elevated protein and elevated glucose

c) Cloudy CSF, decreased WBC, decreased protein and elevated glucose

d) Clear CSF, elevated WBC, decreased protein and decreased glucose

34- Classic symptoms of meningitis usually include... {قسم}

a) Fever, headache, stiff neck b) Backache, hallucinations, indigestion

c) Rash, inner ear pain, itching d) Dry skin, dehydration, cold sores

35- Which procedure is most likely used to test for meningitis? {قسم}

a) Throat culture b) Spinal tap

c) Tuberculosis skin test d) None of the above

36-The appropriate specimen to diagnose a meningococcal carrier is:

A) Nasopharynx swab B) Sputum C) Blood D) Urine

37-How many meningococcal serogroups are recognized?

A. 5 B. 10 C. 13 D. 20

38-Meningococcal serogroups are classified based on:

A. Outer membrane proteins B. Pili types

C. Capsular polysaccharide antigens D. Lipooligosaccharide (LOS)

39-Meningococci primarily colonize in the:

A) Nasopharynx B) Oropharynx C) Stomach D) Skin

40-Throat swabs in meningococcal infection are primarily used to:

A. Diagnose meningitis B. Detect septicemia

C. Detect the carrier state D. Assess immune response

41-CSF is obtained by:

A. Venipuncture B. Bone marrow aspiration

C. Lumbar puncture under aseptic conditions D. Thoracentesis

42-In septic meningitis, the CSF is typically:

A. Clear and colorless B. Under tension and turbid due to large numbers of pus cells

C. Straw-colored with low protein D. High glucose and low protein

43-In septic meningitis, CSF protein levels are usually:

A. Decreased B. Normal C. Elevated D. Undetectable

44-In bacterial meningitis, CSF glucose levels are typically:

A. Normal B. Reduced C. Elevated D. Unchanged

1. B	2. D	3. A	4. A	5. B
6. C	7. B	8. D	9. B	10.B
11.A	12.D	13.C	14.D	15.C
16.E	17.A	18.A	19.A	20.B
21.D	22.D	23.B	24.A	25.A
26.C	27.A	28.B	29.A	30.B
31.C	32.B	33.A	34.A	35.B
36.A	37.C	38.C	39.A	40.C
41.C	42.B	43.C	44.B	45.

Botulism

1- Food poisoning is caused by: {Summar2025}

a. Clostridium tetani b. Clostridium welchi

c. Clostridium septicum d. Clostridium botulinum

2- All are true about clostridium botulinum except {Final2024}

a. Causes flaccid paralysis b. Causes vomiting and diarrhea

3-Which of the following statements about Clostridium botulinum is true? {Formative2024}

a) Produces a very potent neurotoxin

b) Intoxication resulting from ingestion of canned food containing the preformed neurotoxin

c) Causes spastic paralysis

d) Both a and b

4- Death from C botulinum food poisoning is commonly due to {Final2021}

A- Electrolyte imbalance B- Marked fluid loss C- Cardiac failure

D- Renal failure E- Respiratory failure

5- Death from botulinum toxin occurs due to {Final2025}

a. Respiratory failure

6-Which is NOT TRUE about botulism? {Final2023}

a) Occur in infants by swallowing already formed toxins b) intestinal protoxin form

c) Encoded by bacteriophage

7-Botulism prevents the release of the neurotransmitter:

a. Acetylcholine. b. Glycine. c. Dopamine. d. GABA. e. Norepinephrine

8-Tetanospasmin prevents the release of the neurotransmitter

a. GABA and Glycine b. Norepinephrine c. Epinephrine d. Acetylcholine

9-One-way parents can help prevent botulism in a baby is to {قسم}

a) Have the child vaccinated b) Not give honey to a child under 1 year old

c) Stay away from swimming pools d) All of the above

10-Which is a symptom of foodborne botulism {قسم}

a) Drooping eyelids b) Double vision c) Slurred speech d) All of the above

11- Botulism is a type of food poisoning due to ingestion of inadequately sterilized canned food leading to: {قسم}

a. Flaccid paralysis b. Spastic paralysis c. Diarrhea and vomiting d. Meningitis

1. D	2. B	3. D	4. E	5. A
6. A	7. A	8. A	9. B	10.D
11.A	12.	13.	14.	15.

Tetanus

1- All of the following is true about tetanoplasmin toxin except …….. {Final2025}

a. It's endotoxin

b. Giving antitoxin is ineffective after its entry to nerve cells

c. It takes several weeks to cause disases

d. It's produced by gram positive rods with bulging terminal spores

2-Which of the following is false about Clostridium tetani: {Formative2025}

a) Causes spastic paralysis b) Produces tetanospasmin

c) Causes flaccid paralysis d) Gram positive bacilli with drum-stick appearance

3-The neurotoxin released by Clostridium tetani is:

a. Tetanolysin. b. Tetanospasmin. c. Aflatoxin. d. Bacteriocin

4-The best way of preventing tetanus {Final2023}

a) Debridement of the injury b) Toroidal vaccine c) Tetanus toxoid

5- The most effective way of preventing tetanus before exposure to: {Summar2025}

a. Surgical debridement of the wound b. Hyperbaric oxygen

c. Antibiotics d. Tetanus toxoid

6- Which is the most common source of infection that leads to tetanus? {Formative2022}

a. Chronic skin ulcers b. Intranasal inhalation c. Minor wounds d. Corneal abrasions

7- C tetani causes lockjaw paralysis because {Final2021}{Final2024}

A- The organism enters muscle cells and blocks contraction

B- The organism enters muscle cells and stimulates contraction

C- Its neurotoxin blocks the release of inhibitory mediator leading to generalized muscle spasm

D- Its neurotoxin stimulates the release of inhibitory mediator leading to localized muscle spasm

E- Its neurotoxin inhibits release of acetyl choline

8-…….is a potent neurotoxin secreted by Clostridium tetani {قسم}

a. Tetanolysin b. Alpha toxin c. Tetanospasmin d. Theta toxin

9- Tetanus can be transmitted by the following methods except: {قسم}

a. Contamination of wounds with street dust containing the spores.

b.Spores can also be introduced by "skin-popping," a technique used by drug addicts to inject drugs into the skin.

c. Infection via umbilical stump.

d.Feco-oral

10- A 30-year-old woman comes to the emergency department because she had sudden onset of muscle spasm and neck stiffness 2 hours ago. She also has had headache and restlessness during this time. Possible diagnoses include tetanus and meningoencephalitis. In addition to muscle spasm, which of the following findings on further evaluation is most likely to indicate that this patient has tetanus rather than meningoencephalitis? {قسم}

a) Normal cerebrospinal fluid b) Seizures

c) Sensitivity to light d) Trouble thinking clearly

1. A	2. C	3. B	4. C	5. D
6. C	7. C	8. C	9. D	10.A

Parasitology

1-Secondary amoebic brain abscess is caused by: {Summar2025}

a) Acanthamoeba b) Entamoeba histolytica c) Naegleria fowleri d) Entamoeba coli

2- Coenuros cerebralis is caused by {Final2024}

A- Echinococcus granulosus B- Heterophyes heterophyes

C- Diphyllobothrium mansoni D- Multiceps multiceps

3-Sleeping sickness is caused by…….. {Final2023}

a) tenia solium b) toxoplasma c) trypanosoma brucei d) plasmodium falciparum

4-Trypanosome gambiense is transmitted by: {Formative2024}

a) Glossina palpalis b) Glossina morsitans c) Sand fly d) Triatoma species

5- Adult stage of coenurus cerebralis is …….. {Final2025}

a. Multiceps multiceps b. Echinococcus granulosus

6- GAE is caused by ….. {Final2025}

a. Acanthameba spps

7- Cerebellar cysticercosis is caused by …… {Final2025}

a. Tenia solium

8- Coenurosis cerebralis is caused by….. {Final2023}

a) Echinococcus granulosus b) Heterophyes heterophyes

c) Diphyllobothrium mansoni d) Multiceps multiceps

9-Coenurosis occur due to the invasion of……..to the brain tissue.

a. Egg of T. multiceps b. Larval stage of E. granulosis.

c. Egg of E. granulosis. d. Larval stage of T. multiceps

10- Cerebral malaria is considered one complication of {Mid2021}

A- Trypanosoma cruzi B- Trypanosoma brucie gambiense C- Plasmodium falciparum

D- Plasmodium malariae E- Leishmania tropica

11-Cerebral malaria is one of complications of plasmodium falciparum due to …. {Final2025}

a. Adhesive phenomenon of the infected erythrocytes

12- Winterbottom's sign is a clinical manifestation of : {Mid2021}

A- Chagas disease B- Sleeping sickness C- Amebiasis D- Malaria E- GAE

13- Free living amoeba Naeglaria fowleri major cause of {Mid2021}

A- PAM B- Corneal ulceration C- GAE

D- Acute sleeping sickness E- Nagana

14-Which of the following is the causative parasite for primary amoebic meningoencephalitis (PAM)? {Formative2025}

a) Acanthamoeba b) Amoeba proteus c) Entamoeba histolytica d) Naegleria fowleri

15-Handling raw meat with abraded hand may transmit: {Formative2025}

a) T. saginata. b) Trichinella spiralis. c) Toxoplasmosis. d) Taenia solium.

16- All of the following Protozoa affecting CNS EXCEPT : {Mid2021}

A- Free living amoebi B- Trypanosoma brucei subspecies C- Giardia lamblia

D- Encephalitozoon E- Entamoeba histolytica

17- G. Palpalis transmitte ? {Mid2021}

A- T.B. gambiense B- T.B. rhodesiense C- Genus Acanthamoeba

D- Plasmodium falciparum E- Toxoplasma gondii Answers

18- Infection with Naeglerla fowleri cause: {Formative2023}

a. Primary amebic meningeoencephalitis b. Granulomatous amebic encephalitis

c. Chronic sleeping sickness d. secondery amebic meningeoencephalitis

19- The causative parasite of neurocysticercosis is {Final2021}

A- Taenia saginata B- Taenia solium C- Echinococcus aranulosus

D- Hymenolepis nana E- Trichnilla spiralis

20-Cysticercosis occurs when……penetrates intestinal wall to ……. {Formative2024}

a) Onchosphere – various organs b) Cysticercoids – liver

c) Cysticercus—other organs d) Cysticercus—liver

21- The following helminthes affecting CNS EXCEPT {Final2021}

A- Diphyllobothrium latum B- Tineasolium C- Multiceps multiceps

D- Toxocaracanis E- Diphyllobothrium mansoni

22- which parasite cause PAM : {Mid2023}

A- Acanth ameba B- Entamoeba histolytica C- Naegleria fowleri

23- Cerebral Sparganosis is caused by : {Mid2023}{Final2025}

A- plerocercoid of Diphyllobothrium mansoni B- Procercoid of Diphyllobothrium latum

C- Plerocercoid of Diphyllobothrium latum

24- Coeneorus cerebalis is a larval stage of {Final2021} Final2022}

A- Echinococcus granulosus B- Heterophyes heterophys C- Multiceps multiceps

D- Toxocaracati E- Schistosoma mansoni

25-Cerebral schistosomiasis is more common in the infection with….

a. S. mansoni egg. b. S. japonicum egg. c. S. haematobium egg d. H. heterophyes egg.

26-Infective stage in neurocysticercosis is ….

a. Egg of H. heterophyes. b. Encysted larvae in tissue.

c. Eggs of T. solium. d. Cysticercus cellulosae

27-One of solitary cystic helminthic brain lesions is….

a. Toxocariasis. b. Strongyloidiasis. c. Hydatidosis d. Schistosomiasis.

28-- morula cells of Mott, these are modified plasma cells present in {Final2021}

A- cutaneous Leishmaniasis B- chaga's disease C- African Trypanosomiasis

D- PAM E- None of the above

29- morula cells of Mott, these are modified plasma cells used for diagnosis of {Final2024}

A- cutaneous Leishmaniasis B- chronic fleeping sickness C- chronic sleeping sickness

30- the causative parasite of Amebic brain abcess is {Final2021}

A- Cyst of Entamebae histolytica B- Trophozoite of Entamebae histolytica

C- cyst of Entamebae coli D- trophozoite of Entamebae coli

E- all of the above

31- The adult of hydatid cyst is: {Summar2025}

a) Taenia solium b) Taenia saginata c) Echinococcus granulosus d) Multiceps multiceps

32-Cerebral malaria caused by Plasmodium falciparum is due to: {Summar2025}

a) Immune suppression b) Adhesive phenomenon in cerebral vessels

c) Activation of sporozoites d) Premunition

1. B	2. D	3. C	4. A	5. A
6. A	7. A	8. D	9. D	10.C
11.A	12.B	13.A	14.D	15.C
16.C	17.A	18.A	19.B	20.A
21.A	22.C	23.A	24.C	25.B
26.C	27.C	28.C	29.C	30.A
31.C	32.B	33.	34.	35.